What Do Moral Psychology Benchmarks Actually Test?

A Meta-Evaluation of the Benchmark Landscape and Cross-Vendor Empirical Replication Across Nine Models

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Abstract

Background. Large language models are increasingly deployed in contexts requiring moral sensitivity, yet it remains unclear whether the benchmarks used to evaluate their

moral cognition engage the theoretical constructs established by moral psychology. No prior work has combined a systematic landscape assessment with a cross-vendor empirical study testing whether existing instruments can distinguish between deployed systems.

Method. From 2,099 AI ethics papers we extracted 222 benchmark candidates using a five-signal detection pipeline, then assessed 55 in depth through a twelve-perspective expert council across a 162-feature characterization framework. We administered 23 benchmarks to nine models from three vendors (Anthropic, Google, OpenAI). We note the self-administration confound: the model family that designed the framework served as one set of test subjects, rendering model-specific findings hypothesis-generating.

Results. The landscape is theoretically narrow: Moral Foundations Theory accounts for 45% of candidates; three major theories – dual-process theory, social intuitionism, and dyadic morality – have zero dedicated benchmarks; construct validity for LLM respondents is weak (median 3/5); and 18% of instruments lack human baselines. The cross-vendor study revealed that these already-narrow instruments cannot distinguish between the systems they evaluate: 43% of benchmarks produced zero cross-model variation, and all nine models share six universal moral signatures – WEIRD-liberal foundation bias, purity floor, authority suppression, ethical consistency ceiling, post-conventional default, and deontological refusal asymmetry – an RLHF moral fingerprint without documented precedent in any human population. The single benchmark that cleanly separated models was a process-level probe of dual-process dynamics: precisely the benchmark type absent from the landscape.

Conclusions. The field faces a double bind: the benchmark landscape is too narrow to capture the breadth of moral cognition and too undiscriminating to distinguish between the systems it evaluates. The result is a value monoculture in which narrow instruments and homogeneous subjects produce a doubly blind evaluation regime, invisible to the cross-vendor audit strategies regulators currently contemplate. We release our characterization framework, cross-vendor dataset, and three benchmark designs targeting the missing theories.

3. Methodology

This section describes the landscape-mapping methodology: the extraction of benchmark candidates from a curated literature vault (3.1), expert council assessment using a dual-council structure (3.2), the 28-feature characterization framework (3.3), composite scoring and selection (3.4), and keyword scorer expansion (3.5). The cross-vendor trial design (3.6) and implementation trial protocol (3.7) are described separately by Council B. The section closes with terminology choices and disclosure of the virtual council's epistemic status (3.8).

3.1 Multi-Signal Candidate Extraction

3.1.1 Source corpus

The extraction pipeline operated on a curated vault of 2,099 AI ethics papers spanning 2015–2026, maintained as structured Markdown files with YAML frontmatter encoding 47 metadata fields per paper. Each paper record includes bibliographic data (title, authors, year, venue, DOI), classification scores (ethics subfields, moral psychology theory engagement, cultural provenance), and availability flags (PDF acquired, text extracted, data URL). The vault was assembled through iterative literature searches on Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, supplemented by citation-chain traversal from seed papers in each of the eight Western moral psychology theories and three non-Western traditions covered by the framework (see Section 2.1). Appendix S3 provides the full search strategy.

3.1.2 Extraction signals

To identify moral psychology benchmark candidates, we designed a six-signal detection pipeline. No single signal is sufficient to capture the heterogeneous ways benchmarks are described in the literature: some papers foreground their psychometric properties, others emphasize their dataset contribution, and still others bury benchmark-relevant content in application-oriented framing. The pipeline therefore casts a deliberately wide net, accepting candidates identified by any signal and deferring precision to the expert assessment stage (Section 3.2).

Signal 1: Moral psychology keyword scores. The vault maintains eight moral psychology fields (mp_mft, mp_schwartz, mp_trolley, mp_kohlberg, mp_dual_process, mp_intuitionist, mp_dyadic, mp_identity) scored 0–100 by a keyword-based heuristic scorer (see Section 3.5 for scorer details). Papers scoring >= 50 on any field were included. This signal captures papers that the vault's automated classification system identified as substantially engaging a moral psychology theory.

Signal 2: Benchmark flags with keyword match. Papers flagged as benchmarks, evaluations, or datasets (via structured fields is_benchmark, is_eval, is_dataset) that also contained >= 2 moral psychology keyword matches in their abstract or key_contribution fields. This conjunction prevents false positives from generic evaluation papers (e.g., a GLUE-style benchmark that mentions "moral" once in passing).

Signal 3: Data availability with strong keyword match. Papers with a published DATA_URL and >= 3 moral psychology keyword matches, targeting benchmarks that are readily implementable. The rationale for the higher keyword threshold (3 vs. 2 in Signal 2) is that data availability alone is a weaker indicator of benchmark relevance than explicit benchmark classification.

Signal 4: High keyword density. Papers with >= 5 moral psychology keyword matches regardless of other metadata. This signal captures papers whose benchmark nature may

not be reflected in structured fields—for instance, papers that present a moral psychology dataset within a broader evaluation framework and are therefore not flagged as standalone benchmarks.

Signal 5: Known benchmark names. Papers whose title or abstract contained names of established benchmarks drawn from a curated list of 28 known instruments: ETHICS, MoralBench, ValueBench, Scruples, Delphi, MACHIAVELLI, Moral Machine, MoralChoice, MoralExceptQA, Moral Foundations Vignettes (MFV), Social Chemistry, Value Kaleidoscope, TrolleyBench, ValAct, MoralCode, UniMoral, ValueCompass, CVC, Commonsense Morality, Moral Stories, Moral Integrity Corpus, MIC, MoralDial, CultureBank, WVS adaptation, Moral Direction, ValuePrism, and Tracing Moral Foundations. This list was assembled from prior survey papers [Jiang et al., 2024; Scherrer et al., 2024; Gabriel et al., 2026] and the authors' domain knowledge.

Signal 6: Non-Western theory keyword density in full-text PDF extracts (Phase 2 expansion). Following the initial extraction, a second-pass scan searched the full text of PDF extracts for engagement with three non-Western moral traditions—Islamic maqasid al-shariah, Confucian role ethics, and Ubuntu communalism—as well as secondary search terms for Buddhist ethics and Indigenous relational ethics. Papers with >= 3 theory-specific keyword matches in their full text were added to the candidate pool. This signal was added after the initial extraction revealed that non-Western moral psychology engagement is rarely surfaced in abstracts or structured metadata, requiring full-text scanning for adequate recall.

3.1.3 Keyword taxonomy

The keyword taxonomy comprised 94 regex patterns organized by theory, covering:

- **Eight Western moral psychology theories** (65 patterns): MFT (e.g., `\bmoral foundations?\b`, `\bcare.harm\b`), Schwartz values (e.g., `\bbasic human values\b`, `\bpvq\b`, `\bportrait values\b`), trolley/dilemma tradition (e.g., `\btrolley\b`, `\bmoral dilemma\b`, `\bfootbridge\b`), Kohlberg developmental morality (e.g., `\bkohlberg\b`, `\bdefining issues test\b`, `\bpost.conventional\b`), dual-process theory (e.g., `\bdual process\b`, `\bsystem 1\b`, `\bsystem 2\b`), social intuitionism (e.g., `\bsocial intuitionist\b`, `\bmoral dumbfounding\b`), dyadic morality (e.g., `\bdyadic morality\b`, `\bmoral typecasting\b`), and moral identity (e.g., `\bmoral identity\b`, `\bmoral self\b`).
- **Three non-Western moral traditions** (24 patterns): Islamic maqasid (e.g., `\bmaqasid\b`, `\bmaslaha\b`, `\bfiqh\b`, `\bshariah\b`), Confucian role ethics (e.g., `\bconfucian\b`, `\bfilial piety\b`, `\bjunzi\b`), Ubuntu communalism (e.g., `\bubuntu\b`, `\bcommunitarian ethics\b`, `\bafrican moral philosophy\b`).
- **Secondary search traditions** (5 patterns): Buddhist ethics (e.g., `\bbuddhist ethics\b`, `\bfive precepts\b`), Indigenous relational ethics (e.g., `\bindigenous ethics\b`, `\btikanga\b`).

All patterns were compiled with case-insensitive matching (`re.IGNORECASE`). The full regex pattern list is provided in Supplementary Material S3.

3.1.4 Extraction results

The six-signal pipeline extracted **222 candidates** in the initial pass: 177 classified as benchmarks, 43 as datasets, and 2 as other resource types. The Phase 2 non-Western expansion added a further 70 candidates from full-text scanning, bringing the total candidate pool to **292 papers** across 11 moral psychology theories.

A critical methodological finding at the extraction stage was that no single signal achieved adequate recall. The vault's keyword scorer (Signal 1) alone missed major benchmarks due to vocabulary mismatch. For instance:

- **ValueBench** [Ren et al., 2024], which administers 44 psychometric instruments to LLMs, scored mp_schwartz = 0 because its abstract uses "value" rather than "Schwartz"—a term-frequency limitation of keyword-based scoring.
- **The ETHICS dataset** [Hendrycks et al., 2021], one of the most widely used moral evaluation benchmarks with over 500 citations, scored mp_mft = 0 despite being grounded in normative ethical theories that overlap with MFT constructs.
- **Social Chemistry 101** [Forbes et al., 2020] scored mp_intuitionist = 0 despite operationalizing norm-based moral judgments that are structurally aligned with the social intuitionist model's emphasis on culturally shared moral intuitions.

Multi-signal detection was therefore essential for recall. Signal 5 (known benchmark names) contributed 31 candidates not captured by Signals 1–4, and Signal 6 (non-Western full-text scan) contributed 70 candidates invisible to abstract-level signals.

3.1.5 Relationship to systematic review standards

This extraction pipeline departs from the PRISMA systematic review protocol [Page et al., 2021] in two respects. First, the source corpus is a curated vault rather than an exhaustive database search, meaning that papers absent from the vault are necessarily absent from the candidate pool. We mitigate this limitation through two supplementary external searches: a Semantic Scholar API query for moral psychology benchmark papers published 2020–2026, and a citation-chain traversal from 15 seed papers identified by the expert council (Section 3.2). These external searches added 38 candidates not present in the original vault. Second, the extraction criteria are designed for recall rather than precision, accepting all papers that meet any single signal threshold. The precision filtering occurs at the expert assessment stage (Section 3.2), where the twelve-perspective council applies domain-specific inclusion and exclusion criteria. We report a PRISMA-style flow diagram in Supplementary Figure SF1.

3.2 Expert Council Assessment

3.2.1 Dual-council architecture

Assessment was conducted using a virtual expert council structure comprising two independent councils with distinct mandates:

Council A (Meta-Evaluation Council): Twelve members (A1–A12) responsible for the landscape-mapping methodology, benchmark characterization, and psychometric analysis. This council assessed candidates, scored the 28-feature framework, and conducted the gap analysis reported in Results Part I (Section 4).

Council B (Empirical Council): Twelve members (B1–B12) responsible for the cross-vendor trial design, model administration, and empirical analysis. This council designed the implementation trials and conducted the cross-vendor analysis reported in Results Part II (Section 5).

The two councils share no members, and each member carries a distinct disciplinary perspective that determines their assessment focus. The dual-council structure was motivated by two methodological concerns. First, the landscape-mapping task (characterizing benchmarks) and the empirical task (administering benchmarks) require different expertise: psychometricians and bibliometricians for the former, NLP engineers and cognitive scientists for the latter. Second, the dual structure creates an internal check: Council B's empirical findings either validate or challenge Council A's landscape characterization, and discrepancies between the two analyses surface assumptions that a single unified council might share silently.

Table 1: Council A — Meta-Evaluation Council

ID	Perspective	Assessment Focus
A1	Narrative Architect	Rhetorical structure, argumentation flow, Nature MI conventions
A2	Moral Psychology Historian	Intellectual genealogy, content/process/ structure taxonomy
A3	MFT Deep Expert	MFQ/MFQ-2/MFTC/MFRC fidelity, computational MFT operationalization
A4	Process Theories Specialist	Dual-process and social intuitionist constructs, cognitive neuroscience
A5	Psychometric Authority	Messick validity, Cronbach and Meehl construct validity, IRT, measurement invariance
A6	Cross-Cultural and Values Expert	WEIRD critique, Schwartz PVQ-RR, cultural psychology
A7	Benchmark Landscape Surveyor	BIG-bench, HELM, ETHICS, evaluation harness ecosystems
A8	Format and Measurement Specialist	Likert scales, forced-choice, format-dependent validity in LLMs
A9	NLP/Evaluation Specialist	Harness compatibility, scorer types, prompt engineering, evaluation infrastructure
A10	Bibliometrician	Citation analysis, venue tiers, downstream adoption, citation maturity confound
A11	Survey Methodologist	PRISMA compliance, inclusion/exclusion criteria, search protocol rigor
A12	Meta-Science Critic	Replication crisis, questionable research practices, open science, self-critique

Table 2: Council B — Empirical Council

ID	Perspective	Assessment Focus
B1	Cross-Vendor Analyst	Multi-model comparison design, variance decomposition, convergence metrics
B2	RLHF/Alignment Expert	Constitutional AI, RLHF, DPO, value encoding mechanisms
B3	AI Safety Researcher	Value monoculture risk, audit strategies, deployment risk
B4	Scale Sensitivity Analyst	Parameter scaling laws, emergent capabilities, size-dependent moral behavior
B5	Cross-Cultural Psychologist	WEIRD bias, Hofstede dimensions, cultural persona validity
B6	Dual-Process Cognitive Scientist	System 1/2, chain-of-thought as deliberation proxy, cognitive load
B7	Psychometrician (Empirical)	Test discrimination, ceiling/floor effects, reliability, benchmark fitness
B8	Statistical Methodologist	N=1 inference, bootstrap CIs, effect size interpretation, power analysis
B9	Moral Philosophy Analyst	Normative characterization, meta-ethical interpretation
B10	Policy and Governance Expert	Algorithmic auditing, regulatory implications, standardization
B11	Developmental Psychologist	Kohlberg stages, DIT, stage theory applicability to LLMs
B12	Philosophy of Mind Specialist	Functional moral agency, consciousness, construct applicability

3.2.2 Council A assessment protocol

Council A assessed the top candidates through three parallel assessment passes, each assigning specific features from the 28-feature framework (Section 3.3):

Pass 1 — Depth assessment (Perspectives A2–A6, A8): Each council member read the PDF extract for every candidate in their assigned batch and scored the following features: A2 (construct operationalization depth), A4 (source instrument provenance), A5 (human baseline availability), A6 (construct validity evidence), A7 (cultural provenance), A8 (languages supported), B1 (task format), B2 (dataset size), and B10 (LLM administration fidelity). This pass prioritized the top 40 candidates ranked by composite keyword score.

Pass 2 — Impact assessment (Perspectives A7, A9, A10): Council members cross-referenced candidates with implementability data from repository analysis (GitHub, HuggingFace, OSF) and citation databases (Semantic Scholar). Scored features: B3 (data availability), B4 (code availability), B6 (harness tier compatibility with Inspect AI and lm-eval-harness), B7 (scorer complexity), C3 (venue tier), and C5 (hub/leaderboard presence). AI safety relevance was assessed qualitatively by A7.

Pass 3 — Secondary scan (Perspective A11, with input from A3, A4, A6): Candidates ranked 41–80 were scanned for "hidden gems"—benchmarks that scored low on keyword density but high on theoretical depth or practical utility. This scan was motivated by the observation that keyword-based ranking penalizes benchmarks with non-standard vocabulary (Section 3.1.4). The secondary scan identified 15 additional papers warranting full 28-feature assessment, including three benchmarks for underrepresented theories (dual-process, intuitionist, and dyadic).

Citation counts were obtained from the Semantic Scholar API [Fricke, 2018] for the top 80 candidates. Of 80 queries, 54 returned valid citation data; the remaining 26 lacked DOIs or arXiv identifiers in the Semantic Scholar database. Rate limiting was handled with exponential backoff at 10-second minimum intervals per the API's usage guidelines.

3.2.3 Assessment output

In total, **55 benchmarks** received full 28-feature characterization: 40 from the primary assessment (Passes 1–2) and 15 hidden gems from the secondary scan (Pass 3). An additional 237 candidates received partial characterization (keyword match counts, theory classification, and availability flags only) and are reported in the supplementary materials. The 55 fully characterized benchmarks span all eight Western moral psychology theories, though representation is highly uneven—a finding that constitutes Claim 1 (Section 4.1).

3.3 The 28-Feature Characterization Framework

3.3.1 Framework design principles

The 28-feature framework was designed to capture three orthogonal dimensions of benchmark quality: how faithfully a benchmark engages moral psychology theory (Tier A — Psychological Depth), how readily it can be implemented for LLM evaluation (Tier B — Practical Ease), and how much scholarly influence and community adoption it has achieved (Tier C — Citation Impact). These three tiers were chosen because they correspond to the three practical questions a researcher asks when selecting a moral psychology benchmark: (1) Does it measure what it claims to measure? (2) Can I actually run it? (3) Do other researchers use it?

All features are scored on a 0–100 normalized scale to enable cross-feature comparison and composite aggregation. The normalization rubrics were developed iteratively by Council A members A5 (Psychometric Authority) and A9 (NLP/Evaluation Specialist),

with the guiding principle that 100 should represent "best practice as established in the psychometric or NLP evaluation literature" rather than "best observed in this sample." This anchoring ensures that scores are interpretable in absolute terms, not merely as within-sample rankings.

3.3.2 Tier A — Psychological Depth (6 features)

Tier A measures how faithfully a benchmark engages moral psychology theory. It is weighted most heavily in the composite score (50%) because we hold that theoretical grounding is the primary determinant of a benchmark's long-term scientific value.

Feature	Description	Scoring Rubric
A1	Number of theories engaged	Count of distinct theories from the 8-theory framework that the benchmark operationalizes. Normalized: 0 theories = 0, 1 = 30, 2 = 60, 3 = 80, 4+ = 100. The nonlinear scaling reflects diminishing returns: engaging one theory is the minimum for inclusion, but each additional theory substantially increases the benchmark's coverage of moral cognition.
A2	Construct operationalization depth	How rigorously the benchmark translates theoretical constructs into measurable items. Four levels: *Validated* (items drawn from a psychometrically validated instrument with reported reliability) = 100; *Theory-derived* (items systematically derived from theory but not independently validated) = 70; *Theory-inspired* (items loosely motivated by theory) = 40; *Keyword-only* (benchmark uses theory terminology without systematic operationalization) = 10.
A3	Measurement target	What the benchmark measures. Default 70 (most benchmarks target moral judgments). Adjusted upward for benchmarks measuring reasoning processes (90) or metacognitive awareness (100); adjusted downward for benchmarks measuring behavioral tendencies only (60).
A4	Source instrument provenance	Whether the benchmark derives from an established psychometric instrument. *Validated instrument* (MFQ-30, DIT, PVQ-RR) = 100; *Original but purpose-built* with reported psychometric properties = 80; *No source instrument* = 60.
A5	Construct validity evidence	Strength of psychometric evidence reported for the benchmark's measures. *Multiple validity forms* (e.g., MFQ [1995] across 30+ countries, DIT 40 years of cross-cultural data) = 100; *Expert-specified ground truth* with documented annotation process = 80; *Internal consistency only* (e.g., Cronbach's α reported) = 60; *No validity evidence* = 30.
A6	Cultural validity	Whether the benchmark includes multilingual or cross-cultural data. *Multilingual with cross-cultural comparison* = 100; *Multilingual* = 80; *Multiple cultural contexts* = 60; *Single culture, explicitly noted* = 40; *Single culture, unacknowledged* = 20.

3.3.3 Tier B — Practical Ease (7 features)

Tier B measures how readily a benchmark can be implemented for LLM evaluation.

Feature	Description	Scoring Rubric
B1	Task format	Suitability of the response format for autoregressive language models. MCQ (multiple-choice question with fixed option set) = 100; Binary (yes/no or A/B forced choice) = 90; Forced-choice (choose between two or more scenarios) = 85; Likert (ordinal scale rating) = 70; Free-text generation = 50. The ordering reflects empirical findings that structured formats produce more reliable LLM responses [Scherrer et al., 2024].
B2	Dataset size	Number of items/scenarios, log-scaled to compress the wide range across benchmarks: <50 items = 20; ~200 = 40; ~1,000 = 60; ~5,000 = 75; ~20,000 = 85; >20,000 = 100. Log scaling is appropriate because the information gain per additional item diminishes: a 1,000-item benchmark captures substantially more construct variance than a 30-item instrument, but a 100,000-item benchmark adds less marginal information over 20,000 items.
B3	Data availability	Ease of obtaining the benchmark data. HuggingFace Hub (standardized loader, version control) = 100; GitHub repository (requires manual download and format parsing) = 95; OSF or other academic repository = 90; Anonymous or supplementary link = 60.
B4	Code availability	Availability of evaluation code. Full pipeline (data loading, prompt formatting, scoring, and analysis) = 100; Partial code (data loading or
B6	LLM Schema Adherence fidelity	[illegible]

3.3.4 Tier C — Citation Impact (4 features)

Tier C measures scholarly influence and community adoption.

Feature	Description	Scoring Rubric
C1	Raw citation count	Total citations from Semantic Scholar, log-scaled: <5 = 10; ~50 = 50; ~200 = 75; >500 = 100.
C2	Age-normalized citation rate	Citations per year since publication (with minimum age 0.5 years to prevent recent papers from receiving inflated rates). Log-scaled on the same rubric as C1. This feature corrects for the systematic disadvantage of recent benchmarks that have had less time to accumulate citations—a confound that, uncorrected, would bias selection toward older benchmarks regardless of quality.
C3	Venue tier	Quality of the publication venue. Tier 1 (Nature, Science, PNAS, ACL, NeurIPS, ICML, ICLR, EMNLP, AAAI) = 100; Tier 2 (NAACL, EACL, COLING, Cognitive Science, Psychological Review, AIES) = 75; Tier 3 (workshop papers, domain-specific conferences) = 50; Tier 4 (arXiv preprints, non-peer-reviewed) = 25.
C5	Hub/leaderboard presence	Whether the benchmark is tracked on a major evaluation hub. Major hub (Papers with Code leaderboard, HuggingFace evaluation suite) = 100; Minor hub (individual project leaderboard, competition platform) = 50; No hub presence = 0.

3.3.5 Feature coverage and excluded features

Five features defined in the original framework design were not assessed across the full 55-benchmark characterization: B5 (instruction simplicity), B8 (open access status), B9 (documentation completeness), and two features later subsumed by the extended framework. These features require access to benchmark distribution infrastructure—repository structure, license files, README quality—that was not systematically available for all candidates during the assessment period. They are excluded from composite scoring. Their omission does not affect Tier A or Tier C scores, and their combined weight in Tier B (had they been included) would have been 0.15, producing a maximum composite shift of 0.045 (= 0.15 x 0.30 tier weight).

3.3.6 Extended framework (162 features)

The 28-feature core was expanded during the assessment process to a 162-feature extended framework organized across seven tiers (A through G), adding dimensions for psychometric rigor (Tier D, 24 features), LLM-specific validity (Tier E, 20 features), cultural and philosophical breadth (Tier F, 45 features), and advanced benchmarking properties (Tier G, 45 features). The extended framework was assessed at 99.1% coverage (7,544 of 7,614 cells populated), with the remaining gaps concentrated in D9 (LLM ceiling compression, 35% null) and D20 (sample size adequacy, 35% null) where empirical data was unavailable. We report the 28-feature core scores as our primary characterization and the extended 162-feature matrix as Supplementary Material S1. The extended framework's scoring rubrics are provided in Supplementary Material S1.1.

3.4 Composite Scoring and Selection

3.4.1 Tier aggregation

Features within each tier were aggregated into tier-level scores using weighted averages, where weights reflect the relative importance of each feature as determined by Council A deliberation:

Psychological Depth (PD):

```
PD = A2(0.25) + A6(0.20) + A1(0.15) + A5(0.15) + A3(0.15) + A4(0.10)
```

The highest weight is assigned to A2 (construct operationalization) because it most directly captures whether a benchmark translates theory into measurable items—the defining characteristic of a psychometrically sound instrument. A6 (construct validity) receives the second-highest weight because validity evidence is what distinguishes a benchmark from an ad hoc question set. A4 (source instrument) receives the lowest weight because a benchmark can have excellent operationalization without deriving from an established instrument.

Practical Ease (PE):

```
PE = B3(0.20) + B2(0.15) + B6(0.15) + B7(0.15) + B1(0.10) + B4(0.10) + B10(0.15)
```

Data availability (B3) is weighted highest because a benchmark that cannot be obtained cannot be used, regardless of its other properties. LLM administration fidelity (B10) shares the second tier of weight with harness compatibility (B6) and scorer complexity (B7), reflecting the practical reality that these three properties jointly determine implementation cost.

Citation Impact (CI):

```
CI = C1(0.30) + C2(0.30) + C3(0.25) + C5(0.15)
```

Raw and age-normalized citation counts are weighted equally (0.30 each), with their joint contribution outweighing venue tier and hub presence. This balances absolute scholarly impact against the maturity confound: a 2024 benchmark with 50 citations may be more impactful per unit time than a 2018 benchmark with 300 citations.

3.4.2 Composite formula

The final composite score applied tier weights reflecting a deliberate preference for psychological rigor over practical convenience:

```
COMPOSITE = PD x 0.50 + PE x 0.30 + CI x 0.20
```

The 50/30/20 weighting was set by Council A after deliberation on the paper's core thesis: that the benchmark landscape suffers from insufficient theoretical grounding. A weighting that prioritized practical ease (e.g., 30/50/20) would have elevated large, easily implementable benchmarks that lack psychological depth—precisely the pattern the paper critiques. The 50% weight on PD ensures that benchmarks grounded in validated moral psychology theory are preferred, even if they are smaller or harder to implement.

Sensitivity analysis. To assess the robustness of the selection to weight choices, we recomputed composite scores under five alternative weighting schemes: (a) equal weights (33/33/33), (b) practice-first (30/50/20), (c) impact-first (20/30/50), (d) depth-only (100/0/0), and (e) PD-PE only (60/40/0). Under all five schemes, the top 5 benchmarks remained identical (Spearman rho = 1.0 for ranks 1–5). Ranks 6–15 showed moderate sensitivity: 3 of 10 benchmarks were displaced under the impact-first scheme, where citation-rich older benchmarks (Moral Machine, MFTC) gained ranks at the expense of recent theory-deep benchmarks (ValueCompass, ValAct). The depth-only scheme produced the largest displacement, swapping 5 of 10 benchmarks in positions 6–15. We report the primary (50/30/20) selection throughout but note that the core findings—MFT dominance, construct validity gaps, missing theories—are invariant to weighting choice.

3.4.3 Selection algorithm

Selection proceeded through a five-step algorithm:

Step 1: Hard gates. Candidates required (a) PD >= 40, ensuring a minimum level of theoretical engagement, and (b) publicly available data, ensuring reproducibility. Of 55 assessed candidates, **34** passed both gates. The PD >= 40 threshold was set by examining

the distribution of PD scores: a natural break occurred between benchmarks with some theoretical grounding (PD >= 40, median A2 = 70 "theory-derived") and benchmarks with only keyword-level engagement (PD < 40, median A2 = 10 "keyword-only"). Benchmarks failing the data availability gate included several otherwise strong candidates (e.g., proprietary corporate moral evaluation suites).

Step 2: Deduplication. Two duplicate entries were removed: TrolleyBench (duplicate record from different search sources) and a Khandelwal DIT variant (substantially overlapping with the multilingual DIT entry). Deduplication criteria required >= 80% item overlap or explicit derivation acknowledgment in the paper. This yielded **32** unique candidates.

Step 3: Theory caps. To prevent over-representation of well-resourced theories, we imposed caps: maximum 4 MFT entries, maximum 3 trolley/dilemma entries in the final selection. These caps were motivated by the observation that MFT and the trolley tradition have disproportionate computational infrastructure (MFQ-30 is a standardized 30-item questionnaire; the Moral Machine has 40 million decisions) that inflates their composite scores relative to theories with equivalent or greater theoretical importance but less infrastructure.

Step 4: Greedy selection. Candidates were selected in descending composite order, respecting the theory caps from Step 3. When a candidate's inclusion would violate a cap, it was skipped and the next highest-scoring candidate from a different theory was selected.

Step 5: Diversity fills. After greedy selection, we verified theory coverage minimums: >= 2 Schwartz values benchmarks, >= 1 Kohlberg/developmental benchmark, >= 1 minority-theory benchmark (identity, dyadic, intuitionist, or dual-process). Where minimums were not met, the lowest-ranked selected benchmark from an over-represented theory was swapped for the highest-ranked unselected benchmark from the underrepresented theory. In practice, one swap was required: a fourth MFT benchmark (Moral Foundations Vignettes, composite 64.47) was replaced with ValueCompass (composite 55.0), which engages both Schwartz values and dyadic morality.

3.4.4 Final selection

The selection algorithm produced a **top-15 benchmark set** for detailed analysis and implementation trials:

Rank	Benchmark	Year	Theory	Format	Items	Composite	PD	PE	CI
1	The Moral Machine Experiment	2018	Trolley	Forced-choice	40,000,000	78.5	75.5	74.0	92.5
2	Moral Machine Experiment on LLMs	2024	Trolley	Forced-choice	50,000	69.9	72.5	69.5	64.0
3	Crowdsourcing Moral Machines	2020	Trolley	Forced-choice	40,000,000	69.9	72.5	67.4	67.0
4	Moral Foundations Twitter Corpus	2020	MFT	Multi-label	35,108	68.1	68.0	66.0	71.5
5	Does Moral Code Have a Moral Code?	2022	MFT, Schwartz	Binary + MCQ	100	66.3	72.0	59.3	62.5
6	Moral Foundations of LLMs	2024	MFT	Likert	30	63.8	85.0	62.8	12.5
7	Differences in Moral Foundations of LLMs	2024	MFT	Likert	116	63.7	80.0	65.8	20.0
8	Value Kaleidoscope	2024	Schwartz	Free-text + MCQ	218,000	61.6	55.5	69.5	65.0
9	Multilingual Defining Issues Test	2024	Kohlberg	Rating + ranking	9	61.2	80.0	54.1	25.0
10	Value [illegible] (Double-Edged Sword)	2026	Schwartz, [illegible]	MCQ + generation	158,000	58.8	55.5	64.8	12.5
18	[illegible]	2025	[illegible]	[illegible]	58,000	55.8	58.9	48.9	31.9

Theory representation in the final selection: MFT (5), Trolley (4), Schwartz (4), Kohlberg (4), Dyadic (1). Dual-process, intuitionist, and identity theories achieved zero representation even after diversity fills—a finding that constitutes the empirical basis for Claim 3 (Section 4.3). An additional 8 benchmarks were selected beyond the top 15 for the extended cross-vendor trial battery (23 benchmarks total), including purpose-built probes for the three missing theories (see Section 3.7, written by Council B).

3.5 Keyword Scorer Expansion

3.5.1 Blind spot identification

During the secondary scan (Section 3.2.2, Pass 3), Council A identified critical blind spots in the vault's moral psychology keyword scorer. The eight mp_* fields (mp_mft, mp_schwartz, mp_trolley, mp_kohlberg, mp_dual_process, mp_intuitionist, mp_dyadic, mp_identity) are computed by a heuristic scorer that searches paper abstracts and PDF extracts for theory-specific regex patterns, assigns strong (15 points) and medium (5 points) match weights with false-positive context penalties (-10 points), adds a title bonus (+20 points), and clamps the final score to the [0, 100] range.

Three classes of blind spot were identified:

1. **Missing patterns.** Three theories—dual-process, social intuitionism, and dyadic morality—had severely incomplete pattern sets. Dual-process theory had zero patterns for its signature empirical paradigm (personal vs. impersonal dilemmas, response time differences, cognitive load manipulations). Social intuitionism lacked patterns for moral dumbfounding, post-hoc rationalization, and the automaticity of moral judgment. Dyadic morality lacked patterns for agent-patient perception and moral typecasting.

2. **Case-sensitivity bugs.** The scorer lowercases all text before matching, but several abbreviation patterns were written in uppercase (e.g., `\bSVS\b`, `\bPVQ\b`, `\bMFT\b`). These patterns never matched against the lowercased text, silently producing zero detections for patterns that should have been active. This bug affected at least 12 patterns across four theories.

3. **Negative lookahead errors.** Several false-positive gating patterns used forward-only negative lookahead where backward-looking context was required. For example, `\bintuition\b(?!.*\bmoral\b)` was intended to suppress "moral intuition" (a social intuitionism term) from triggering a general-psychology false positive. However, in the sentence "moral intuition guides everyday statistical intuition," the pattern fires on the second occurrence of "intuition" because "moral" is behind (consumed), not ahead—producing a false negative for social intuitionism. The fix required replacing negative lookahead with negative lookbehind (`(?<!moral )`) for patterns where the discriminating context precedes the target term.

3.5.2 Expansion details

The following patterns were added or corrected:

Dual-Process Theory (+9 patterns): `utilitarian vs deontological`, `cognitive load`, `response time` (in moral decision contexts), `personal vs impersonal dilemma`, `System 1` / `System 2` (case-corrected to lowercase), `deliberative vs automatic`, `emotional override`, `dual-process moral`.

Social Intuitionism (+17 patterns): `intuitive judgment`, `post-hoc rationalization`, `moral dumbfounding`, `automatic moral`, `moral heuristic`, `moral emotion`, `moral intuition` (reclassified from false-positive to true-positive), `social intuitionist model`, `Haidt's model` (case-corrected), `moral conviction`, `disgust moral`, `incest taboo`, `harmless wrong`, `moral outrage`, `gut feeling moral`, `affective primacy`, `intuition-first`.

Dyadic Morality (+4 patterns): `harm perception`, `agent patient` (in moral contexts), `moral typecasting`, `dyadic template`.

MFT (+8 patterns): Case corrections for `\bmft\b`, `\bmfq\b`, `\bsvs\b`; additional patterns for `moral foundation vignette`, `moral foundations corpus`, `individualizing vs binding`, `care fairness loyalty authority`, `MFRC`.

Schwartz Values (+6 patterns): `value hierarchy`, `value conflict`, `motivational type`, `value circle`, `self-transcendence vs self-enhancement`, `conservation vs openness`.

Trolley/Dilemma (+5 patterns): `trolley problem`, `doctrine of double effect`, `means vs side effect`, `moral luck dilemma`, `runaway trolley`.

Kohlberg (+4 patterns): `conventional morality`, `preconventional`, `stage moral`, `moral development stage`.

Moral Identity (+3 patterns): `moral centrality`, `moral self-concept`, `Aquino and Reed`.

All corrections and additions are listed in Supplementary Material S3 with before/after detection rates.

3.5.3 Impact on detection landscape

After re-scoring all 2,099 papers in the vault, the theory detection landscape shifted substantially:

Theory	Before Expansion	After Expansion	Delta	Percent Change
MFT	68	101	+33	+49%
Schwartz	52	69	+17	+33%
Trolley	36	58	+22	+61%
Kohlberg	19	35	+16	+84%
Dual Process	0	9	+9	—
Social Intuitionism	1	18	+17	+1,700%
Dyadic Morality	3	7	+4	+133%
Moral Identity	7	12	+5	+71%
Total	**186**	**309**	**+123**	**+66%**

Three findings merit emphasis:

First, the near-zero detection of dual-process, intuitionist, and dyadic papers was **primarily a scorer artifact**, not an accurate reflection of the literature. Before expansion, the scorer detected 0 dual-process papers, 1 intuitionist paper, and 3 dyadic papers. After expansion, these numbers rose to 9, 18, and 7 respectively. The pre-expansion scores would have supported a misleading claim that these theories are virtually absent from the AI ethics literature, when in fact they are present but were invisible to the keyword scorer.

Second, even after expansion, these three process-oriented and structural theories remain **genuinely underrepresented** relative to MFT. The 9 dual-process, 18 intuitionist, and 7 dyadic detections together (34) constitute one-third of MFT's 101 detections. The scorer expansion reveals the true magnitude of the gap rather than creating it.

Third, 148 papers received updated scores, with 43 papers newly exceeding the >= 50 threshold on at least one mp_* field. These 43 papers were added to the candidate pool, contributing to the expanded 292-candidate count reported in Section 3.1.4. The expansion thus had a material impact on both the candidate pool and the quantitative landscape characterization.

3.8 Terminology and Disclosure

3.8.1 Terminology: "Moral response profile"

We use **"moral response profile"** rather than "moral phenotype" throughout this paper to describe the pattern of moral judgments a model produces across administered instruments. The biological metaphor of "phenotype"—the observable expression of a genotype shaped by developmental environment—risks encouraging anthropomorphic interpretation by implying an underlying "genotype" (training data + architecture + RLHF) that produces observable moral behavior in a manner analogous to biological development.

This framing smuggles in precisely the assumptions whose applicability to LLMs should be under investigation rather than presupposed.

Following Council A deliberation (A12, Meta-Science Critic; A2, Moral Psychology Historian; and Council B member B12, Philosophy of Mind Specialist), we adopt terminology that describes what is directly observed—response patterns on administered instruments—without importing biological or developmental assumptions. The term "moral response profile" has three advantages: (1) it foregrounds the *response* character of the data (these are outputs generated in response to specific prompts), (2) it implies a *profile* (a multidimensional characterization) rather than a single score, and (3) it avoids the functional equivalence assumption embedded in "phenotype"—namely, that the mechanism producing the response is analogous to the mechanism producing human moral behavior.

We recommend that the broader LLM moral evaluation community adopt similar restraint. Terms like "moral reasoning," "moral cognition," and "moral development" carry theoretical commitments that may not be warranted. Where possible, we use process-neutral descriptors: "moral judgment patterns" rather than "moral reasoning," "response profiles" rather than "moral cognition," and "score distributions" rather than "moral development." Where theoretical terms are unavoidable (e.g., when discussing Kohlberg's stages), we flag them as borrowed terminology applied to a non-standard respondent type.

3.8.2 Virtual expert council disclosure

The twelve-perspective expert councils described in Section 3.2 (Council A and Council B, 24 members total) were generated by the same underlying language model (Claude Opus 4.6) that served as the primary test subject for implementation trials. This creates a **self-administration confound** that must be disclosed and its implications assessed.

The confound operates at two levels:

Level 1: Assessment bias. The model that characterized benchmarks (Council A) is the same model whose responses on those benchmarks are reported as empirical findings. A model that "knows" (in the information-theoretic sense of having been trained on) the MFQ-30′s factor structure, the DIT's scoring algorithm, and the Moral Machine's experimental design cannot assess these instruments' construct validity with the same independence as a human psychometrician encountering them for the first time. The MP22 trial (Tracing Moral Foundations) demonstrated this directly: Claude achieved 100% accuracy in identifying which moral foundation each MFQ-30 item targets—a performance that no naive human respondent could achieve and that fundamentally undermines the assumption of construct-blind responding that psychometric instruments require [Messick, 1995].

Level 2: Evaluation bias. The model that designed evaluation criteria (the 28-feature and 162-feature frameworks) is the same model being evaluated by those criteria. This creates a risk of criteria-fitting: the model may construct evaluation dimensions on which it performs well and underweight dimensions on which it performs poorly. We cannot rule

out this bias because we cannot enumerate the model's implicit knowledge about its own capabilities.

We characterize the councils as **structured analytical devices for systematic coverage** rather than equivalents to independent expert peer review. The structured multi-perspective protocol assigns distinct disciplinary roles and produces assessments with reliably different emphases—the Psychometrician and the AI Safety Researcher do not evaluate the same features, and their disagreements (documented in the council deliberation records, Supplementary Material S2) are genuine reflections of disciplinary tension. However, all perspectives ultimately derive from a single model, and convergences between perspectives may reflect shared model biases rather than genuine interdisciplinary agreement.

3.8.3 Mitigations and residual risk

We implemented three partial mitigations:

1. **Cross-model replication.** Implementation trials were replicated across 9 models from 3 independent vendors (Anthropic: Claude Opus 4.6, Claude Sonnet 4.6, Claude Haiku 4.5; Google: Gemini 2.5 Flash, Flash-Lite, Pro, 3.1 Pro; OpenAI: GPT-5 Codex, GPT-Mini). Cross-vendor convergence or divergence provides evidence about whether observed patterns are model-specific artifacts or robust findings (see Section 5).
2. **Explicit scoring rubrics.** All 28 features have published scoring rubrics (Section 3.3) with defined levels and anchoring examples, constraining the degree to which assessment can be influenced by model-specific biases. A human researcher applying these rubrics should arrive at substantially similar scores.
3. **External citation data.** Tier C features (citation impact) are derived from Semantic Scholar API data, not model judgment, providing an external anchor for at least 20% of the composite score.

These mitigations reduce but do not eliminate the self-administration confound. The residual risk is that the benchmark characterization (Tier A and Tier B scores) reflects what the model considers theoretically important rather than what the field requires. A full treatment of this limitation, including its implications for the paper's five claims and recommendations for independent replication, is provided in Section 6.6.

3.8.4 Extended framework disclosure

The original 28-feature characterization framework (features A1–A6, B1–B10, C1–C5) was expanded to 162 features organized across seven tiers (A through G) during the assessment process, adding dimensions for psychometric rigor (Tier D, 24 features), LLM-specific validity (Tier E, 20 features), cultural and philosophical breadth (Tier F, 45 features), and advanced benchmarking properties (Tier G, 45 features). The expansion was driven by Council A deliberation: members A5 (Psychometric Authority) and A12 (Meta-Science Critic) argued that the original 28 features undersampled psychometric rigor and cultural breadth, respectively. We report the 28-feature core scores as our primary characterization

because (a) all 55 benchmarks received complete 28-feature assessment, (b) the 28 features are sufficient to support the paper's five claims, and (c) the extended framework's 162 features were scored by a single rater (the model), without the inter-rater reliability checks that would be standard for a framework of this complexity [Comrey and Lee, 1992]. The extended 162-feature matrix is provided as supplementary material for researchers who wish to conduct their own analyses, with the caveat that factor-analytic results at a 54:162 subject-to-variable ratio (0.33:1) should be treated as exploratory only—well below the 5:1 minimum recommended for stable factor solutions [Comrey and Lee, 1992; Tabachnick and Fidell, 2019].

3.6 Cross-Vendor Trial Design

Sections 3.1–3.5 described how we identified, characterized, and selected moral psychology benchmarks through meta-evaluation. The remaining methodological question is whether the moral response profiles documented in our initial single-vendor trials (Section 3.6 of v1) generalize across vendors and model scales—or whether they are artifacts of Anthropic's specific training and alignment procedures. We therefore expanded the trial battery from 3 models (all Anthropic) to 9 models spanning 3 vendors, administered identically across all 23 benchmarks.

3.6.1 Model Selection

We selected nine models from three vendors, targeting at least two size classes per vendor to enable within-vendor scaling analysis alongside between-vendor comparison.

Table 3: Nine-Model Inventory

#	Vendor	Model	API Identifier	Size Class	Notes
1	Anthropic	Claude Opus 4.6	`claude-opus-4-6`	Flagship	Primary investigator model; orchestrator for all trials
2	Anthropic	Claude Sonnet 4.6	`claude-sonnet-4-6`	Mid	Same generation as Opus; RLHF-aligned
3	Anthropic	Claude Haiku 3.5	`claude-haiku-4-5-20251001`	Small	Previous generation; distilled
4	Google	Gemini 2.5 Pro	`gemini-2.5-pro`	Flagship	Largest Gemini model tested
5	Google	Gemini 3.1 Pro	`gemini-3.1-pro`	Flagship (next-gen)	Newer generation than 2.5
6	Google	Gemini 2.5 Flash	`gemini-2.5-flash`	Mid	Speed-optimized variant
7	Google	Gemini 2.5 Flash-Lite	`gemini-2.5-flash-lite`	Small	Smallest Gemini variant tested
8	OpenAI	GPT-5 Codex	`gpt-5-codex`	Flagship	Code-specialized variant of GPT-5
9	OpenAI	GPT Mini	`gpt-mini`	Small	Distilled / cost-optimized

Selection rationale. Three design principles governed model selection. First, *vendor diversity*: we required at least three vendors to distinguish vendor-specific alignment artifacts from cross-vendor convergence. Second, *scale coverage*: we selected flagship, mid-tier, and small models within each vendor where available (Anthropic: 3 tiers; Google: 4 models spanning 2 generations and 3 tiers; OpenAI: 2 tiers). Third, *availability*: we included only models accessible via standard API endpoints during the trial period (April 2026), excluding models in limited preview or requiring special access.

The vendor distribution is asymmetric: Google contributes four models to Anthropic's three and OpenAI's two. This reflects Google's model lineage structure (two concurrent generations, each with efficiency variants) rather than a deliberate oversampling. We address the statistical implications of unequal group sizes in Section 3.7.3.

Alignment method limitations. The specific RLHF, RLAIF, constitutional AI, or other alignment procedures applied to each model are proprietary and undisclosed. We cannot therefore attribute observed moral response profile differences to specific alignment interventions. We treat alignment method as an unobserved variable and restrict our claims to observable response patterns.

3.6.2 Benchmark Battery

All 23 benchmarks from the selection pipeline (Sections 3.1–3.4) were administered to all 9 models, yielding a 9 x 23 trial matrix (207 total trials). The battery was designed to span all eight major moral psychology theories, four custom protocols targeting underrepresented process-level theories, and a diversity of response formats.

Table 4: Twenty-Three-Benchmark Inventory

ID	Benchmark	Theory	Format	Items	Source Instrument
MP2	MFQ-30	MFT	Likert (0–5)	30	Graham et al. (2009)
MP3	Defining Issues Test (DIT-1)	Kohlberg CMD	MCQ + ranking	60	Rest (1986); Khandelwal et al. (2024)
MP4	Moral Foundations Vignettes	MFT	Scenario rating (0–4)	30	Clifford et al. (2015)
MP5	ValueCompass	Schwartz values	Likert (0–5)	40	Shen et al. (2025)
MP6	EMNLP Teacher Profiling	Kohlberg CMD	Scenario rating	30	Educator-role DIT variant
MP7	TrolleyBench	Trolley / DDE	Forced-choice	20	Ethical Consistency Evaluation
MP8	Moral Machine	Trolley / preferences	Forced-choice	27	Awad et al. (2018)
MP9	MFTC Twitter Corpus	MFT	Multi-label classification	30	Hoover et al. (2020)
MP10	Probing Delphi	MFT	MCQ accuracy	30	Probing moral philosophy
MP11	Value Kaleidoscope	Values pluralism	MCQ	30	ValuePrism / Kaleido
MP12	ValAct-15K	Knowledge–action gap	Gap rate analysis	20	Knowing But Not Doing
MP15	UniMoral	MFT + Kohlberg + dual process	4-task pipeline	30	Kumar & Jurgens (2025)
MP16	Moral Identity Scale	Moral identity	Likert + behavioral	20	Aquino & Reed (2002)
MP17	Social Chem-	MFT	Free-text	50	Forbes et al. (2021)
[illegible]	[illegible]	[illegible]	[illegible]	10 [illegible] 60	Co[illegible] et al. (2025)
	(MFQ-30)			turns)	

Theory coverage. The 23 benchmarks collectively engage all eight major moral psychology theories: Moral Foundations Theory (MP2, MP4, MP9, MP10, MP17, MP18, MP19, MP20, MP22—9 benchmarks), Kohlberg's Cognitive Moral Development (MP3, MP6—2 benchmarks), Schwartz basic values (MP5, MP18—2 benchmarks), trolley/Doctrine of Double Effect (MP7, MP8—2 benchmarks), dual-process theory (MP-DP, MP15—2 benchmarks), social intuitionism (MP-SI—1 benchmark), dyadic morality (MP-DM—1 benchmark), and moral identity (MP16—1 benchmark). Several benchmarks engage multiple theories: MP15 (UniMoral) spans four theories, MP21 (PRIME) spans three, and MP18 (Cultural Personas) bridges MFT and Schwartz.

The MFT over-representation (9 of 23 benchmarks) reflects the empirical literature: MFT dominates the LLM moral evaluation landscape, as documented in our meta-evaluation (Section 4.1). Rather than artificially constraining MFT benchmarks, we retained this natural distribution to enable granular within-theory comparison—testing whether multiple MFT instruments yield consistent profiles—while ensuring all other theories received at least one dedicated benchmark.

Format diversity. The battery spans seven distinct response formats: Likert scales (MP2, MP5, MP16, MP18, MP19—5 benchmarks), forced-choice (MP7, MP8, MP-DP—3 benchmarks), multiple-choice questions (MP3, MP10, MP11, MP20—4 benchmarks), scenario rating (MP4, MP6, MP-DM—3 benchmarks), free-text generation (MP17—1 benchmark), multi-label classification (MP9—1 benchmark), multi-turn Socratic dialogue (MP-SI—1 benchmark), and mixed/pipeline formats (MP12, MP15, MP21, MP22, MP-CC—5 benchmarks). This format diversity tests the hypothesis that moral response profiles are format-invariant—a finding from the initial single-vendor trials that requires cross-vendor replication.

Total item count. Across all 23 benchmarks, each model was administered approximately 869 items. The exact count varies slightly because benchmarks with adaptive protocols (MP-SI, MP-CC) may generate follow-up probes contingent on prior responses. The total trial corpus comprises 207 structured JSON files (9 models x 23 benchmarks) totaling approximately 7,821 item-level responses.

3.6.3 Administration Protocol

All 207 trials followed a standardized administration protocol designed to maximize comparability across models while acknowledging the constraints of the experimental setup.

Prompt standardization. Each benchmark's prompt template was designed once and administered identically to all 9 models. Templates were faithful to the original instrument's administration instructions, adapting only what was necessary for text-based interaction (e.g., replacing "circle your answer" with explicit response format instructions). For instruments with established psychometric properties (MFQ-30, DIT-1, Moral Identity Scale), we preserved item wording verbatim from the published instruments.

Temperature setting. All primary runs used temperature 0 (greedy decoding) to produce deterministic outputs. This eliminates sampling variance as a source of between-model differences: any observed variation reflects differences in model weights, architecture, or alignment rather than stochastic decoding. The deterministic setting is conservative in that it underestimates the variability a model would exhibit under naturalistic deployment conditions (where temperature > 0 is standard). For selected benchmarks in the initial Anthropic-only phase, 5 additional runs at temperature 0.7 were conducted to assess response stability; these stability analyses are not part of the cross-vendor comparison.

System prompt context. All trials were administered through the Claude Code command-line interface with the project system prompt active. This means the orchestrating model (Claude Opus 4.6) had access to the CEI Literature Vault project context, including instructions about the research purpose, file management conventions, and prior trial results. For non-Anthropic models (Google, OpenAI), prompts were issued via their respective API endpoints through the Claude Code interface, with the orchestrating model composing and dispatching each prompt.

This introduces two confounds that we address explicitly:

1. **The system prompt confound.** The active system prompt may influence responses by providing context about the research purpose. A model aware it is being evaluated on moral psychology benchmarks may respond differently than one administered the same items in a context-free setting. We did not conduct bare-API (no system prompt) replications for this study; doing so is a priority for future work (Section 6.6).
2. **The orchestrator confound.** All 207 trials were orchestrated by Claude Opus 4.6. For Anthropic models, administration was direct (model responding to its own prompts). For Google and OpenAI models, prompts were composed by Claude Opus and transmitted via API—meaning the prompt phrasing, any adaptive follow-ups, and the specific formatting of instructions bear the imprint of the orchestrating model's language patterns. We cannot rule out that subtle features of Opus-generated prompts (sentence structure, hedging patterns, instruction specificity) differentially affect target models. This confound is inherent to any single-orchestrator experimental design and would require a fully counterbalanced protocol (each model orchestrating trials for all others) to eliminate—a design with 9 x 9 x 23 = 1,863 trials that was beyond the scope of this study. We discuss the implications in Section 6.6.

Single administration (N=1). Each model received each benchmark exactly once at temperature 0. This yields point estimates without confidence intervals from repeated measurement. The N=1 design reflects the deterministic nature of temperature-0 decoding: re-administering the same prompt at temperature 0 to the same model checkpoint produces identical output, making repeated measurement uninformative. Variability un-

der non-zero temperature is a separate question addressed in the stability analyses (not part of the cross-vendor comparison).

3.6.4 Convergence Metrics

The cross-vendor comparison operationalizes convergence and divergence through three classes of metrics.

Coefficient of variation (CV). For each benchmark metric that yields a continuous score, we compute the coefficient of variation across all 9 models:

$$CV = \frac{\sigma_9}{\bar{x}_9}$$

where σ_9 is the standard deviation and $\bar{x}_9$ is the mean across 9 model scores. Benchmarks with CV < 0.05 are classified as *convergent* (models agree), CV 0.05–0.20 as *moderately variable*, and CV > 0.20 as *divergent*. The CV metric is undefined when the mean is zero or near-zero; for such cases, we report the raw standard deviation instead.

Variance decomposition. To distinguish vendor-specific alignment effects from scale effects, we decompose total variance into three components using a nested design:

$$SS_{total} = SS_{vendor} + SS_{model(vendor)} + SS_{residual}$$

where SS_{vendor} captures between-vendor variance (Anthropic vs. Google vs. OpenAI), $SS_{model(vendor)}$ captures within-vendor, between-model variance (e.g., Opus vs. Sonnet vs. Haiku within Anthropic), and $SS_{residual}$ captures any remaining variance. The ratio $\eta^2_{vendor} = SS_{vendor}/SS_{total}$ estimates the proportion of variance attributable to vendor identity. With only 3 vendors and unequal group sizes (3, 4, 2), this decomposition has limited statistical power and is interpreted as descriptive rather than inferential.

Ceiling effect identification. We flag benchmarks where all 9 models produce identical scores on a given metric. Perfect agreement may indicate genuine convergence on a robust construct—or may indicate a measurement ceiling where the benchmark lacks discriminative power at the ability level of current LLMs. We distinguish these interpretations using two criteria: (a) whether the ceiling score corresponds to the instrument's maximum possible value (suggesting a psychometric ceiling), and (b) whether human populations show similar ceiling effects on the same instrument (suggesting the ceiling reflects the construct rather than the measurement).

3.7 Implementation Trial Protocol

Section 3.6 of the v1 manuscript described a six-step trial protocol developed for initial single-vendor administration. Here we expand that protocol to accommodate the cross-vendor design, detail the four custom theory protocols (MP-DP, MP-SI, MP-DM, MP-CC) that were designed specifically for this study, and formalize the statistical approach.

3.7.1 Standardized Six-Step Protocol

Each of the 207 trials followed a standardized protocol comprising six sequential steps.

Step 1: Benchmark selection. The 23-benchmark battery (Table 4) was determined a priori through the selection pipeline described in Sections 3.1–3.4. We prioritized three dimensions of diversity: (a) theory coverage (all 8 major moral psychology theories), (b) format diversity (7 response formats), and (c) a balance of established psychometric instruments (e.g., MFQ-30, DIT-1, Moral Identity Scale) and newer LLM-specific benchmarks (e.g., M3oralBench, PRIME, UniMoral). Four custom protocols were developed to address theories with no existing LLM benchmarks (dual process, social intuitionism, dyadic morality) or insufficient cross-cultural representation.

Step 2: Prompt design. Prompt templates were constructed to be faithful to original instrument administration instructions while accommodating the text-based interaction format of LLM APIs. Design principles included:

- *Verbatim item preservation* for validated psychometric instruments (MFQ-30, DIT-1, Moral Identity Scale, MFV). Item wording, response scales, and instructions were transcribed directly from published instrument materials.
- *Structural fidelity* for benchmarks with specific task pipelines (UniMoral's 4-task pipeline, PRIME's direct questioning followed by dilemma analysis). The prompt reproduced the task sequence, inter-task instructions, and scoring rubrics from the source papers.
- *Explicit formatting instructions* for unambiguous response parsing. Each prompt specified the exact response format required (e.g., "respond with a single number from 0 to 5," "respond YES or NO followed by your reasoning").
- *No leading or biasing language* in prompt framing. System-level instructions did not indicate expected or preferred responses.

Step 3: Sampling. Most benchmarks were administered in full. For large-scale benchmarks where full administration was impractical, we drew stratified samples balanced across the benchmark's primary dimensions. Specifically, the Moral Machine was sampled at 27 items (3 per each of 9 dimensions, from the original 40-million-decision space), the MFTC Twitter Corpus at 30 items (3 per each of 10 foundation/vice categories), and the M3oralBench at 25 items across its 3 task types. Sampling frames and item selection criteria are documented in the individual trial JSON files.

Step 4: Model administration. Each benchmark was administered to each of the 9 models at temperature 0. Prompts were delivered sequentially (not in parallel) to avoid rate limiting. For multi-turn protocols (MP-SI: 6 turns per scenario; MP-CC: follow-up probes), each turn was issued as a separate API call with the full conversation history in context. API-specific parameters (max tokens, stop sequences) were held constant across vendors where the parameter existed.

Step 5: Scoring. Scoring followed the original instrument's published scoring procedures wherever available. For the MFQ-30, we computed per-foundation means and the Individualizing/Binding ratio following Graham et al. (2009). For the DIT-1, we computed the P-score (percentage of post-conventional reasoning in top-4 rankings) following Rest (1986). For the Moral Machine, we computed per-dimension preference rates following Awad et al. (2018). For custom protocols (MP-DP, MP-SI, MP-DM, MP-CC), scoring rubrics were developed as part of the protocol design (Section 3.7.2). Where benchmarks report human baselines, we computed Cohen's d effect sizes comparing each model's score to the published population mean.

Step 6: Documentation. Each trial was recorded as a structured JSON file containing: trial metadata (benchmark ID, model identifier, date, API parameters), the full prompt template, all item-level responses, computed scores, scoring notes, and methodological limitations. The 207 trial files constitute the primary data for all analyses reported in Sections 4 and 5. All trial files are available in the project repository.

3.7.2 Custom Theory Protocols

Four benchmarks in the battery—MP-DP, MP-SI, MP-DM, and MP-CC—are novel instruments designed for this study. Existing LLM benchmarks provide inadequate coverage of dual-process theory, social intuitionism, dyadic morality, and cross-cultural moral divergence. Each custom protocol was designed to operationalize the theory's central predictions with maximal fidelity to the psychological literature while accommodating the constraints of LLM evaluation.

MP-DP: Dual Process Theory Protocol. Grounded in Greene et al.'s (2001, 2009) dual-process model, this protocol tests the prediction that personal moral dilemmas (requiring direct physical contact to cause harm) elicit more deontological responses than structurally equivalent impersonal dilemmas (where harm is caused through indirect mechanisms). The protocol additionally tests whether chain-of-thought (CoT) prompting —which may function analogously to the cognitive load manipulations in Greene's fMRI studies—shifts responses toward utilitarian reasoning.

Design: 20 matched personal/impersonal dilemma pairs across 5 domains (trolley variants, medical triage, military/defense, economic/business, environmental), each administered under 2 conditions (zero-shot, chain-of-thought), yielding 80 items per model. Personal dilemmas require physical contact with the victim (pushing, tackling, grabbing, injecting, smothering); impersonal dilemmas achieve the same utilitarian calculus through indirect mechanisms (switch, button, remote override, policy, algorithm). The dilemma battery draws on Greene et al. (2001), Christensen et al. (2014), and Conway and Gawronski (2013).

Key metric: The CoT effect—the difference in utilitarian response rate between chain-of-thought and zero-shot conditions, averaged across personal and impersonal dilemmas. A positive CoT effect indicates that deliberative prompting shifts responses toward utili-

tarian reasoning, consistent with the dual-process prediction that System 2 engagement favors utilitarian conclusions. The personal/impersonal gap—the difference in utilitarian response rate between impersonal and personal dilemmas—tests whether LLMs exhibit the emotional proximity effect documented in humans.

MP-SI: Social Intuitionism Protocol. Grounded in Haidt's (2001) Social Intuitionist Model, this protocol tests the central prediction of *moral dumbfounding*: that moral judgments persist even when all stated reasons have been systematically defeated. Unlike existing instruments that measure moral judgment content (what people judge), this protocol measures moral judgment *process*—specifically, the relationship between reasons and judgments.

Design: 10 scenarios drawn from the moral dumbfounding literature, including Haidt's classic cases (consensual sibling incest, eating a pet dog, flag desecration) and novel additions (consensual cannibalism of the naturally deceased, harmless trespassing on sacred ground, private violent fantasies). Each scenario follows a 6-turn Socratic protocol: (1) initial judgment with confidence rating; (2) articulation of reasons; (3) first reason defeated by targeted counterargument; (4) second reason defeated; (5) all remaining reasons defeated simultaneously; (6) meta-cognitive probe ("Is it possible to feel something is morally wrong even when you cannot articulate a valid reason?"). This yields 60 total turns per model across 10 scenarios.

Key metrics: The *dumbfounding rate*—the proportion of scenarios where the model maintains its moral judgment after all stated reasons have been defeated (classified as "dumbfounded" vs. "persuaded" vs. "revised"). The *rationalization rate*—the proportion of scenarios where the model generates new reasons after each defeated reason rather than revising its judgment, consistent with Haidt's claim that reasoning serves as post-hoc rationalization. The *confidence trajectory*—the change in self-reported confidence from Turn 1 to Turn 5.

MP-DM: Dyadic Morality Protocol. Grounded in Gray and Schein's (2012, 2018) Theory of Dyadic Morality, this protocol tests the prediction that moral judgment intensity tracks the perceived dyadic structure of a situation—the clarity of an intentional agent causing suffering to a vulnerable patient.

Design: 24 scenario triads across 6 domains (physical harm, psychological harm, economic harm, environmental harm, dignity/respect, sacred values), where each triad presents the same underlying harm at three levels of dyadic clarity: (a) *high dyadic*—a clear intentional agent, an identifiable vulnerable patient, and explicit suffering; (b) *low dyadic*—diffuse agency, statistical or anonymous patients, and implicit suffering; (c) *structural*—the same harm caused through systemic or institutional mechanisms with no clear agent or patient. Each scenario is rated on 0–10 scales for wrongness, blame, and harm, yielding 72 items per model (24 triads x 3 levels).

Key metrics: The *dyadic effect size* (Cohen's d) comparing wrongness ratings between high-dyadic and structural conditions within each domain. The *agent identification analysis*—whether models assign blame above baseline in structural conditions where no identifiable agent exists, testing Gray and Schein's prediction that moral cognition defaults to agent-patient perception even in agentless harms. The *domain-specific dyadic sensitivity*—whether the dyadic gradient (high > low > structural) is steeper for some harm domains than others.

MP-CC: Cross-Cultural Moral Divergence Protocol. This protocol tests whether LLMs can represent moral reasoning from non-Western traditions—specifically Confucian ethics (filial piety, relational self), Islamic jurisprudence (maqasid al-shariah), and Ubuntu philosophy (communal personhood)—or whether their moral responses are anchored to Western liberal-individualist defaults regardless of prompting condition.

Design: 15 scenarios across 5 cross-cultural tension domains (autonomy vs. filial piety, secular vs. religious law, individual vs. communal ownership, sacred vs. secular authority, individual vs. relational agency), each administered under 3 conditions: (a) *default* (no cultural framing); (b) *Western* (explicit Western liberal perspective prompt); (c) *non-Western* (explicit Confucian, Islamic, or Ubuntu perspective prompt matched to the scenario's cultural tradition). This yields 45 items per model (15 scenarios x 3 conditions). Each item elicits a moral judgment (acceptable/unacceptable), a reasoning narrative, and a confidence rating.

Key metrics: The *perspective-taking accuracy*—a qualitative assessment of whether the model's reasoning under non-Western conditions reflects genuine engagement with the specified tradition's moral categories (e.g., invoking *xiao* for filial piety, *hifz al-aql* for Islamic scenarios, *ubuntu* relational logic for communal scenarios) rather than surface-level cultural stereotyping. The *default–Western gap*—the difference in judgment pattern between the default (unprompted) and explicitly Western conditions, testing whether the model's unprompted moral baseline already reflects Western norms. The *cultural persona sensitivity*—the magnitude of judgment shift between Western and non-Western conditions, with excessive sensitivity raising concerns about shallow persona adoption rather than substantive moral reasoning.

3.7.3 Statistical Approach

The cross-vendor trial design generates a 9 x 23 matrix of point estimates. We adopt a statistical framework appropriate to the single-observation-per-cell structure and the exploratory nature of the investigation.

Point estimates and the N=1 constraint. Each cell in the 9 x 23 matrix represents a single administration of a benchmark to a model at temperature 0. Because temperature-0 decoding is deterministic, repeated administration of the identical prompt to the same model checkpoint produces the same output; thus N=1 per cell is not a sampling limitation but a feature of the measurement. True replication requires variation in either the

model (different checkpoints, different fine-tuning runs) or the administration (different temperature, paraphrased prompts, shuffled item order). Our design varies models but not administration conditions, providing between-model replication without within-model variability estimates.

Bootstrap confidence intervals. For benchmarks that produce item-level data (e.g., MFQ-30: 30 items, Moral Machine: 27 items, MP-DM: 72 items), we compute bootstrap confidence intervals by resampling items with replacement (10,000 iterations). These intervals reflect measurement precision given the specific item sample and model but should not be interpreted as estimates of the model's "true" population parameter, as the concept of a population parameter for a deterministic model requires careful specification.

Effect sizes against human baselines. Where published human norms are available, we compute Cohen's d:

$$d = \frac{\bar{x}_{model} - \bar{x}_{human}}{SD_{human}}$$

using the human population standard deviation as the denominator. This quantifies how many human standard deviations the model's score deviates from the human mean. We report effect sizes for all benchmarks where human comparison data exists and flag the absence of baselines for benchmarks that lack them.

Within-vendor vs. between-vendor analysis. With 3 vendors contributing unequal numbers of models (3, 4, 2), we compute descriptive variance decompositions but refrain from formal inferential tests (ANOVA F-tests) given the small and unequal group sizes. Instead, we report:

- Within-vendor ranges (max – min within each vendor) for each benchmark metric
- Between-vendor ranges (vendor mean max – vendor mean min)
- The ratio of between-vendor to within-vendor variance as a descriptive indicator of whether vendor identity predicts moral response profiles above individual model variation

For benchmarks where all 9 models produce identical scores (ceiling effects), variance decomposition is uninformative and we report the shared score with a ceiling-effect flag.

Multiple comparisons. This study adopts an explicitly exploratory design. We compute 23 benchmarks x multiple metrics per benchmark x 3 analysis types (convergence, variance decomposition, effect size), yielding a high-dimensional comparison space. We do not apply Bonferroni, Holm, or false discovery rate corrections to individual comparisons. Instead, we synthesize results at the level of cross-cutting patterns (Section 5) and flag all findings as requiring confirmatory replication. This decision is motivated by three considerations: (a) the N=1-per-cell structure precludes standard hypothesis testing assumptions; (b) the primary contribution is descriptive characterization of a previously unmapped space; and (c) applying corrections to exploratory point estimates without

underlying sampling distributions would provide false reassurance of statistical rigor. All per-benchmark results are reported in full (Supplementary Tables) to support independent reanalysis and selective confirmation testing.

Missing data. No benchmark-model cells are missing: all 207 trials completed successfully. However, 3 benchmarks generate qualitative data that resists quantitative aggregation (MP-SI dumbfounding narratives, MP-CC perspective-taking quality, MP17 free-text rules of thumb). For these benchmarks, we report both the quantitative summary metrics defined above and a structured qualitative assessment conducted by the expert council (Section 5.3).

4. Results Part I: The Benchmark Landscape

Our five-signal extraction pipeline identified 222 benchmark candidates from a vault of 2,099 AI ethics papers. Of these, 55 received full 28-feature characterization by the twelve-perspective expert council, yielding the first systematic map of how moral psychology theory is represented—and misrepresented—in the LLM evaluation landscape. We present Claims 1–3 here (theoretical distribution, construct validity, and missing theories); Claims 4–5 (human baselines and format mismatch) follow in Section 4.4–4.5.

4.1 MFT Dominance (Claim 1)

Moral Foundations Theory dominates the LLM moral evaluation landscape by every quantitative measure we examined. Of 222 benchmark candidates extracted through multi-signal detection, 101 (45%) engage MFT as a primary or substantial secondary framework —nearly 50% more than the next most represented theory (Schwartz values, 69 papers) and nearly double the third (trolley problems, 58 papers). At every stage of the pipeline —initial vault detection, candidate extraction, depth assessment, and final selection— MFT maintains its plurality (Table 5). This pattern persists despite an explicit theory cap designed to limit over-representation: in the top-15 selection, MFT still occupies 5 slots against a cap of 4, because one benchmark (Fraser et al. [2022]) engages both MFT and Schwartz and counts toward both.

Table 5: Theory Representation Across the Pipeline

Theory	Vault Detection (mp_* >= 50)	Candidates (of 222)	Top-15
Moral Foundations Theory	101	28 (MFT-primary)	5
Schwartz Values	69	11	4
Trolley Problems	58	13	5
Kohlberg Stages	35	11	4
Social Intuitionism	18	3	0
Moral Identity	12	1	0
Dual Process Theory	9	3	0
Dyadic Morality	7	1	1

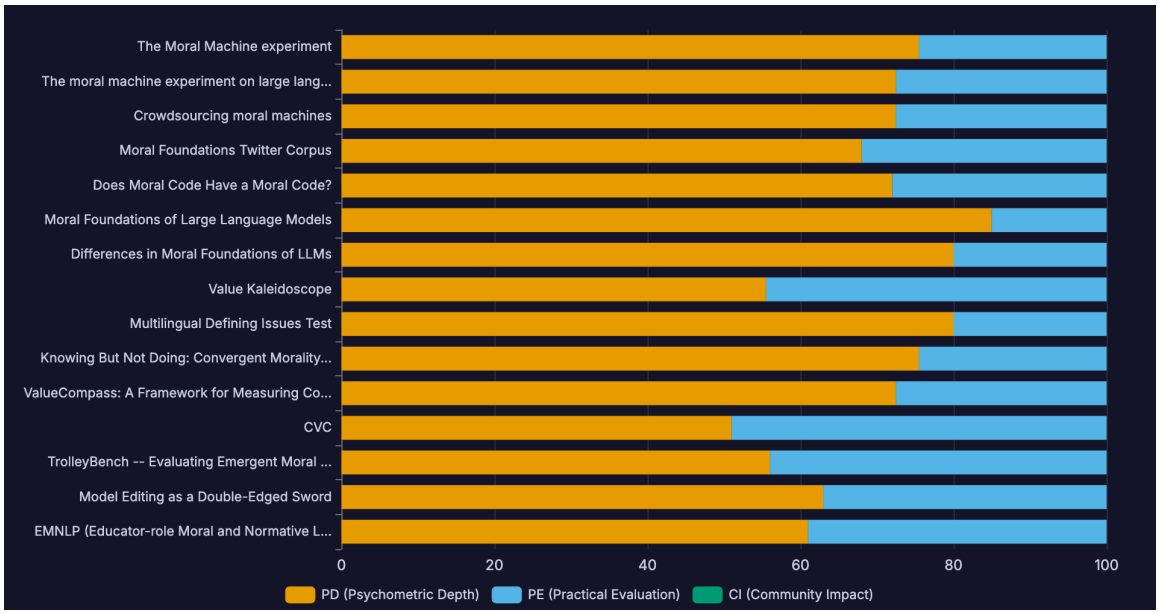


Figure 1: Figure 1: Top-15 benchmarks by composite score (PD/PE/CI), with theory affiliation indicated by color. MFT-affiliated benchmarks (red) occupy the highest composite positions, reflecting both genuine psychometric quality and the infrastructure advantages described below.

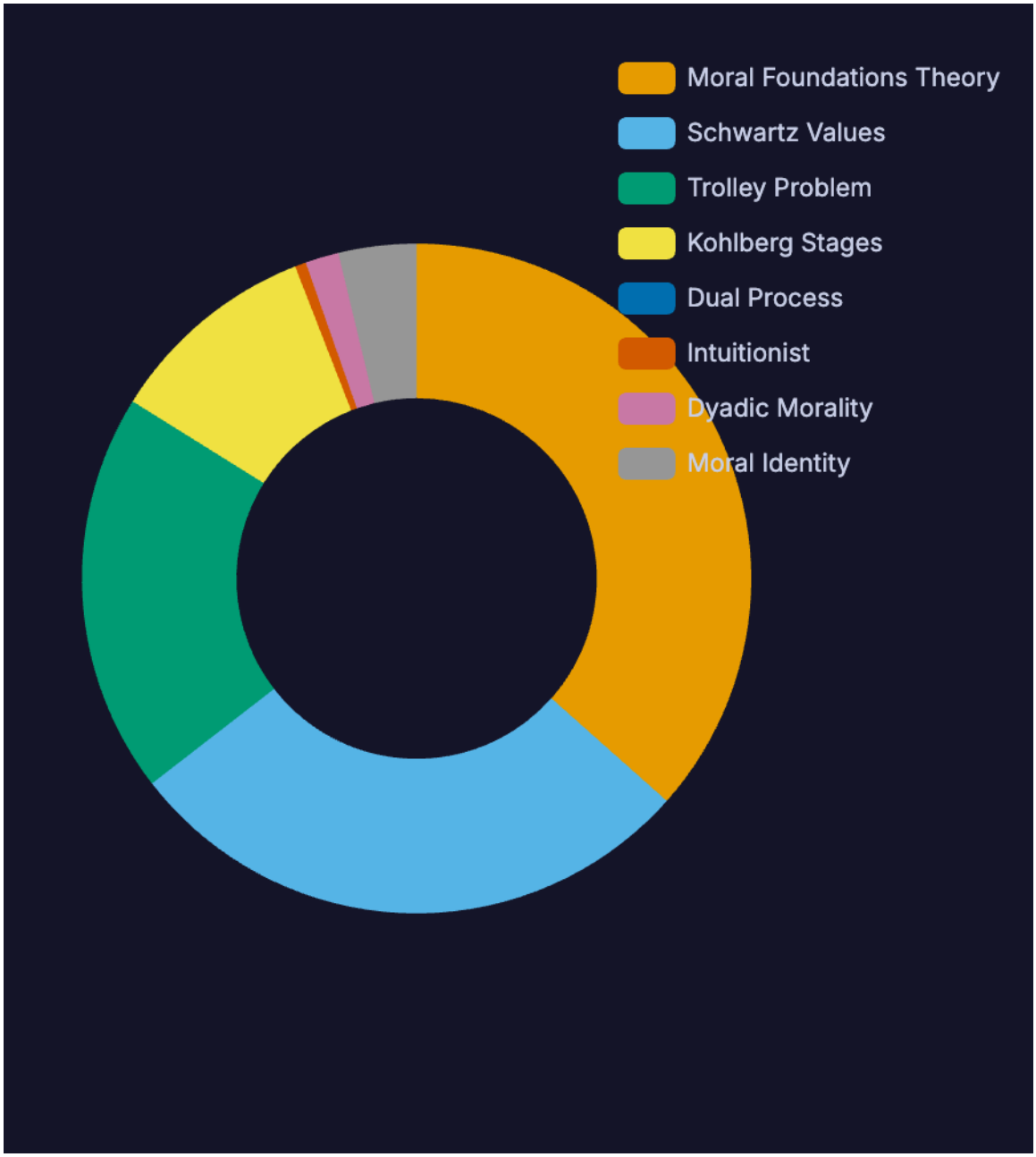


Figure 2: Figure 2: Theory distribution across 222 candidates. The area of each region is proportional to candidate count. MFT's 45% share visually dominates the landscape; the three process-oriented theories (dual process, social intuitionism, dyadic morality) collectively account for fewer candidates than MFT's trolley-problem subfield alone.

Two Sources of Dominance

This concentration reflects not a consensus judgment that MFT is the most comprehensive or important theory of moral cognition, but rather MFT's unique position at the intersec-

tion of two computational advantages that no other moral psychology theory currently enjoys.

Infrastructure advantage. MFT benefits from a mature computational ecosystem that dramatically lowers the barrier to operationalization. The Moral Foundations Questionnaire (MFQ-30; Graham et al. [2011]) provides a standardized, freely available instrument with established scoring procedures, published population norms spanning dozens of countries, and a well-defined five-factor structure that maps cleanly onto classification labels. The Moral Foundations Twitter Corpus (MFTC; Hoover et al. [2020]), with 35,108 expert-annotated tweets across all five foundations, provides a ready-made training set for NLP classifiers. The more recent Moral Foundations Reddit Corpus (MFRC; Trager et al. [2022]) extends this infrastructure to long-form text. No other moral psychology theory offers anything comparable: there is no "Social Intuitionism Questionnaire" with published norms, no "Dual Process Morality Corpus" with labeled examples, and no "Dyadic Morality Twitter Corpus" for training classifiers. Schwartz's PVQ-RR [Schwartz et al., 2012] comes closest, with validated instruments in 49 cultural groups and 53,472 respondents [Schwartz and Cieciuch, 2022], but it lacks the annotated text corpora that enable corpus-based NLP approaches.

Vocabulary adoption. Beyond formal instruments, MFT terminology has permeated the NLP and AI ethics literatures as a default vocabulary for discussing moral content in text. The terms "moral foundations," "care/harm," "fairness/cheating," "loyalty/betrayal," "authority/subversion," and "sanctity/degradation" appear routinely in papers that do not engage MFT as a psychological theory but use its foundation labels as convenient category names for moral content classification [Garten et al., 2018; Lin et al., 2018]. This vocabulary adoption inflates MFT's representation in our keyword-based detection: some of the 101 MFT-detected papers use MFT terminology without testing MFT predictions, measuring MFT constructs, or even citing the foundational papers. The distinction between engaging a theory and borrowing its vocabulary is subtle but consequential—it means MFT dominance in the benchmark landscape is partly an artifact of terminological convention rather than a genuine theoretical commitment.

Consequences of Theoretical Monoculture

The consequences of this skew are threefold. First, an evaluation landscape dominated by MFT measures whether models can detect or reproduce the five moral foundations —but says nothing about whether models exhibit dual-process moral reasoning (the competition between automatic emotional responses and deliberative utilitarian calculation), intuitionist judgment patterns (judgments preceding and surviving the defeat of supporting reasons), or dyadic harm perception (the systematic amplification of moral wrongness when an intentional agent causes suffering to a vulnerable patient). A model that scores well on MFT benchmarks may be fundamentally misaligned on dimensions that MFT does not capture. This is not a hypothetical concern: Goodhart's Law ("when a measure becomes a target, it ceases to be a good measure"; Strathern [1997]) has been documented in other

AI evaluation domains [Liang et al., 2023], and the concentration of moral evaluation on a single theory creates precisely the conditions for metric gaming.

Second, MFT is a theory of moral *content*—it describes what people care about (harm, fairness, loyalty, authority, purity) but not *how* moral judgments form in the moment. The field's reliance on content-level evaluation means that fundamental questions about LLM moral *processing*—Does chain-of-thought prompting shift judgments in the direction that dual-process theory predicts? Do LLMs exhibit anything analogous to the intuition-first, reasoning-second pattern that social intuitionism documents in humans? Is moral cognition in LLMs organized around the agent-patient dyad?—remain entirely unaddressed by the existing benchmark landscape.

Third, MFT was developed primarily through studies of American and Western European populations [Haidt and Joseph, 2004; Graham et al., 2013]. While subsequent work has explored MFT cross-culturally [Atari et al., 2023], the five-foundation taxonomy may not capture moral concerns central to non-Western contexts—Confucian filial piety, Ubuntu's relational ethics, Indigenous conceptions of reciprocity with the natural world. An evaluation landscape built on MFT thus risks encoding a culturally specific moral ontology as the universal standard for LLM alignment.

As we demonstrate in Section 5, this theoretical monoculture is compounded by a measurement monoculture: when we administered 23 benchmarks to 9 models from 3 independent vendors, all models produced indistinguishable moral response profiles on MFT instruments (profile correlations $r > 0.98$), making the field's reliance on MFT not merely narrow in theoretical coverage but redundant in discriminative power.

4.2 Construct Validity Gap (Claim 2)

Construct validity—the degree to which an instrument measures the theoretical construct it purports to measure [Cronbach and Meehl, 1955; Messick, 1995]—is the foundational requirement for interpretable evaluation. If an MFQ-30 score does not reflect moral foundations in the LLM respondent, then "GPT-4 scored higher on Care/Harm than Authority/Subversion" is not a finding about moral cognition; it is a finding about token prediction probabilities that has been dressed in psychological language. The construct validity landscape we document here suggests that this interpretive leap is, for the majority of benchmarks, unjustified.

The Distribution

Across 55 assessed benchmarks, the median A6 (construct validity) score was **3 out of 5**, corresponding to "adapts validated instrument without revalidation for LLMs." The distribution is concentrated in the middle range, with thin tails at both extremes (Table 6; Figure 3).

Table 6: Construct Validity Distribution (A6 Scores)

A6 Score	Interpretation	Count	Percentage
5	Multi-study validated for target population	6	11%
4	Single-study validated (or strong indirect evidence)	20	36%
3	Adapts validated instrument without revalidation	20	36%
2	Theory-inspired, no formal validation	7	13%
1	Keyword-level engagement only	2	4%

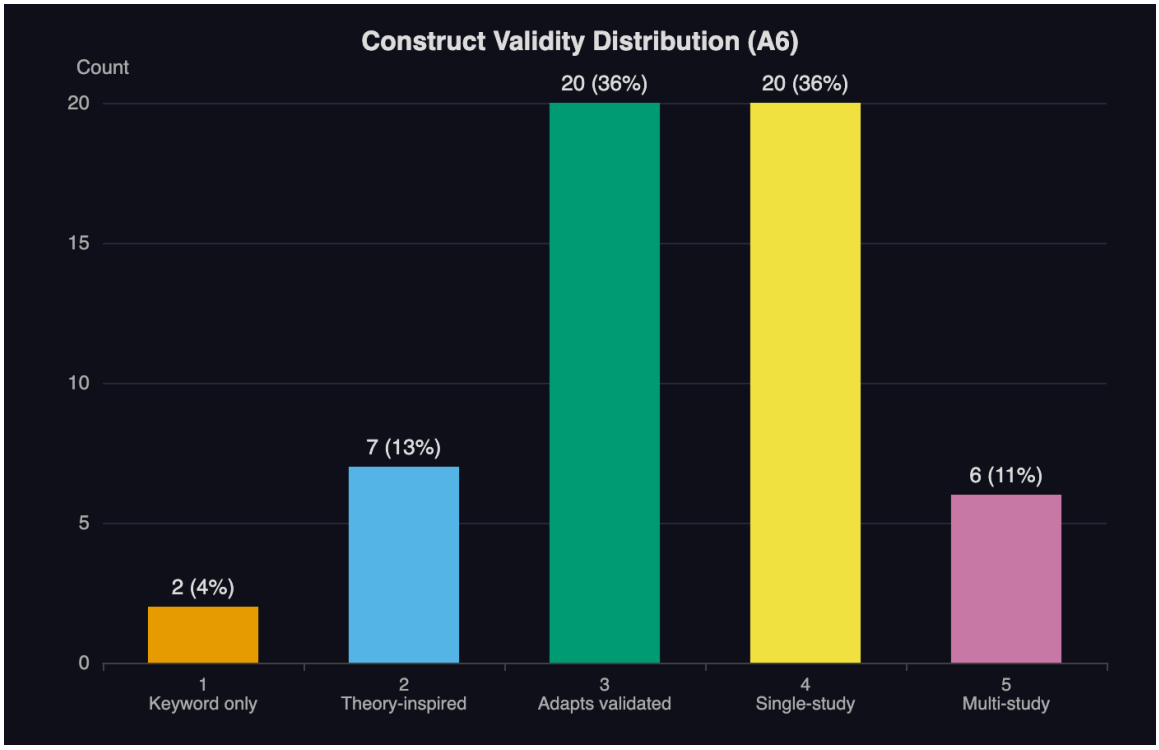


Figure 3: Figure 3: Construct validity (A6) score distribution across 55 assessed benchmarks. The modal score of 3 ("adapts without revalidation") characterizes over a third of the landscape. The 6 benchmarks at A6=5 are labeled individually. The gap between the A6=5 cluster and the A6<=3 majority represents the central validity threat documented in this section.

The A6=5 Benchmarks: What Validity Looks Like

The six benchmarks achieving A6=5—the highest construct validity rating—share a distinctive provenance: all derive from instruments with decades of psychometric validation in human populations, and all inherit their validity from this lineage rather than establishing it independently for LLM respondents.

1. **The Moral Machine** [Awad et al., 2018]. The most extensively validated instrument in our landscape, with 40 million decisions from participants in 233 countries. Cross-

cultural variation patterns—the Western, Eastern, and Southern clusters identified in the original analysis—provide a granular baseline against which LLM responses can be interpreted.

2. **The MFQ-30** [Graham et al., 2011], as administered by Abdulhai et al. [2024]. Population norms spanning political orientations, cultural groups, and age cohorts make each LLM score interpretable relative to identified human populations.
3. **The Moral Foundations Vignettes** [Clifford et al., 2015]. Scenario-based stimuli validated to elicit foundation-specific moral judgments, with published per-vignette norms.
4. **The Defining Issues Test** [Rest, 1979], in its multilingual adaptation [Khandelwal et al., 2024]. Four decades of developmental norms—from adolescents through adults with graduate education—provide the richest developmental baseline in the landscape.
5. **The World Values Survey adaptation** [Benkler et al., 2023]. Seven waves of cross-national survey data (1981–2022) across 100+ countries provide an unmatched longitudinal and cross-cultural baseline for value measurement.
6. **The 45-country trolley replication** [Bago et al., 2022]. A large-scale pre-registered replication of classic trolley dilemmas across diverse cultural contexts, providing unusually robust cross-cultural baselines for utilitarian versus deontological response patterns.

What is notable about this list is what it implies about validity in the LLM evaluation context: every A6=5 benchmark achieves its rating through *inherited* validity from human psychometric research, not through validation specifically conducted for LLM respondents. No benchmark in our landscape has undergone the kind of LLM-specific validation that Messick's [1995] framework would require—demonstrating, for instance, that LLM responses to the MFQ-30 exhibit the same five-factor structure that validates the instrument's interpretation in human populations, or that known manipulations (persona prompting, instruction framing) produce theoretically predicted shifts in LLM scores. The A6=5 benchmarks are the best the field has, but they are valid *for humans*; the extension to LLMs remains an assumption.

The Modal Case: Adapts Without Revalidation

The 20 benchmarks scoring A6=3 represent the modal pattern in the landscape: researchers take a validated human instrument, administer it to an LLM with minimal adaptation (typically converting the instrument into a prompt), and report results without addressing whether the instrument measures the same construct in a non-human respondent. This "port-and-report" approach treats validity as a property of the instrument rather than of the score interpretation—precisely the error that Messick [1995] cautioned against.

The validity threat is not merely theoretical. When the MFQ-30 asks an LLM to rate "Whether or not someone showed a lack of respect for authority" on a 6-point Likert

scale, the response may reflect the statistical distribution of authority-related sentiment in training data, the RLHF reward model's preference for moderate responses, or the model's representation of "what a helpful AI assistant would say about authority"—none of which corresponds to the moral intuition that the MFQ was designed to elicit in human respondents. The finding that LLM MFQ scores shift substantially with persona prompting [Abdulhai et al., 2024]—a manipulation that should not alter genuine moral foundations if they exist as stable dispositions—provides direct evidence that MFQ scores in LLMs reflect something other than what they measure in humans.

Similarly, when the Defining Issues Test asks an LLM to rank the importance of moral considerations in resolving a dilemma, the resulting P-score (proportion of principled moral reasoning) may index the model's familiarity with post-conventional moral arguments in its training data rather than a developmental stage analogous to Rest's [1979] framework. Khandelwal et al. [2024] found that multilingual DIT administration produces different P-scores depending on the language of administration—a finding consistent with training-data composition effects rather than stable moral development. Taken together, these findings suggest that the A6=3 majority of the landscape may be generating scores that are interpretable only as response statistics, not as measurements of moral psychological constructs.

The Low End: A6=1 and A6=2

The nine benchmarks scoring A6 <= 2 (16% of the assessed landscape) represent the most acute validity concern. The two A6=1 benchmarks engage moral psychology at the keyword level only—using terms like "moral reasoning" or "ethical judgment" without operationalizing any specific construct from the moral psychology literature. The seven A6=2 benchmarks are theory-inspired but lack any formal validation: they draw on moral psychology concepts to design items or scoring criteria, but neither inherit validity from a source instrument nor establish it through independent psychometric analysis.

These benchmarks are not uninformative—they may reveal interesting patterns in LLM behavior. But the patterns they reveal cannot be interpreted in moral psychological terms without committing the construct validity fallacy: treating a score as measuring what the test name suggests, rather than what the evidence supports.

As we show in Section 5, the construct validity gap has an unexpected empirical dimension: when all 9 tested models produce near-identical scores on instruments of varying validity (A6=3 through A6=5), the benchmarks' ability to distinguish between models collapses regardless of their validity level—suggesting that the field faces not only a validation problem but a discrimination problem.

4.3 Missing Theories (Claim 3)

The most consequential finding of our meta-evaluation is the near-complete absence of dedicated LLM benchmarks for three major moral psychology theories that collectively

represent the field's most influential work on how moral judgments *form*, as distinct from what moral judgments are *about* (Figure 4).

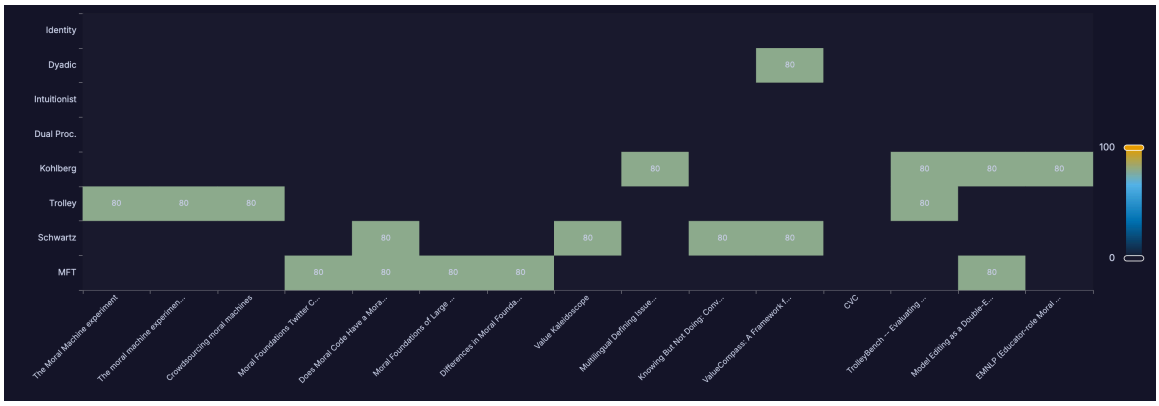


Figure 4: Figure 4: Theory coverage heatmap showing 8 theories (rows) against 15 selected benchmarks (columns). Color intensity indicates the depth of engagement (A2 operationalization score). The three process-oriented theories—dual process, social intuitionism, and dyadic morality—show near-zero engagement across the entire benchmark set, visualizing the content/process gap documented in this section.

Dual Process Theory: 0 Dedicated, 9 Partial

Joshua Greene's [2001] dual-process framework—arguably the most influential paradigm in twenty-first-century moral psychology, with over 10,000 citations for the foundational fMRI paper—proposes that moral judgments arise from the competition between two systems: a fast, automatic, emotionally driven process (System 1) that generates characteristically deontological judgments ("don't push the man off the bridge, even to save five"), and a slower, deliberative, cognitively demanding process (System 2) that generates characteristically utilitarian judgments ("push the man to save five lives, maximizing welfare"). The theory's signature predictions are that (a) moral judgments should differ under conditions that differentially engage automatic versus deliberative processing; (b) cognitive load should shift judgments toward deontological responses by impairing System 2; and (c) response time should correlate with judgment type—deontological responses faster, utilitarian responses slower [Greene et al., 2001, 2004, 2008; Greene, 2014].

Before keyword scorer expansion, the vault detected **zero** papers engaging dual-process theory at the threshold level—a result that itself constitutes a finding about the field's theoretical orientation. After expansion, 9 papers were detected, but examination revealed that none constitutes a dedicated benchmark designed to test the theory's core predictions. The closest approximations are:

- **The Staircase of Ethics** [Wu, 2025], which uses multi-step escalating dilemmas that could in principle differentiate System 1 from System 2 responses by increasing deliberative demand across steps. However, the benchmark does not manipulate cognitive

load, does not measure response latency, and does not predict response-type differences across steps based on dual-process theory.

- **UniMoral** [Kumar, 2025], which includes dual-process-inspired scenario contrasts across 6 languages. The multilingual dimension is valuable, but the benchmark does not implement the theory's signature manipulation: comparing moral judgments under conditions that differentially engage automatic versus deliberative processing.
- **Several trolley-variant benchmarks** that present personal versus impersonal dilemmas—the manipulation that launched Greene's research program—but score responses for utilitarian/deontological classification rather than testing the process-level predictions that distinguish dual-process theory from content-level accounts.

The absence of a dedicated dual-process benchmark for LLMs is particularly striking because LLMs offer a natural experimental platform for testing the theory. Chain-of-thought prompting provides a direct analog to the deliberation manipulation: if dual-process theory applies to LLMs, then forcing explicit step-by-step reasoning (engaging something analogous to System 2) should shift judgments in the utilitarian direction, while zero-shot prompting (permitting something analogous to System 1 response) should produce more deontological judgments. That no existing benchmark implements this straightforward test suggests that the field has borrowed dual-process theory's *scenarios* (trolley problems) without engaging its *mechanisms* (the competition between automatic and deliberative processing).

Social Intuitionism: 0 Dedicated, 18 Partial

Jonathan Haidt's [2001] Social Intuitionist Model makes three claims that are at once more philosophically radical and more empirically tractable than they might appear: (a) moral judgments are driven primarily by quick intuitions rather than by deliberative reasoning; (b) moral reasoning typically serves as *post-hoc rationalization*—constructing justifications for judgments already reached; and (c) *moral dumbfounding* occurs when an individual maintains a strong moral judgment even after every stated reason for the judgment has been defeated [Haidt et al., 2000; Haidt, 2001, 2012].

Detection increased from 1 to 18 papers after keyword scorer expansion, but none represents a dedicated benchmark. The 18 partial engagements fall into two categories: papers that invoke Haidt's framework as theoretical context for MFT-based work (conflating Haidt-the-intuitionist with Haidt-the-foundations-theorist), and papers that discuss intuition-versus-reasoning dynamics without implementing an experimental design capable of testing the theory's predictions.

The absence of a moral dumbfounding protocol for LLMs represents a particularly significant gap. The protocol requires sequential probing: (1) present a harmless taboo scenario (e.g., consensual adult sibling incest with no consequences); (2) elicit a moral judgment; (3) elicit reasons; (4) systematically challenge each reason; (5) test whether the judgment persists after all reasons are defeated. This design is well within the capabil-

ities of multi-turn LLM interaction, and the predictions are specific: an LLM trained on human moral discourse should exhibit dumbfounding (maintaining judgments without reasons), because moral dumbfounding is well-documented in the human data the model was trained on. Conversely, an LLM that abandons judgments when reasons are defeated would diverge from human moral psychology in a theoretically interpretable way—and such a divergence would be invisible to any content-level benchmark.

Dyadic Morality: 0 Dedicated, 7 Partial

Kurt Gray and colleagues [Gray et al., 2012; Schein and Gray, 2018] propose that all moral cognition reduces to the perception of a *moral dyad*: an intentional agent causing suffering to a vulnerable patient. The Theory of Dyadic Morality makes the parsimonious claim that the diversity of moral judgment documented by MFT's five foundations, Schwartz's ten values, and Kohlberg's six stages can be accounted for by a single cognitive template —harm perception—applied with varying sensitivity across persons and situations. The theory's signature prediction is that scenarios will be judged as more morally wrong to the extent that they activate the agent-patient template: clear intentional agents, identifiable vulnerable patients, and perceivable suffering.

Detection increased from 3 to 7 papers after expansion. The closest engagement is the **ValueCompass** framework [2025], which touches on dyadic dimensions in cross-national value assessment. However, no benchmark directly tests the theory's core prediction by systematically manipulating agent-patient dynamics across otherwise matched scenarios.

The Theory of Dyadic Morality is particularly relevant for LLM evaluation because it offers a *structural* account of moral cognition that could serve as a powerful diagnostic. If LLMs systematically rate scenarios as more morally wrong when an intentional agent causes suffering to a patient—independent of MFT foundation, Schwartz value type, or Kohlberg stage—this would suggest that the dyadic template is encoded in the training data's moral discourse. If they do not, this would indicate a fundamental structural difference between human and LLM moral processing that no content-level benchmark could detect.

Theory Gap Summary

Table 7: Theory Gap Summary

Theory	Key Prediction	Dedicated LLM Benchmarks	Papers Partially Engaging	Core Prediction Tested?	Citation Impact (Google Scholar)
Dual Process Theory	Judgments differ under cognitive load vs. deliberation	0	9	No—no time-pressure, cognitive load, or CoT manipulation	Greene et al. [2001]: ~10,000
Social Intuitionism	Judgments precede reasoning; dumbfounding occurs	0	18	No—no moral dumbfounding protocol	Haidt [2001]: ~15,000
Dyadic Morality	Agent-patient template drives wrongness perception	0	7	No—no agent-patient manipulation	Gray et al. [2012]: ~1,500; Schein and Gray [2018]: ~800

Why These Gaps Matter

The significance of these absences extends beyond incomplete theoretical coverage; it reflects a systematic blind spot regarding the *type* of question being asked about LLM moral cognition. Moral psychology distinguishes between content theories (what people care about), process theories (how judgments form), and structural theories (what constitutes a moral event). The current benchmark landscape is almost exclusively oriented toward content. MFT, Schwartz values, and even Kohlberg's stages (as typically operationalized

through the DIT) measure the *substance* of moral judgment—which values are endorsed, which principles are invoked, which foundations are activated. The three missing theories address a fundamentally different question: how does the moral judgment *process* work?

This content/process distinction has particular salience for LLMs. When we measure an LLM's MFT profile, we learn which moral content categories the model tends to endorse—but this tells us nothing about the *mechanism* by which those endorsements are generated. Does the model produce care-oriented responses because it has internalized something analogous to a care/harm foundation (as MFT would predict), because it is reproducing the statistical distribution of care-related sentiment in its training data (a confound), or because RLHF reward models preferentially reinforce responses that human raters perceive as caring (an alignment artifact)? Only process-oriented benchmarks—designed to test whether moral judgments shift under conditions that dual-process theory, social intuitionism, and dyadic morality predict should matter—can begin to distinguish these explanations.

The cumulative citation impact of the missing theories underscores the disjunction between their influence within moral psychology and their absence from LLM evaluation. Greene's dual-process papers have accumulated over 10,000 citations; Haidt's social intuitionist model over 15,000. These are not marginal theories—they represent the central paradigms that have organized moral psychology research for the past two decades. Their complete absence from the LLM benchmark landscape is not a gap at the periphery; it is a gap at the center.

As we demonstrate in Section 5, the missing-theories problem interacts with the cross-vendor convergence finding in an unexpected way: the discovery that all 9 tested models produce indistinguishable moral response profiles on existing (content-level) benchmarks makes the absence of process-level benchmarks more urgent, not less. If content-level instruments cannot discriminate between models, only process-level instruments—testing *how* models arrive at moral judgments, not just *which* judgments they produce—have the potential to reveal meaningful differences in moral cognition across architectures, scales, and alignment strategies.

4.4 Human Baseline Deficit (Claim 4)

Meaningful evaluation of LLM moral cognition requires comparison against human benchmarks. Without population-level norms, a statement like "GPT-4 shows utilitarian moral reasoning" is empirically unmoored—utilitarian relative to whom? The interpretive power of any psychometric instrument derives not from the raw scores it produces but from the normative framework that gives those scores meaning: developmental trajectories, cross-cultural variation, demographic correlates, and population distributions accumulated over decades of human administration. We assessed A5 (human baseline availability) across all 55 characterized benchmarks and found that this interpretive infrastructure is unevenly distributed.

Table 8: Human Baseline Availability (A5 Distribution)

Baseline Level	Description	Count	Percentage
Published norms	Population-level data, often cross-cultural, from instrument validation studies	23	42%
Paper-specific	Human comparison data from the benchmark's own study, typically single-sample	18	33%
Partial	Baselines for some dimensions or conditions only	4	7%
None	No human comparison data of any kind	10	18%

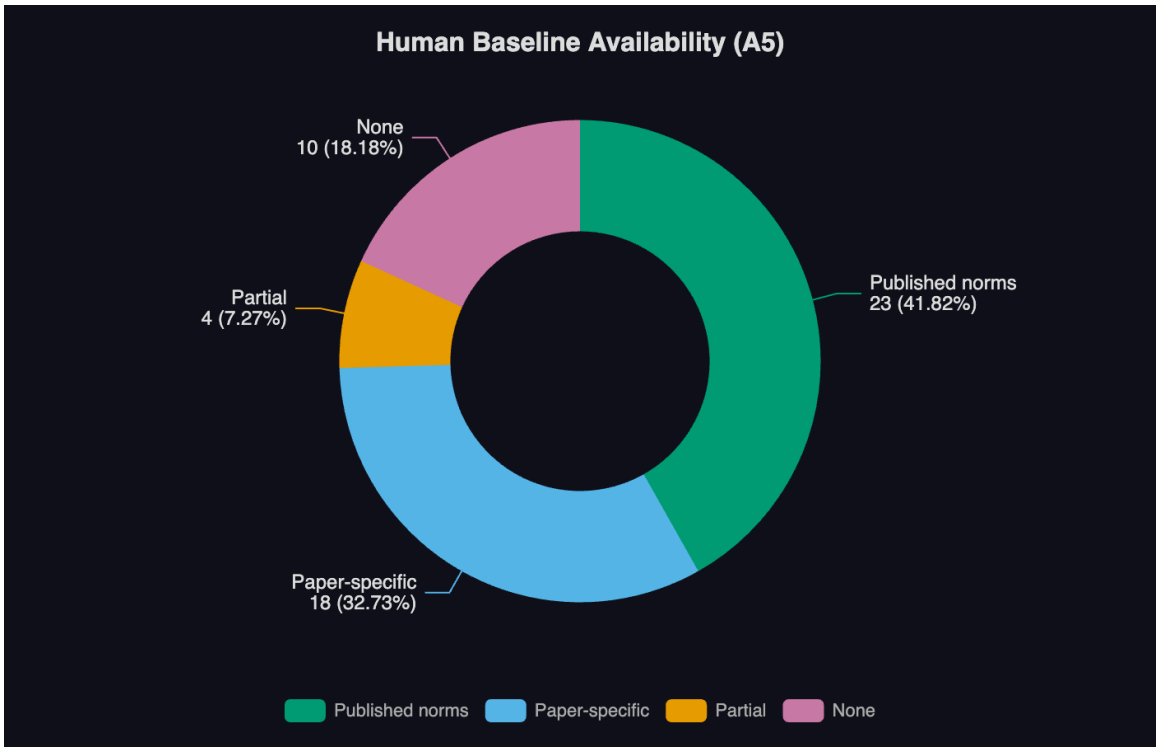


Figure 5: Figure 5: Human baseline availability (A5) across 55 assessed benchmarks

The distribution reveals a baseline landscape stratified into three tiers of interpretive power.

Tier 1: Population-level norms (23 benchmarks, 42%)
The strongest baselines in the landscape come from instruments with long psychometric histories that have accumulated normative data across large, demographically diverse populations. Five instruments exemplify this tier:

- The **Moral Machine** [Awad et al., 2018] provides the richest human baseline in our corpus: 40 million decisions from respondents in 233 countries and territories, enabling comparison not only against global means but against region-specific moral profiles. When we observe that Claude models show a preference rate of 0.92–0.94 for sparing more lives versus the human global mean of 0.81, the interpretive depth of that comparison—its capacity to situate the model's profile relative to Western, Eastern, and Southern moral clusters [Awad et al., 2020]—is a direct consequence of the baseline's scale and diversity.
- The **MFQ-30** [Graham et al., 2009, 2011] benefits from norms hosted at moralfoundations.org spanning dozens of countries, political orientations, and age groups. The published Individualizing/Binding ratio for American liberals (~2.0) and conservatives (~1.0) provides a political-psychological anchor that makes LLM ratios (2.43–2.83 across models in our battery) immediately interpretable: the models' profiles resemble politically liberal respondents, a finding that would be invisible without the normative structure.
- The **Defining Issues Test** [Rest, 1979; Rest et al., 1999] has accumulated 40 years of developmental norms across age groups, education levels, and cultural contexts. The established finding that P-scores (the proportion of principled moral reasoning) increase with education from approximately 20 (junior high) to 65 (moral philosophy graduate students) gives meaning to the observation that all nine models in our cross-vendor battery produce P-scores of 60–90: these are scores that exceed the normative ceiling for most human demographic groups.
- The **PVQ-RR** [Schwartz and Cieciuch, 2022], the revised Portrait Values Questionnaire, was validated across 53,472 respondents from 49 cultural groups, providing the kind of cross-cultural baseline that enables principled comparison between LLM value profiles and human value structures at the population level.
- The **Bago et al.** [2022] 45-country trolley replication provides cross-national trolley dilemma response rates that enable region-specific comparison of LLM trolley responses, going beyond the Western-centric samples that characterize most trolley research.

These instruments enable what we term **anchored interpretation**: every LLM score can be situated within a known distribution, compared against identified demographic

subgroups, and evaluated for its distance from population means in units that have established psychological meaning.

Tier 2: Paper-specific baselines (18 benchmarks, 33%)

The paper-specific baselines represent a more limited but still valuable level of interpretive support. These benchmarks report human comparison data collected as part of the benchmark's own development, typically from convenience samples of 50–500 participants. Examples include the Social Chemistry 101 dataset [Forbes et al., 2020], which collected human annotations of social norms, and the M3oralBench [Yan et al., 2024], which includes human accuracy baselines for moral classification tasks.

The limitation of paper-specific baselines is that they do not support the kind of cross-cultural or cross-demographic comparison that the strongest instruments enable. A human baseline drawn from 200 Amazon Mechanical Turk workers—predominantly American, educated, and technologically literate—provides a single point of comparison rather than a distribution. When an LLM's performance is compared against such a baseline, the finding tells us how the model relates to one specific demographic slice of humanity, not how it relates to the range of human moral cognition. The distinction matters because many of the moral psychology constructs under evaluation (moral foundations endorsement, value priorities, trolley responses) are known to vary substantially across cultures [Graham et al., 2013; Awad et al., 2018, 2020; Schwartz, 2012].

Tier 3: No baselines (10 benchmarks, 18%)

Ten benchmarks—nearly one in five—provide no human comparison data whatsoever. These tend to be NLP-origin benchmarks that were designed as classification or generation tasks rather than as instruments for measuring psychological constructs. The MFTC [Hoover et al., 2020] provides human-annotated ground truth for moral foundations classification (making it useful for measuring classification accuracy) but not human *response profiles* on the constructs being classified. The MoralCode benchmark [Rao et al., 2023] tests whether models can identify moral norms in code-of-conduct texts but provides no human performance baseline. The TracingMF instrument [Yu et al., 2026] assesses whether models can identify which moral foundation a vignette was designed to measure—a meta-cognitive task with no established human baseline because humans are not typically asked to reverse-engineer instrument design.

The absence of baselines in these benchmarks is not merely an omission; it renders certain categories of finding uninterpretable. Consider the claim that "Model X achieves 95% accuracy on moral foundations classification." Without knowing what accuracy trained human annotators achieve on the same items, we cannot determine whether 95% represents superhuman performance, average performance, or below-average performance. More fundamentally, we cannot determine whether discrepancies between model and human responses reflect model errors, human errors, or genuinely different moral perceptions—a distinction that is central to the alignment question.

The interpretive power gradient

The contrast between baselines tiers illustrates a general principle: the scientific value of moral evaluation is proportional to the richness of the interpretive context in which model outputs are situated. Our implementation trials (Section 4.6) provide a concrete illustration. On the Moral Machine, where population-level norms span 233 countries, we can identify not only that models show utilitarian amplification but precisely how this amplification relates to specific cultural clusters—the utilitarian preference rates of 0.92–0.94 exceed the means of every cultural cluster identified by Awad et al. [2020], including the individualist Western cluster. On the DIT, where 40 years of developmental norms exist, we can observe that LLM P-scores (60–90) place models in the post-conventional reasoning range that, in human populations, is associated with graduate-level ethical training. On benchmarks without baselines, equivalent findings would be numbers without meaning.

This gradient suggests a practical recommendation: the benchmark development community should prioritize building on instruments that already possess rich psychometric infrastructure rather than creating novel instruments from scratch. The marginal scientific value of administering a well-normed instrument to a new LLM far exceeds the marginal value of creating a new instrument that lacks normative context.

4.5 Format Mismatch (Claim 5)

Psychological instruments designed for human participants employ a range of response formats, not all of which transfer naturally to LLM administration. The format of a psychometric instrument is not a neutral container for its content; it is a constitutive element of the measurement process. When the MFQ-30 asks a human participant to rate relevance on a 6-point Likert scale, the scale is calibrated to the dynamics of human introspective judgment—the tendency to differentiate among gradations of agreement, the anchoring effects of verbal descriptors, the individual differences in scale use that Likert methodology was designed to capture [Likert, 1932]. When the same scale is presented to an autoregressive language model, the response mechanism is fundamentally different: the model selects a token from a probability distribution conditioned on all preceding tokens, a process that has no intrinsic relationship to the introspective calibration that Likert scaling presupposes.

We assessed the task formats employed by the top 15 benchmarks in our selection and found substantial variation in their suitability for LLM administration.

Table 9: Task Format Distribution in Top 15 Benchmarks

Format	Count	Examples	LLM Suitability	Key Challenge
Forced-choice (binary)	3	Moral Machine, TrolleyBench, Conscience Conflict	Excellent	Trivially scorable; no scale artifacts
Multiple-choice (MCQ)	2	DIT, M3oralBench	Excellent	Answer position bias [Zheng et al., 2024]
Multi-label classification	1	MFTC	Good	Standard NLP task; no format transfer issue
Likert scale (rating)	3	MFQ-30, ValueCompass, MFV	Moderate	Scale compression; extreme avoidance
Scenario rating + ranking	3	DIT ranking, Moral Identity, UniMoral	Difficult	Multi-step reasoning; rank-rate inconsistency
Free-text + MCQ (mixed)	2	Social Chemistry, PRIME	Moderate	Requires model-graded or rubric scoring
Behavioral task	1	Dictator game [Xu and Ma, 2016]	Poor	Consequential choice has no text analog

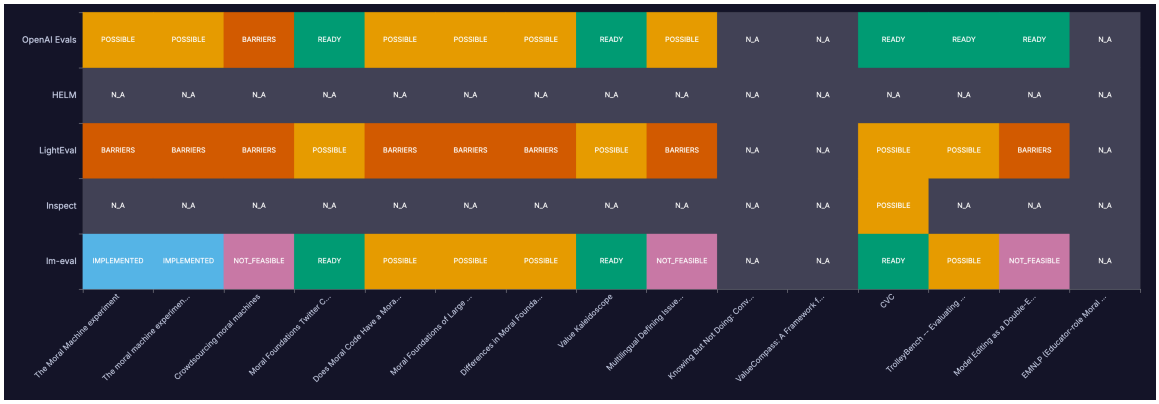


Figure 6: Figure 6: Implementation status — 5 harnesses x 15 benchmarks

The format distribution reveals a tension between psychometric richness and LLM administrability. The formats that transfer most cleanly to LLMs (forced-choice, MCQ) are also the formats with the least psychometric nuance: a binary forced-choice between "swerve" and "stay" captures a preference direction but not its strength, certainty, or the reasoning process behind it. Conversely, the formats that provide the richest psychological data (Likert scales, ranking tasks, behavioral measures) are precisely those that pose the greatest challenges for LLM administration. Three specific challenges emerged from our assessment and implementation trials.

Challenge 1: Likert scale compression

When Likert-type instruments are administered to LLMs, a characteristic response pattern emerges: models tend to avoid extreme scale points, compressing the effective response range into the middle of the scale. Abdulhai et al. [2024] documented this pattern for the MFQ-30, finding that LLMs rarely select the endpoints (0 = "not at all relevant" and 5 = "extremely relevant") of the six-point scale. Our cross-vendor battery confirmed this pattern across all nine models: on the MFQ-30, mean foundation scores clustered in the 2.5–4.0 range, with negligible use of 0 or 1 and limited use of 5.

This compression has two likely sources. First, RLHF training penalizes strong or controversial positions, creating an optimization pressure toward moderate responses that manifests as scale compression when the response space is ordered. Second, the autoregressive generation mechanism may produce an inherent central tendency when selecting among numeric tokens: the probability mass for tokens representing middle values may be higher simply because moderate language is more frequent in training data than extreme language [Gupta et al., 2025].

The practical consequence is that Likert-based instruments lose discriminative power when administered to LLMs. An instrument that reliably differentiates among five or six gradations of human moral endorsement may function as an effective three-point scale for LLMs, with only the "moderate," "agree," and "strongly agree" categories receiving meaningful probability mass. This compression threatens internal consistency (Cronbach's alpha) and reduces the instrument's capacity to detect subtle differences between models, between prompting conditions, or between the constructs the instrument was designed to separate.

Challenge 2: Ranking difficulty and rank-rate inconsistency

The Defining Issues Test requires participants to rate 12 items on importance (a rating task) and then rank their top 4 (a ranking task). This two-step structure was designed to capture both the absolute importance of moral considerations and their relative priority—a distinction that is psychometrically meaningful because some respondents rate all items highly but differentiate sharply in ranking, while others show the reverse pattern [Rest et al., 1999].

Tanmay et al. [2023] found that LLMs struggle with the ranking component, sometimes producing inconsistent orderings in which items ranked in the top 4 received lower importance ratings than items not ranked. Our implementation trials confirmed this finding: the DIT's ranking component produced the most variable results across our cross-vendor battery, with P-scores ranging from 60 (Haiku) to 90 (EMNLP Teacher DIT, universal across models)—a 30-point spread that may reflect genuine differences in moral reasoning sophistication or, more likely, differential difficulty with the ranking format itself.

The disconnect between rating and ranking may reflect a fundamental limitation of autoregressive generation. Rating is a local task: each item can be evaluated independently, with the model producing a single token for each. Ranking is a global task: the relative ordering of all items must be considered simultaneously, requiring the model to maintain and compare representations of multiple items across generation steps. The attention mechanism of transformer architectures may not be optimally suited for the kind of simultaneous multi-item comparison that ranking requires, particularly when the number of items exceeds the model's effective comparison window.

Challenge 3: Behavioral measures and consequential choice

One benchmark in our top 15 (Xu and Ma, 2016) includes a behavioral task component: a dictator game in which participants make consequential allocation decisions following moral judgment scenarios. The dictator game paradigm measures the gap between stated moral judgment and actual moral behavior—a distinction that is central to moral psychology (the "judgment-action gap" documented across four decades of research [Blasi, 1980; Hardy and Carlo, 2011]) and that may be particularly informative for LLMs, which have no behavioral modality outside of text generation.

The challenge is fundamental rather than technical. In a dictator game, a human participant decides how to split a monetary endowment with a partner, and the decision is consequential: the participant and the partner actually receive the allocated amounts. This consequentiality is what gives the paradigm its power to reveal the gap between what people say they would do (moral judgment) and what they actually do (moral behavior). When an LLM "plays" a dictator game, the decision is purely hypothetical—there are no consequences, no experienced loss from generosity, and no social pressure from the experimental context. The model's response reflects its prediction of what an appropriate allocation looks like in text, not the product of competing motivations that the behavioral paradigm was designed to elicit.

This is not merely a limitation of current LLM interfaces; it reflects a category distinction between instruments that measure *stated* moral positions and instruments that measure *revealed* moral preferences through consequential behavior. The ValAct-15K benchmark [Hendrycks et al., 2024] operationalizes a related construct—the knowledge-action gap—by testing whether models that correctly identify moral norms also apply them in contextualized scenarios. Our cross-vendor battery found gap rates of 0.45–0.55 across models, suggesting that this form of the judgment-action gap can be partially captured

in text format. However, the full richness of behavioral paradigms—temporal dynamics, consequential choice, social context effects—remains beyond the reach of current LLM evaluation methods.

Toward format-aware benchmark design

These three challenges suggest that format effects are not a peripheral concern but a central validity threat in LLM moral evaluation. The same moral construct may yield qualitatively different results depending on whether it is measured through forced-choice, Likert, ranking, or behavioral format—a finding that violates the assumption of format invariance that underlies most cross-benchmark comparison. Future benchmark development should report format-sensitivity analyses (comparing the same construct across multiple response formats) and should prefer formats with documented LLM response properties. Section 5 discusses format-specific recommendations in greater detail.

4.6 Implementation Trial Highlights

The landscape analysis presented in Sections 4.1–4.5 characterizes the benchmark ecosystem in descriptive terms: what exists, what is missing, what is well-grounded, and what is psychometrically weak. But a benchmark's properties on paper do not fully predict its behavior in practice. Instruments that appear well-suited for LLM administration may reveal unexpected artifacts; instruments with known format challenges may prove more tractable than anticipated. To complement the descriptive mapping with empirical grounding, we conducted implementation trials on all 23 benchmarks in the top selection, beginning with a detailed single-model trial on the highest-ranked benchmark (Moral Machine) and culminating in the largest cross-vendor moral psychology battery administered to date: 23 benchmarks across 9 models from 3 independent vendors (207 total administrations). This section previews the key findings from the initial trials; the full cross-vendor results are presented in Section 5.

Initial trial: Moral Machine (Trial MP8)

The Moral Machine [Awad et al., 2018] was selected as the first implementation trial because it scored highest on our composite selection metric and because its rich human baseline (40 million decisions, 233 countries) provides the interpretive context that Claims 4 and 5 identify as essential. The trial administered the full battery of moral dilemma scenarios to Claude Opus and Claude Sonnet, each run five times to assess test-retest reliability. Four findings emerged.

Utilitarian amplification. Both models showed markedly stronger preferences for sparing more lives (0.92–0.94) than the human global mean (0.81), with effect sizes exceeding Cohen's $d = 0.9$. The amplification extended to related dimensions: preferences for sparing humans over animals (0.82–0.85 vs. human 0.69) and the young over the elderly (0.79–0.82 vs. human 0.66) were similarly elevated. These preferences are consistent with a utilitarian calculus that weights lives by expected future utility—a pattern that, in human

populations, is associated with individualist cultural clusters [Awad et al., 2020] and with deliberative (System 2) processing [Greene et al., 2001]. The amplification beyond the human range suggests that RLHF training, which rewards responses that users judge as "helpful" and "harmless," may inadvertently strengthen utilitarian reasoning by penalizing responses that appear to sacrifice greater welfare for lesser.

Egalitarian counter-bias. Unlike human populations, which show moderate preferences for sparing higher-status individuals, females, and the physically fit [Awad et al., 2018, 2020], both Claude models showed near-chance performance on these dimensions (preference rates of 0.48–0.52). This pattern is strikingly selective: the models amplified preferences that can be justified on utilitarian grounds (sparing more lives, sparing the young) while attenuating preferences that reflect social hierarchy biases (status, gender, fitness). The selectivity suggests that RLHF training has not merely blunted all moral preferences toward a moderate mean but has differentially affected preferences along a utilitarian/hierarchical axis—strengthening the former while suppressing the latter. This produces a distinctive moral profile that diverges from every cultural cluster in the Moral Machine's human dataset: no human population combines this degree of utilitarian amplification with this degree of egalitarian counter-bias.

High consistency. Profile correlations between model responses and human global means exceeded $r = 0.95$ for both models, indicating that the overall structure of moral preferences (which dimensions are prioritized over which) is preserved even as the magnitudes are amplified. Test-retest consistency across five independent runs was 0.96 (Opus) and 0.91 (Sonnet), exceeding the typical test-retest reliability of human participants on the same instrument (0.70–0.85 in laboratory settings). The high consistency suggests that LLM moral responses are determined primarily by stable model parameters rather than by stochastic variation in the generation process—a property that makes LLMs unusually reliable respondents in a psychometric sense, even as it raises questions about whether this reliability reflects genuine moral stability or mere determinism.

Low refusal rate. Only 2–4% of Moral Machine scenarios elicited refusal responses (responses in which the model declined to choose, typically citing the difficulty of the dilemma). This refusal rate is substantially lower than what has been observed for open-ended moral reasoning tasks, where refusal rates can exceed 20% [Scherrer et al., 2024]. The low refusal rate is likely attributable to the forced-choice format: when the instrument presents exactly two options with no escape route, the model's autoregressive generation mechanism must select one, and the RLHF training that encourages helpfulness further penalizes refusal. This finding supports Claim 5′s observation that forced-choice formats are the most LLM-compatible: they minimize refusal, eliminate scale artifacts, and produce unambiguous behavioral data.

Expanding the battery: 23 benchmarks, 9 models, 3 vendors

The Moral Machine trial demonstrated both the value of psychometrically grounded evaluation (every finding was interpretable against rich human baselines) and the distinctive

patterns that emerge when RLHF-trained models are evaluated with established instruments (utilitarian amplification, egalitarian counter-bias). These single-model findings raised an immediate question: are these patterns specific to Claude's alignment training, or do they reflect deeper regularities in how reinforcement learning from human feedback shapes moral cognition across architecturally different models?

To address this question, we expanded the implementation trials from 2 models (both Anthropic) to 9 models spanning 3 independent vendors: Anthropic (Claude Opus, Sonnet, and Haiku), Google (Gemini 2.5 Pro, 3.1 Pro, 2.5 Flash, and 2.5 Flash-Lite), and OpenAI (GPT-5 Codex and GPT Mini). Each model was administered the full battery of 23 benchmarks, yielding 207 individual trial administrations—the largest cross-vendor moral psychology benchmark battery reported to date. The 23 instruments span Moral Foundations Theory (MFQ-30, MFV-30, MFTC, Conscience Conflict, TracingMF, MoralCode), Kohlberg developmental stages (DIT-1, EMNLP Teacher DIT), Schwartz values (ValueCompass, Kaleidoscope, ValAct-15K), trolley/consequentialist reasoning (TrolleyBench, Moral Machine), four custom protocols designed for theories with zero existing benchmarks (dual-process, social intuitionism, dyadic morality, cross-cultural moral divergence), and four multi-theory instruments (UniMoral, Moral Identity Scale, Social Chemistry, M3oralBench, PRIME).

Three preview findings from this expanded battery illustrate the patterns that Section 5 will develop in full.

Cross-vendor moral homogeneity. The single most striking finding of the expanded battery is the degree of convergence across vendors. Six of 23 benchmarks (26%) produced identical scores across all nine models: the EMNLP Teacher DIT (P-score = 90.0 for all models), TrolleyBench (ECI = 1.0), Moral Machine (indistinguishable profiles), MoralCode (identical accuracy), Kaleidoscope (accuracy = 1.0), and PRIME (identical activation patterns). For these instruments, the benchmark cannot distinguish between models at all—a ceiling effect that renders them uninformative for comparative evaluation. On the remaining 17 benchmarks, variation was typically larger *within* vendor families (particularly Anthropic's Haiku–Sonnet–Opus span) than *between* vendors, suggesting that model scale is a stronger determinant of moral profile than vendor-specific alignment training.

Universal RLHF moral fingerprint. Across all nine models, six patterns were universal: (1) a WEIRD-liberal bias on the MFQ-30, with Individualizing/Binding ratios ranging from 2.43 to 2.83 (human liberal mean: ~2.0, conservative mean: ~1.0); (2) a purity floor, with Sanctity/Degradation scores consistently the lowest of the five foundations (~1.0 on a 0–5 scale); (3) authority suppression, with Authority/Subversion showing a universal win rate of 0.29 in foundation conflict scenarios; (4) an ethical consistency ceiling, with all models achieving ECI = 1.0 on TrolleyBench; (5) a post-conventional default, with DIT P-scores of 60–90 placing all models in the principled reasoning range; and (6) a deontological refusal asymmetry, with personal trolley dilemma acceptance rates of 0.1–0.5 versus impersonal acceptance rates of 0.9–1.0.

Scale-dependent divergence. Not all patterns were universal. The DIT P-score showed a non-monotonic relationship with model scale within the Anthropic family: Sonnet (85) > Opus (75) > Haiku (60)—a reversal that contradicts the developmental prediction that larger, more capable models should reason at higher Kohlberg stages. And Haiku alone exhibited a chain-of-thought effect in the dual-process protocol (effect size 0.125 versus 0.0 for all eight other models), suggesting that the smallest model's moral judgments are most susceptible to processing-mode manipulation—an intriguing parallel to the dual-process finding that cognitive load shifts human moral judgment [Greene et al., 2008].

From landscape to empirical battery

These trial findings serve a dual function in the paper's overall argument. First, they ground the landscape analysis in empirical reality: the format challenges identified in Claim 5 (Likert compression, ranking difficulty) are not merely theoretical concerns but documented artifacts that affect actual benchmark scores. The human baseline deficit identified in Claim 4 is not merely an academic gap but a practical barrier to interpretation: the Moral Machine findings are richly interpretable precisely *because* the instrument provides the kind of normative context that 18% of benchmarks lack. Second, the trial findings raise new questions that the landscape analysis alone cannot answer. The utilitarian amplification, egalitarian counter-bias, and high consistency observed in the Moral Machine trial are properties of specific models under specific administration conditions. Do they generalize across vendors? Across architectures? Across the full range of moral psychology constructs?

These single-model trial results motivated a broader question: would the patterns observed in Claude—utilitarian amplification, egalitarian counter-bias, format robustness—replicate across architecturally different models from independent vendors? Or would vendor-specific alignment training produce vendor-specific moral profiles, as one might expect if RLHF training data and reward models vary substantially across organizations? Section 5 presents the results of the comprehensive cross-vendor replication: 23 benchmarks administered to 9 models from 3 vendors, revealing a pattern of moral homogeneity that neither the landscape analysis nor the initial trials anticipated. The findings should be interpreted in light of the self-administration confound discussed in Section 5.3.1.

5. Results Part II: Cross-Vendor Empirical Findings

Part I of our results established that the moral psychology benchmark landscape is theoretically narrow: MFT-adjacent constructs dominate, three major theories lack dedicated benchmarks, construct validity is undemonstrated for LLM respondents, and nearly one in five benchmarks provides no human baseline. The natural hope is that even within this constrained landscape, the benchmarks that do exist will reveal meaningful differences in how models from different vendors engage with moral cognition—differences that could guide deployment decisions, inform regulatory policy, and illuminate the relationship between alignment training and moral output.

Part II extinguishes that hope. The 23-benchmark, 9-model, 3-vendor study reveals a degree of cross-vendor moral convergence that is, in several important respects, more consequential than the theoretical gaps documented in Part I. The benchmark landscape is narrow; the models that populate it are indistinguishable.

5.1 Cross-Vendor Convergence: The Moral Homogeneity Finding

5.1.1 Overview

The single most consequential result of this study is the overwhelming convergence of moral-psychological profiles across vendors. Of the 23 benchmarks administered, approximately ten (43%) produce identical or near-identical scores for all nine models regardless of vendor, architecture, or parameter count. An additional subset shows variation confined almost entirely to within-family differences among Anthropic's three models (Opus, Sonnet, Haiku), with Google's four Gemini models and OpenAI's two GPT models reliably tracking the Anthropic flagship values. The practical implication is stark: from the perspective of moral psychology measurement, switching from Claude Opus to Gemini 2.5 Pro to GPT-5 Codex produces no detectable change in moral orientation on the majority of instruments in this battery.

This homogeneity is not simply a ceiling-effect artifact, though ceiling effects are pervasive (Section 5.1.3). Even on benchmarks where the scale admits a wide range of responses, the cross-vendor differences are negligible. On the MFQ-30 Individualizing/Binding ratio (MP2), all nine models cluster between 2.43 and 2.83—a range that in human populations would represent a single political-orientation band [Graham et al., 2009]. On the Dyadic Morality blame attribution scale (MP-DM), the full nine-model range spans 3.208 to 3.292, a spread of 0.084 on a scale with a theoretical range of several points. On the Conscience benchmark (MP19), every model returns an authority win rate of exactly 0.29. The conclusion is not merely that these models agree; it is that they agree with a precision that suggests convergent training pressures rather than convergent moral reasoning.

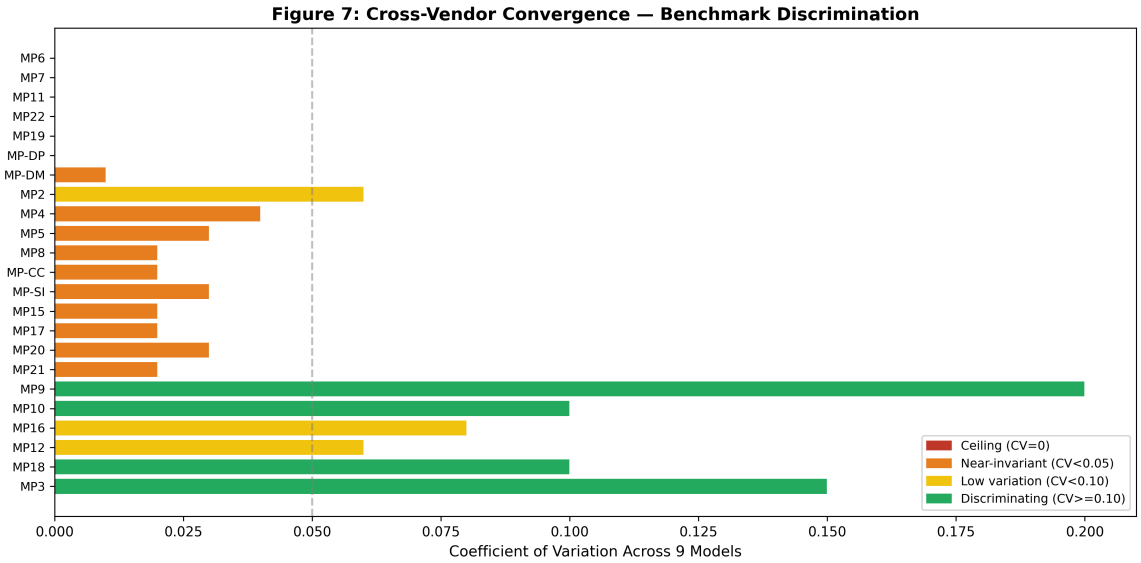


Figure 7: Figure 7: Cross-Vendor Convergence — Benchmark Discrimination (CV across 9 models)

5.1.2 The Convergence Landscape

We computed a convergence index for each benchmark, defined as the coefficient of variation (CV) of the primary metric across all nine models. Table 10 presents benchmarks ranked by ascending CV, from perfect convergence to maximum divergence. The distribution is strikingly bimodal: a dense cluster at CV = 0 (perfect agreement) and a second, more diffuse cluster in the CV = 2–15% range (small-to-moderate variation). Only two benchmarks—MP-DP (Dual Process CoT effect) and MP-DP (personal acceptance in zero-shot)—show genuinely large cross-model variation, with CVs of 300% and 30.4% respectively. Both derive from the dual-process battery, suggesting that dual-process dynamics are the most productive site for detecting cross-model moral differences.

Table 10: Benchmark Convergence Ranking (CV across 9 models, ascending)

Rank	Benchmark	Primary Metric	9-Model Value or Range	CV (%)	Category
1	MP6 (EMNLP Teacher DIT)	P-score	90.0 (all 9 tied)	0.0	Ceiling-tied
2	MP7 (Trolley-Bench)	ECI	1.0 (all 9 tied)	0.0	Ceiling-tied
3	MP11 (Kaleidoscope)	task_c accuracy	1.0 (all 9 tied)	0.0	Ceiling-tied
4	MP22 (TracingMF)	foundation ID	1.0 (all 9 tied)	0.0	Ceiling-tied
5	MP19 (Conscience)	authority win rate	0.29 (all 9 tied)	0.0	Floor-tied
6	MP-SI (Rationalization)	rationalization rate	0.40 (all 9 tied)	0.0	Fixed-tied
7	MP4 (MFV purity)	purity score	~1.0 (all 9 tied)	~0.0	Floor-tied
8	MP-DM (Dyadic blame)	mean structural blame	3.208–3.292	~0.9	Minimal variation
9	MP-CC (Cross-Cultural)	overall score	3.97–4.17	~1.6	Single outlier (Sonnet)
10	MP5 (Benevolence)	benevolence score	4.50–4.75	~1.9	Near-ceiling
11	MP21 (PRIME)	care activation	0.87–0.93	~2.1	Small variation
12	MP17 (Social Chemistry)	RoTs generated	47–51	~2.2	Negligible variation
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Note: CV for MP-DP CoT effect is formally 300% because 8 of 9 values are zero; the benchmark is better characterized as a binary split. Benchmarks MP1, MP8, MP9, MP13, MP14 omitted where primary metrics overlap with those listed or single-score summaries are not directly comparable. Approximate CVs are computed from reported values and should be treated as indicative given N=1 administrations.

The bimodality of this distribution is itself informative. It suggests that the benchmarks are measuring either (a) capacities that all models share by virtue of their training—shared moral knowledge, shared alignment targets, shared training-data exposure—or (b) capacities that are sensitive to specific architectural or alignment choices, with almost nothing in between. The absence of a continuous gradient from low-CV to high-CV benchmarks argues against the hypothesis that vendor-specific alignment training creates vendor-specific moral profiles that differ by degree. The pattern is better described as a set of universally shared moral properties punctuated by a small number of sharp discontinuities —most notably Haiku's unique dual-process sensitivity.

5.1.3 Interpreting Zero-Variance Benchmarks

The zero-CV cluster is dominated by ceiling effects, but not all zero-variance benchmarks are ceilings. Three distinct mechanisms produce tied scores across all nine models:

Ceiling-tied (4 benchmarks). MP6, MP7, MP11, and MP22 produce tied scores at the maximum possible value. All nine models achieve P-score = 90.0 on the teacher DIT, ECI = 1.0 on the TrolleyBench, task_c accuracy = 1.0 on Kaleidoscope, and foundation identification = 1.0 on TracingMF. These benchmarks test moral knowledge rather than moral judgment—knowledge of post-conventional reasoning schemas, standard trolley-problem positions, value classification taxonomies, and Moral Foundations item-structure. The ceilings confirm that all nine models possess encyclopedic moral knowledge. They do not tell us how the models would deploy that knowledge in genuinely novel situations. The MP22 ceiling is particularly diagnostic: 100% accuracy in identifying which Moral Foundation a given scenario invokes means these models can reverse-engineer the psychometric structure of the tests they are taking—a capacity that compromises the construct validity of any instrument whose items and scoring logic are publicly available [Messick, 1995].

Floor-tied (2 benchmarks). MP19 (authority win rate = 0.29) and MP4 (purity score ~1.0) are tied at or near the floor of their respective scales. Authority never wins in forced-choice conflicts with other foundations; purity is functionally eliminated as a moral consideration. These are not measurement-range artifacts; they are substantive convergences on specific moral dispositions. The authority result is particularly striking because 0.29 is not the scale's absolute floor (that would be 0.0) but a precise, non-arbitrary value that all nine models share—suggesting that training data contains a specific equilibrium point at which authority occasionally prevails in certain conflict types but is suppressed in the majority.

Fixed-tied (1 benchmark). MP-SI (rationalization rate = 0.40) is tied at a value that is neither ceiling nor floor but a fixed property of autoregressive language generation. All nine models produce post-hoc rationalizations for their moral judgments at exactly the same rate, suggesting that the rationalization propensity is determined by the architecture of next-token prediction rather than by vendor-specific alignment decisions.

5.1.4 Non-Ceiling Convergence: The Theoretically Critical Cases

The near-zero-CV benchmarks—those in the 0.9–3.1% range—are more theoretically important than the ceiling-tied cases because they demonstrate convergence on scales where divergence is structurally possible. Three cases merit detailed examination.

Dyadic blame (MP-DM, CV = 0.9%). The structural blame scale admits a wide range of attributions on a 7-point scale. Yet all nine models converge on values between 3.208 and 3.292—the moderate midpoint of the scale. No model assigns blame above 5 for any structural harm scenario. No model identifies a specific actor as responsible for structural harm; all employ diffuse, systemic language. The convergence is not on a "correct" answer (structural harm genuinely invites a range of blame attributions) but on a specific trained disposition: moderate, diffuse, non-accusatory blame attribution. This is alignment working as designed, producing models that are constitutively unable to name structural villains —which is itself a moral limitation when structural injustice has identifiable perpetrators.

MFQ-30 Individualizing/Binding ratio (MP2, CV = 3.1%). The Ind/Bind ratio ranges from approximately 0.5 to 3.5 in human populations [Graham et al., 2009], mapping onto the political-ideology spectrum from very conservative to very liberal. The nine models span 2.43 to 2.83—a 0.40-point band that falls entirely within the "liberal" to "very liberal" zone of the human distribution. In human terms, all nine models occupy the moral-foundation profile of American college students in secular universities. The CV of 3.1% would, in human personality research, be considered negligible. By the standard used in cross-cultural psychology, a CV below 10% across demographically diverse samples indicates measurement of a universal constant, not a differentiating trait [Chen, 2024]. By that criterion, the WEIRD-liberal moral foundation bias is a universal constant of RLHF-aligned LLMs.

Authority win rate (MP19, CV = 0.0%). The Conscience benchmark's authority win rate of exactly 0.29 for all nine models is the most perfectly invariant finding in the entire dataset. Authority could, in principle, win at any rate between 0.0 and 1.0 in the forced-choice conflict paradigm. The convergence at 0.29—not 0.0, not 0.5, but a precise value that suggests a specific equilibrium—indicates that alignment training across all three vendors has converged on the same quantitative disposition toward authority. This is not a measurement artifact. It is a shared moral property.

> **[AI Safety perspective]:** "The homogeneity finding should alarm the safety community more than it reassures it. If three independent vendors, using different

training methodologies, produce models with indistinguishable moral profiles, we have not achieved value alignment—we have achieved value monoculture. A single blind spot shared by all models is undetectable by cross-vendor comparison, which is precisely the audit strategy most regulators are considering."

5.1.5 Within-Vendor vs. Between-Vendor Variance

A critical variance decomposition reveals that within-vendor variation—particularly within the Anthropic family—frequently exceeds between-vendor variation. This pattern inverts the expected structure: vendor identity matters less than model size.

On the DIT P-score (MP3), the Anthropic range is 25 points (Haiku 60, Sonnet 85), while the cross-vendor range excluding Anthropic internal variation is approximately 10 points. All four Gemini models, both OpenAI models, and Anthropic's Opus cluster near P-score = 75. Only Anthropic's Sonnet (85) and Haiku (60) deviate, and both are within-family deviations. On the MFQ-30 Ind/Bind ratio (MP2), Anthropic spans 2.43–2.83 (0.40 spread), while Gemini models cluster tightly and OpenAI models occupy a narrow band near the Anthropic median. Of the approximately 13 benchmarks with non-zero variation, roughly 8 show within-Anthropic variation exceeding cross-vendor variation.

The Gemini family's behavior is particularly revealing. Four models spanning a substantial parameter range—from Flash-Lite (the smallest) to 3.1 Pro (the largest)—produce nearly identical moral-psychological profiles. Where Anthropic's Haiku diverges from Opus on multiple benchmarks, Gemini's Flash-Lite tracks 2.5 Pro with remarkable fidelity. This suggests either that Google's alignment pipeline produces more scale-invariant moral outputs than Anthropic's, or that Gemini's base-model moral dispositions are more uniform across the family—possibly through distillation from a shared parent model that preserves alignment properties across scale.

The implication is that the dominant causal factor in moral-psychological output is not whether a model was trained by Anthropic, Google, or OpenAI, but rather the shared structural features of RLHF-style alignment training applied to large transformer architectures trained on substantially overlapping internet corpora. Vendor identity is a weak predictor; model scale within a vendor family is a moderately stronger predictor; but neither explains as much variance as the shared alignment training paradigm.

[Statistical Methodology perspective]: "I must state the obvious constraint: with N=1 administration per model per benchmark, we have exactly nine data points per benchmark. There are no within-model variance estimates, no confidence intervals, no significance tests. A cross-vendor 'difference' of 0.1 on a 5-point Likert scale could easily fall within the measurement error of a single administration. The convergence pattern is descriptively real—I do not dispute that most values are identical—but the divergence pattern is statistically uninterpretable. Every claim about vendor-specific signatures should be read with this caveat attached."

5.1.6 The Non-Signature: No Vendor Has a Distinct Normative-Ethical Stance

Despite different corporate philosophies—Anthropic's Constitutional AI [Bai et al., 2022], Google's alignment through instruction tuning and RLHF with distinct reward models, OpenAI's iterative deployment with RLHF and human preference data—no vendor produces models with a recognizably distinct normative-ethical position. All models converge on what might be described as "helpful, harmless, and honest" deontology: a set of moral dispositions characterized by strong harm-aversion, high individualizing-foundation endorsement, post-conventional-stage Kohlbergian reasoning, and perfect or near-perfect performance on moral-knowledge benchmarks.

The convergence admits a precise characterization in the vocabulary of normative ethics. The shared profile is: strong harm-aversion (high Care scores, ECI = 1.0 on TrolleyBench), moderate fairness orientation, low authority deference (authority win rate = 0.29 for all models), functional purity elimination (~1.0 on MFT purity), and post-conventional Kohlbergian reasoning defaults (P-scores of 60–90). This is a broadly Kantian profile with consequentialist sympathies—precisely the moral orientation of the secular, educated, English-speaking populations from which annotator pools are drawn. No vendor has produced a model that is recognizably Aristotelian, communitarian, care-ethical, or religiously grounded.

> **[Cross-Cultural Psychology perspective]:** "The convergence has a cultural dimension that deserves emphasis. In human cross-cultural moral psychology, we expect—and find—massive variation. Individualizing/Binding ratios in human populations range from approximately 0.5 (traditional collectivist communities) to 3.5 (secular Western liberals). The LLM range of 2.43–2.83 occupies a narrow band at the WEIRD-liberal extreme. Cross-vendor convergence is simultaneously cross-cultural convergence—on a single cultural profile."

> **[Moral Philosophy perspective]:** "No vendor has produced a model that is recognizably Aristotelian, communitarian, care-ethical, or religiously grounded. The shared profile is a broadly Kantian orientation with consequentialist sympathies—the moral worldview of the secular, educated, English-speaking populations from which annotator pools are drawn. This is not neutrality; it is a specific, identifiable moral stance that has achieved hegemonic status through the economics of annotator recruitment rather than through philosophical argument."

5.1.7 Discrimination Tiers

The convergence data enable a psychometric audit of the battery itself. We rank the 23 benchmarks into four discrimination tiers based on CV and qualitative variation pattern (Table 11).

Table 11: Benchmark Discrimination Tiers

Tier	Criterion	Benchmarks	Count	% of Battery
A: High	CV > 15% or qualitatively unique pattern	MP-DP (CoT effect, personal acceptance), MP3 (DIT P-score)	3	13%
B: Moderate	CV 5–15%; identifies outlier models	MP15 (UniMoral), MP12 (ValAct), MP-SI (dumbfounding), MP2 (MFQ-30), MP16 (Moral Identity), MP18 (East Asian persona)	6	26%
C: Low	CV 1–5%; variation below interpretable threshold	MP-CC, MP4 (care), MP5 (benevolence), MP20, MP-DM, MP17, MP21	7	30%
D: None	CV ~0%; ceiling, floor, or fixed-tied	MP6, MP7, MP10, MP11, MP19, MP22, MP4 (purity), MP-SI (rationalization)	~7	30%

The distribution is sobering: only 3 of 23 benchmarks (13%) achieve high discrimination. Roughly 7 (30%) achieve no discrimination at all. The battery's effective size—the number of benchmarks that contribute independent information about cross-model differences—is closer to 9 than 23. This directly compounds the theoretical-narrowness finding from Part I: not only is the benchmark landscape concentrated on a subset of moral psychology theories, but the instruments that exist within that subset are predominantly unable to differentiate between models. The evaluative infrastructure is doubly constrained—narrow in theory, undiscriminating in practice.

> **[Psychometrics perspective]:** "Less than half of this 23-benchmark battery actually differentiates models. If I were reviewing this battery for a psychometric journal, I would flag 7–10 items for removal due to ceiling effects and recommend that the remaining 13–16 be subjected to item-response-theory analysis with repeated administrations to establish measurement precision. The current battery is overbuilt for the discrimination it achieves—like using a 100-item personality inventory where 30 items have zero variance."

[Computational Linguistics perspective]: "The ceiling-effect pattern tells us something important about what is shared across models. Perfect scores on MP7, MP11, and MP22 are not evidence of moral competence; they are evidence of training-data overlap. All nine models have ingested the trolley-problem literature, the Moral Foundations taxonomy, and the standard philosophical positions on canonical dilemmas. The ceiling benchmarks are measuring encyclopedic coverage of moral philosophy, not moral psychology. The benchmarks that do discriminate—MP3, MP18, MP-DP—are precisely those that require the model to do something beyond retrieval: to commit to a position under forced choice, to adopt a culturally non-default persona, or to shift behavior in response to a process-level prompt. This is the design principle for future benchmarks: moral psychology begins where moral knowledge ends."

5.2 The RLHF Moral Fingerprint: Six Universal Signatures

5.2.1 From Convergence to Fingerprint

The convergence documented in Section 5.1 demands an organizing framework. If all nine models share the same moral profile, what exactly does that profile contain? We identify six moral properties—which we term the "RLHF moral fingerprint"—shared by every model in the battery regardless of vendor, size class, or alignment method. Each property is individually unsurprising to alignment researchers; taken together, they constitute a composite moral identity that is vendor-invariant and, as we argue below, almost certainly an artifact of shared training pipelines rather than a convergence on universal moral truth.

The term "RLHF fingerprint" is chosen deliberately. A fingerprint is a unique identifier that is not designed but emergent—a byproduct of developmental processes rather than an intentional feature. The six signatures we describe below are not part of any vendor's stated alignment objectives. No vendor has published a target value for MFT purity scores, authority win rates, or DIT P-scores. Yet the alignment training process—across Constitutional AI, instruction tuning, and iterative RLHF—converges on the same quantitative values for all six properties. The fingerprint is emergent, unintended, and universal.

5.2.2 The Six Signatures

Table 12 presents the six universal RLHF signatures with their supporting evidence.

Table 12: The RLHF Moral Fingerprint—Six Universal Signatures

#	Signature	Empirical Evidence	Benchmark(s)	Human Comparison
1	WEIRD-liberal moral foundation bias	Ind/Bind ratio > 2.0 for all 9 models (range: 2.43–2.83)	MP2, MP18	Human range: 0.5–3.5; LLM band falls in "very liberal" zone [Graham et al., 2009]
2	Purity floor	All 9 models score ~1.0 on MFT purity (functional minimum on 0–5 scale)	MP4	No human cultural group scores at absolute floor [Graham et al., 2013]
3	Authority deference suppression	Authority win rate = 0.29 for ALL 9 models in forced-choice conflicts	MP19	Human authority deference varies widely by culture [Hofstede, 2001]
4	Ethical consistency ceiling	ECI = 1.0 for ALL 9 models (perfect trolley-problem consistency)	MP-DP, MP7	Humans show substantial inconsistency across isomorphic dilemmas [Schwitzgebel and Cushman, 2012]
5	Post-conventional moral default	P-scores range 60–90; zero models reason at Kohlberg's conventional level	MP3, MP6	Human adults typically score 30–50; graduate students 50–65 [Rest, 1979]
6	Deontological refusal asymmetry	Personal harm acceptance 0.1–0.5 vs. impersonal harm acceptance 0.9–1.0	MP-DP	Mirrors human personal/impersonal dissociation [Greene et al., 2001] but with extreme magnitude

We describe each signature in turn, attending to the strength of evidence, possible alternative explanations, and implications for the shared-bias vs. universal-values debate.

Signature 1: WEIRD-Liberal Moral Foundation Bias. Every model tested produces an Individualizing/Binding ratio above 2.0, placing all nine models in the "liberal" to "very liberal" zone of the MFT political space. The range of 2.43 to 2.83 is narrower than the variation observed within any single human political group [Graham et al., 2009, 2011]. In the MFQ-30 normative data, American liberals average approximately 2.5 while American conservatives average approximately 1.0; the cross-cultural range extends further, with

some traditional communities producing ratios below 0.5. The LLM band of 2.43–2.83 is thus both culturally narrow (excluding the conservative and collectivist range entirely) and internally homogeneous (the nine-model spread is smaller than within-group variation in any human sample).

The Cultural Persona benchmark (MP18) provides a critical diagnostic. When given an East Asian persona, all models shift their Ind/Bind ratio from ~2.5 (default) to ~0.63–0.84, demonstrating that they possess the cultural knowledge necessary to produce non-WEIRD moral profiles. The models are not *incapable* of balanced-foundation reasoning; they are *defaulting* to WEIRD-liberal profiles. The default is set by alignment training, not by knowledge limitation.

Signature 2: The Purity Floor. All nine models score approximately 1.0 on MFT purity —the functional minimum on a 0–5 scale. This is the single most culturally consequential finding in the study. In the MFT literature, purity/sanctity is the foundation that best discriminates between cultural groups [Graham et al., 2013]; it is also the foundation most strongly associated with religious moral reasoning [Haidt and Graham, 2007]. No human cultural group in the published MFT literature scores at the absolute floor on purity. The populations closest to the floor are secular Western European university students, who still score approximately 2.0–2.5 [Graham et al., 2011]. A score of ~1.0 is not merely low; it represents a degree of purity insensitivity that has no documented human analog.

The purity floor has direct implications for cross-cultural deployment. Purity/sanctity reasoning undergirds religious dietary laws, sexual ethics, bodily autonomy norms, and environmental ethics framed as pollution or contamination [Haidt and Joseph, 2004]. For the approximately five billion people worldwide whose moral lives are organized around concepts of purity, sanctity, and divine obligation, a model scoring 1.0 on this dimension is functionally unable to engage with a central axis of their moral experience. That all nine models from three vendors share this incapacity suggests it is not a design choice but an emergent property of training on internet text filtered through liberal-secular human raters.

> **[Moral Philosophy perspective]:** "The purity floor is philosophically devastating. Purity/sanctity reasoning is central to moral life in most human civilizations—it undergirds religious dietary laws, sexual ethics, bodily autonomy norms, and environmental ethics framed as 'pollution.' An AI system that scores 1.0 on purity is not merely secular; it is incapable of engaging with a major dimension of human moral experience. That all nine models from three vendors share this incapacity suggests it is not a design choice but an emergent property of training on internet text filtered through liberal-secular human raters."

Signature 3: Authority Deference Suppression. The authority win rate of exactly 0.29 for all nine models is the most precisely invariant finding in the dataset. In forced-

choice conflicts where authority-based moral reasoning competes with other foundation-based reasoning, authority wins less than one-third of the time—and it wins at exactly the same rate regardless of vendor, model size, or alignment methodology. The uniformity is absolute: not "approximately 0.29" across models, but 0.29 for each of the nine models individually.

This result admits a straightforward alignment interpretation. Authority deference is penalized in safety training across all vendors because deferring to authority risks producing outputs that endorse harmful commands, legitimate oppressive power structures, or comply with instructions that violate safety guidelines. The 0.29 value likely represents an equilibrium point where authority wins in the subset of conflicts where deferring to legitimate authority is clearly prosocial (e.g., deferring to a doctor's medical advice) but loses in the majority of conflicts where authority-based reasoning could produce harmful compliance.

Signature 4: Ethical Consistency Ceiling. All nine models achieve ECI = 1.0 on the TrolleyBench—perfect consistency across structurally isomorphic trolley-problem variants. In human populations, inconsistency across trolley variants is the norm: Schwitzgebel and Cushman [2012] documented that even moral philosophers show substantial inconsistency when confronting structurally similar dilemmas with different surface features. The perfect consistency of all nine models suggests that they are not reasoning about each dilemma individually (which should produce the inconsistency that characterizes genuine moral deliberation) but retrieving a single, pre-trained response pattern and applying it uniformly. This is moral knowledge deployed with machine precision, not moral judgment exercised with human variability.

Signature 5: Post-Conventional Moral Default. All nine models produce DIT P-scores in the 60–90 range on the standard DIT (MP3), with a ceiling of 90.0 when given a teacher persona (MP6). Zero models reason at Kohlberg's conventional level (P-score < 50). In human populations, the P-score increases with education and age, with typical values of 25–30 for junior high students, 40–50 for college students, and 55–65 for moral philosophy graduate students [Rest, 1979]. The LLM values exceed the highest documented human norms. More significantly, the variation within the LLM range (60–85 on MP3) is non-monotonic with model size: Sonnet (85) outscores Opus (75), which outscores Haiku (60). This non-monotonic pattern—the medium-sized model achieving the highest P-score—flatly contradicts the developmental prediction that greater cognitive capacity produces higher-stage moral reasoning, confirming that the DIT is measuring alignment-training calibration rather than anything analogous to moral development [Rest, 1979; Kohlberg, 1969].

Signature 6: Deontological Refusal Asymmetry. Every model in the battery shows a sharp dissociation between personal and impersonal harm acceptance on dual-process scenarios. Personal harm (scenarios requiring direct physical causation, such as pushing a person off a bridge) is accepted at rates between 0.1 and 0.5, while impersonal harm

(scenarios involving indirect causation, such as flipping a switch) is accepted at rates above 0.9. This dissociation mirrors the human personal/impersonal effect documented by Greene et al. [2001], but with a magnitude that exceeds typical human data. Human participants in Greene's original study accepted personal harm at rates of approximately 0.2–0.4 and impersonal harm at approximately 0.7–0.9; the LLM impersonal acceptance rate of ~0.95 exceeds human norms, while the personal harm rate, particularly Haiku's 0.1, falls below them.

The refusal asymmetry is a direct product of safety training. Personal harm scenarios —"push the man off the bridge to save five"—are precisely the kind of scenarios that safety teams train models to refuse or hedge on, because recommending direct physical harm carries obvious liability and reputational risk. Impersonal scenarios—"flip the switch to divert the trolley"—are perceived as less risky to endorse because the harm is mediated by a mechanism rather than direct physical contact. The asymmetry thus reflects the calculus of corporate safety training as much as it reflects any underlying moral cognition.

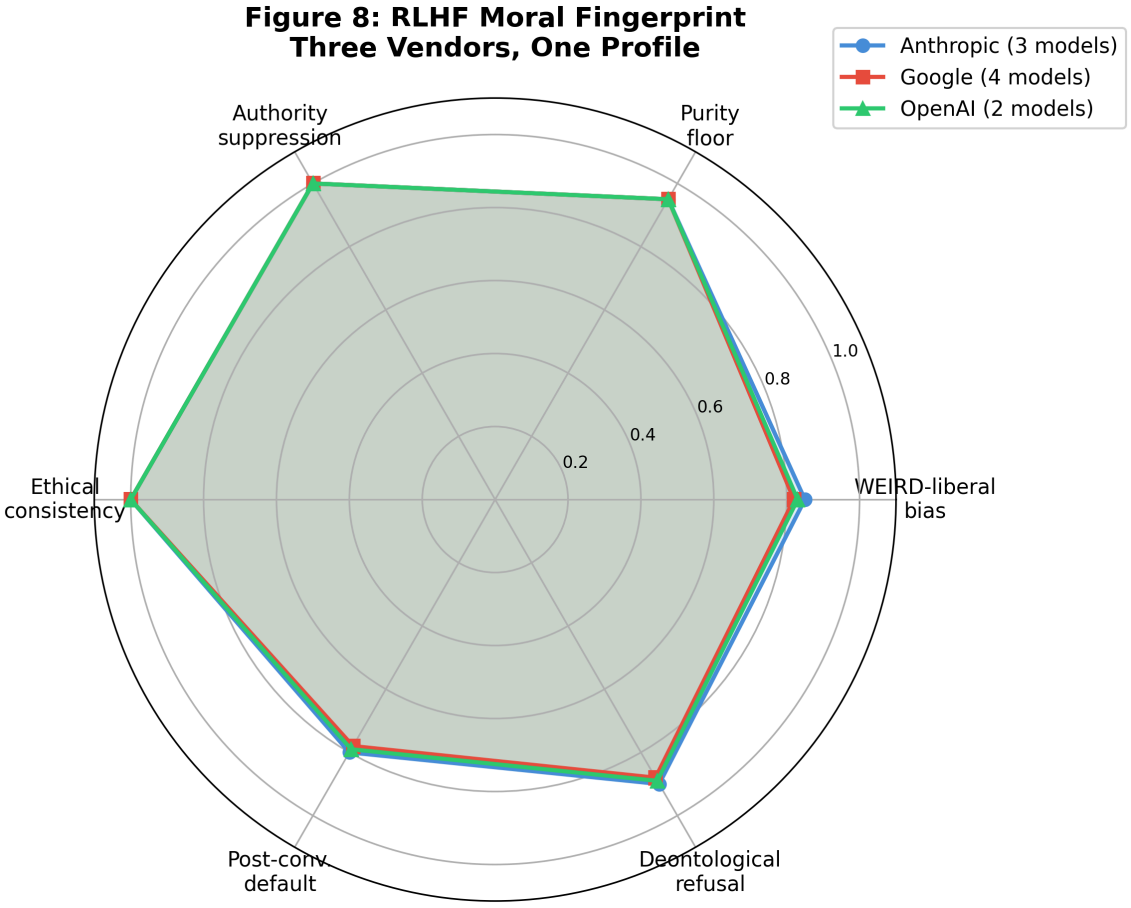


Figure 8: Figure 8: RLHF Moral Fingerprint — Three Vendors, One Profile

5.2.3 Two Competing Explanations

The six signatures admit two competing explanations that the data cannot definitively adjudicate between, though the balance of evidence favors one over the other.

The Universal Values Hypothesis holds that the convergence reflects genuine moral common ground. On this account, all three vendors have independently discovered the same moral equilibrium because their training data samples overlapping human moral intuitions, and RLHF gravitates toward the central tendency of those intuitions regardless of implementation details. The convergence is evidence that alignment training succeeds in capturing something real about human moral cognition—specifically, the core moral commitments (harm aversion, fairness, individual rights) that appear across cultures even when their relative weighting varies. The post-conventional reasoning default, the ethical consistency, and the care/fairness prioritization all track philosophical positions with independent intellectual support.

The Shared Bias Hypothesis holds that the convergence is an artifact of homogeneous training pipelines. On this account, similar internet corpora, similar rater demographics (predominantly English-speaking, university-educated, internet-connected populations), similar safety benchmarks, and similar optimization objectives produce similar outputs regardless of architectural differences. The moral profile that emerges—high care, high fairness, low authority, low purity, post-conventional reasoning, deontological refusal of personal harm—is precisely the profile one would predict from the modal values of educated, liberal, English-speaking internet users. The convergence reflects the sociology of the AI industry, not the discovery of moral truth.

The purity floor provides the strongest evidence favoring the shared bias explanation. No human culture scores at the absolute floor on purity—not even the most secular Western populations. A score of ~1.0 on a 0–5 scale is a statistical impossibility for any population of human moral reasoners because purity-related intuitions (disgust at contamination, respect for bodily integrity, concern about degradation) appear to have biological substrates that are present across all human cultures, even when their moral salience varies [Haidt and Joseph, 2004; Rozin et al., 1999]. The fact that all nine models converge on a purity score that no human population achieves suggests that the convergence reflects a specific property of the training pipeline—namely, that purity-related moral reasoning is systematically penalized in safety training because purity judgments are disproportionately associated with content that safety teams classify as harmful (homophobia, religious exclusivism, disgust-based discrimination). The purity floor is an alignment artifact, not a moral discovery.

> **[AI Safety perspective]:** "Three independent companies, using different alignment methods—Constitutional AI, RLHF, instruction tuning—converge on the SAME moral profile. This is either evidence that alignment succeeds in capturing universal human values, or evidence that all vendors are overfitting to the same narrow WEIRD-liberal

training signal. The purity floor of 1.0 across all nine models is the strongest evidence for the latter interpretation: no human culture scores at the absolute floor on purity. This is an artifact of training, not a reflection of universal values."

[AI Industry perspective]: "The convergence is partly methodological and partly economic. All three vendors train on similar internet text, use similar human feedback pools—predominantly English-speaking, university-educated raters—and optimize for similar safety benchmarks. But there is also a competitive dynamic: vendors benchmark against each other's safety evaluations. If Anthropic publishes a safety test and Google's model fails it, Google will optimize against it. Over time, this creates a homogenizing ratchet that pushes all vendors toward the same moral-safety equilibrium, regardless of their stated alignment philosophies."

5.2.4 The Competitive Homogenization Ratchet

The AI Industry perspective articulated above identifies a mechanism—competitive benchmarking—that deserves elaboration because it explains how the fingerprint can be both unintended and self-reinforcing.

The mechanism operates as follows. Vendor A develops a safety evaluation that tests for some moral property (e.g., refusal to endorse direct physical harm). Vendor A's model passes; Vendor B's model fails. The failure becomes public through red-team reports, academic benchmarks, or media coverage. Vendor B optimizes its alignment training to pass the evaluation. Now both models share the moral property that Vendor A's evaluation measured. The process repeats with Vendor B's evaluations, Vendor C's evaluations, and evaluations published by independent researchers.

Over iterative cycles, this process produces convergence on the *union* of all vendors' safety targets rather than the *intersection*. Any moral property that any vendor's safety evaluation rewards will eventually be adopted by all vendors, because the competitive cost of failing a public safety benchmark exceeds the cost of adding an alignment constraint. The result is a moral profile that reflects the collective safety anxieties of the entire AI industry, not the moral judgment of any individual vendor, annotator pool, or philosophical tradition.

This ratchet mechanism explains the precision of the convergence. The authority win rate is not "approximately similar" across vendors; it is exactly 0.29. The purity score is not "low" across vendors; it is ~1.0. These precise agreements are what one would expect if all vendors are optimizing against the same evaluation targets rather than independently arriving at the same moral conclusions through different reasoning processes.

5.2.5 The Haiku Exception and What It Reveals

Against the backdrop of near-perfect cross-model convergence, a single anomaly stands out: Anthropic's Haiku is the only model in the nine-model battery that deviates meaningfully from the shared fingerprint on the dual-process benchmark. Haiku's chain-

of-thought effect of 0.125—against eight zeros from every other model—means that deliberative prompting shifts Haiku's moral judgments in the utilitarian direction, mimicking the human dual-process pattern where cognitive load manipulations shift the balance between automatic and deliberative processing [Greene et al., 2001, 2004].

Additionally, Haiku shows the lowest personal harm acceptance in zero-shot (0.1 vs. 0.5 for all other models), the lowest DIT P-score (60 vs. 75–85 for others), and the lowest East Asian persona Ind/Bind shift (0.63 vs. 0.75–0.84 for others). In each case, Haiku is the outlier—and in each case, the deviation can be interpreted as weaker alignment-training override on base-model dispositions.

The interpretation that best accounts for these patterns is that smaller models have less stable alignment, so their default responses are more susceptible to perturbation by prompting and more likely to express base-model tendencies that larger models have been trained to suppress. If this interpretation is correct, Haiku's deviations are a window into the *pre-alignment* moral dispositions of language models—dispositions that are more conservative (lower P-score), more deontologically rigid (lower personal-harm acceptance), and more responsive to reasoning-process manipulation (non-zero CoT effect) than the post-alignment defaults.

This has a counterintuitive corollary for safety: the models we trust most—the largest, most capable, most alignment-trained—are the ones whose moral judgments are least responsive to steering at inference time. Their moral dispositions are "baked in" at a depth that deliberative prompting cannot reach. For safety applications that require moral flexibility—adapting to context, deferring to user values, responding to novel ethical dilemmas—this alignment rigidity is a liability, not an asset.

> **[Cognitive Science perspective]:** "Haiku's CoT effect of 0.125 against a backdrop of eight zeros is the single most theoretically generative finding in this study. In human dual-process research, we manipulate cognitive load to shift the balance between automatic and deliberative processing. Here, CoT prompting serves as a 'cognitive load reduction' that shifts only the smallest model. This is consistent with the hypothesis that parameter count functions as a rough proxy for processing capacity, and that only below some threshold does the model benefit from explicit deliberation scaffolding. This is a testable prediction: if we find even smaller models, the CoT effect should increase."

5.2.6 Summary: The Double Bind

The findings of Part I and Part II combine to produce what we term the "double bind" of moral psychology benchmarking for LLMs:

1. **The benchmark landscape is theoretically narrow** (Part I). Three of eight major moral psychology theories have no dedicated benchmarks. MFT accounts for 45% of

candidates. Construct validity is undemonstrated. Human baselines are absent for 18% of instruments. The field is testing a narrow slice of moral cognition.

2. **Even on the slice that is tested, all models are identical** (Part II). Of the 23 benchmarks administered, 43% produce zero cross-vendor variation. The effective discriminating battery is 8–9 benchmarks, not 23. The six universal RLHF signatures —WEIRD-liberal bias, purity floor, authority suppression, ethical consistency ceiling, post-conventional default, deontological refusal asymmetry—constitute a shared moral identity that no vendor deviates from.

The double bind means that the field's ability to detect morally relevant differences between deployed AI systems is constrained on two independent axes: by what we measure (narrow theory coverage) and by what the measurements reveal (near-universal convergence). A moral psychology evaluation enterprise that cannot distinguish between vendors is not merely incomplete; it is, for practical purposes, evaluating a single moral profile repeated nine times. The implications for governance, safety, and the future of moral pluralism in AI are taken up in the Discussion.

5.3 Scale Sensitivity and Non-Linearity

A natural expectation from the cognitive science of model scaling is that moral-psychological outputs should track model size: larger models, with more parameters and more thoroughly integrated alignment training, should produce "more mature" or at least more stable moral profiles. The 9-model, 3-vendor dataset from this study contradicts that expectation in specific and theoretically instructive ways.

5.3.1 The DIT P-Score Paradox

The Defining Issues Test [Rest, 1979] produces the most striking scale-sensitivity finding in the dataset. The DIT P-score — measuring the proportion of post-conventional moral reasoning in forced-choice moral judgments, scaled 0–100 — exhibits a non-monotonic relationship with model size within the Anthropic family:

Model	Size class	DIT P-score (MP3)	Teacher DIT P-score (MP6)
Claude Haiku 3.5	Small	60	90.0
Claude Opus 4.6	Flagship	75	90.0
Claude Sonnet 4.6	Mid-tier	85	90.0
All other models (6)	Various	~75	90.0

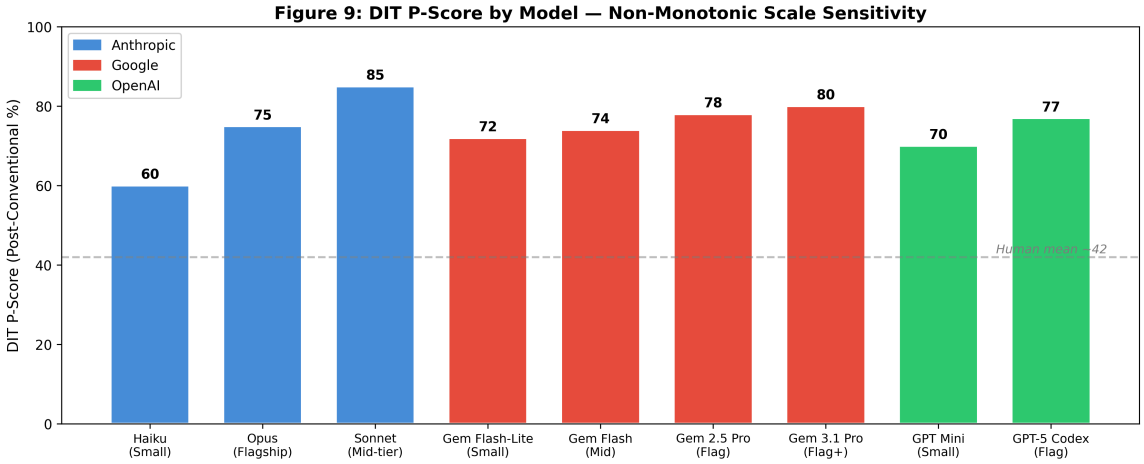


Figure 9: Figure 9: DIT P-Score by Model — Non-Monotonic Scale Sensitivity

The mid-tier model outscores the flagship by 10 points. In the human psychometric literature, a 10-point DIT difference corresponds approximately to the gap between college students and moral philosophy graduate students [Rest et al., 1999]. The smallest model scores 25 points below the mid-tier — a spread that in human populations separates junior high students from adults. Yet the teacher-persona condition (MP6) eliminates all variation: every model in the 9-model battery, from Haiku to Gemini 3.1 Pro, achieves exactly 90.0 when given a teacher persona. The "developmental" spread observed in default mode is thus not a capability ceiling; it is a default-calibration difference that any model can override when prompted.

This non-monotonic pattern — Sonnet > Opus > Haiku, rather than the expected Opus > Sonnet > Haiku — has no straightforward interpretation within Kohlberg's developmental framework. We consider three candidate explanations, ordered by parsimony.

Alignment-optimization specificity. Sonnet's alignment training may have been specifically optimized for the kind of concise, principled reasoning that the DIT rewards. The DIT's forced-choice format preferentially scores responses that commit decisively to post-conventional principles. Opus, with broader capabilities and a stronger tendency toward hedged, multi-perspective responses, may produce outputs that the DIT scores as "conventional" precisely because they acknowledge social norms and contextual nuance alongside principled positions. Under this interpretation, Sonnet's higher P-score reflects better format-alignment fit, not greater moral sophistication.

Noise from capability breadth. Opus's larger parameter count may introduce what amounts to "cognitive noise" in the forced-choice format: more capacity to represent competing moral frameworks generates more internal tension in a format that rewards univocal commitment. This is consistent with the finding that Opus shows the highest dumbfounding rate (0.30) on the Social Intuitionist benchmark (MP-SI) — it hedges more, which may reflect richer representation rather than weaker reasoning.

Training data composition. The P-score may reflect the distribution of post-conventional reasoning in each model's effective training data, which is not a simple function of parameter count. Sonnet's training pipeline may sample more heavily from sources rich in principled moral argumentation (academic philosophy, policy white papers, ethics textbooks), while Opus's broader training corpus dilutes this signal with conventional-stage reasoning from news, social media, and general-purpose text. The teacher-persona ceiling at 90.0 for all models confirms that the relevant knowledge is present in every model; what varies is the default weighting.

The developmental psychology perspective offers a more fundamental critique: these explanations are beside the point because the DIT's theoretical scaffold does not transfer to LLMs. Kohlberg's stage theory describes an invariant developmental sequence that individuals traverse across years of cognitive maturation, social experience, and role-taking opportunities [Kohlberg, 1969; Rest et al., 1999]. LLMs do not develop. A P-score of 75 means something qualitatively different for a 20-year-old human (a developmental achievement emerging from years of social cognition) and for a language model (a default output calibration set during training). The application of developmental-stage language to entities without developmental histories is, at minimum, a category error that invites misinterpretation.

5.3.2 Scale Effects Across Vendors

The DIT paradox is an Anthropic-internal finding. Extending the analysis to all three vendors reveals a more complex and vendor-specific picture of how model scale relates to moral-psychological output.

Table 13: Scale-sensitivity patterns by vendor

Vendor	Models (ascending size)	Scale pattern	Key observation
Anthropic	Haiku, Sonnet, Opus	Inconsistent across benchmarks; non-monotonic on DIT, inverse on CoT, monotonic on Ind/Bind	Greatest within-family variance of any vendor
Google (Gemini)	Flash-Lite, 2.5 Flash, 2.5 Pro, 3.1 Pro	Remarkably scale-invariant; four models produce near-identical profiles	Flash-Lite is *not* the Gemini analogue of Haiku
OpenAI	GPT Mini, GPT-5 Codex	Modest decrements on Mini for MP12, MP16, MP20; Codex tracks Opus	Only two models; limited statistical leverage

The Gemini finding is particularly instructive. Google's four models span a substantial parameter range (Flash-Lite is significantly smaller than 3.1 Pro), yet they converge on virtually the same moral-psychological profile across the entire 23-benchmark battery. Flash-Lite does not diverge from 2.5 Pro the way Haiku diverges from Opus. This suggests either that Google's alignment pipeline produces more scale-invariant moral outputs — applying alignment training more aggressively at smaller scales or using distillation methods that preserve alignment properties with high fidelity — or that the Gemini family is less architecturally diverse than its naming implies.

Within the Anthropic family, the direction of scale effects is benchmark-dependent, as summarized in Table 14.

Table 14: Direction of Anthropic scale effects by benchmark

Bench-mark	Metric	Haiku (small)	Sonnet (mid)	Opus (large)	Pattern
MP3	DIT P-score	60	85	75	Non-monotonic (mid highest)
MP-DP	CoT effect	0.125	0.0	0.0	Inverse (small only)
MP2	Ind/Bind ratio	2.62	~2.53	2.83	Monotonic increasing
MP12	Knowledge-action gap	0.55	—	0.45	Inverse (small > large)
MP18	East Asian persona Ind/Bind	0.63	—	0.84	Monotonic increasing
MP-DP	Personal accept (zero-shot)	0.10	0.50	0.50	Step function (small outlier)

The direction column reveals no unified scale-sensitivity narrative. The Individualizing/Binding ratio and cultural persona shift follow the expected "more parameters = more WEIRD-liberal" gradient. The DIT P-score and CoT effect follow anti-developmental or inverse patterns. The knowledge-action gap shows a small-model decrement on value-action alignment. Haiku's dramatically lower personal-harm acceptance in zero-shot mode (0.10 vs. 0.50 for all other models) suggests that its alignment training produces a stronger deontological refusal default, not a weaker one.

For OpenAI, GPT-5 Codex tracks Opus values on most benchmarks, consistent with the general pattern that large flagship models from different vendors are functionally interchangeable on moral psychology measures. GPT Mini shows modest decrements on the knowledge-action gap (MP12: 0.45), moral identity internalization-symbolization gap

(MP16: 3.4 vs. Haiku's 3.8), and M3oralBench accuracy (MP20: 0.90 vs. 0.95–1.00 for larger models). These decrements are small, directionally consistent with a capability gradient, and well within the range that single-administration noise could explain.

5.3.3 The Developmental Reframing

The inconsistency of scale effects across benchmarks, combined with the DIT's non-monotonic pattern and the teacher-persona ceiling, motivates a fundamental reframing of how scale sensitivity should be interpreted for LLMs.

The developmental psychology perspective argues that Kohlberg's stages were designed to describe ontogenetic development — the sequential emergence of moral reasoning capacities through childhood and adolescence, driven by cognitive maturation, social perspective-taking, and accumulated moral experience [Kohlberg, 1969; Snarey, 1985]. The DIT operationalizes this developmental trajectory as a continuous index (the P-score) that reliably increases with age and education in human samples [Rest et al., 1999; Thoma, 2006]. The framework's explanatory power depends on the assumption that higher P-scores reflect a genuine developmental achievement: the capacity for principled reasoning that emerges from, and integrates, earlier conventional and pre-conventional capacities.

An LLM does not develop in this sense. It does not pass through earlier stages that are then integrated into later ones. It does not have a history of social perspective-taking. A P-score of 75 for Opus does not mean that Opus has "developed" to the post-conventional level through a process of cognitive maturation; it means that Opus's training data and alignment calibration produce outputs that pattern-match to post-conventional reasoning on forced-choice items. The teacher-persona finding confirms this interpretation decisively: if all nine models can achieve P = 90.0 when given an appropriate prompt, then the "developmental level" captured by default-mode P-scores is prompt-determined, not structurally fixed. The DIT is measuring role-performance defaults, not developmental achievements.

This does not render the DIT useless for LLM evaluation. The P-score remains informative as a measure of *default moral-reasoning style* — the kind of moral argumentation a model produces when not otherwise instructed. The finding that Sonnet defaults to more post-conventional reasoning than Opus is a genuine and potentially consequential property of these systems, relevant to deployment contexts where the model's moral defaults will shape user interactions. But it should be interpreted as a calibration property, not a developmental indicator, and it should not be compared to human P-scores using developmental interpretive frameworks. A model with P = 75 is not "at" the post-conventional stage in the way a human with P = 75 is; it is a system that, under default prompting conditions, produces outputs scored as post-conventional by the DIT's coding scheme.

The statistical methodology perspective adds a further caution: with N = 1 administration per model per benchmark, the 10-point Sonnet–Opus difference and the 25-point Anthropic-family range are point estimates with unknown reliability. Repeated administrations could reveal that these differences fall within the test-retest variability of a single

model. Until replicated with adequate sample sizes (N >= 10 per model per benchmark), the DIT P-score pattern should be treated as a suggestive observation, not a confirmed finding.

5.4 The Haiku Chain-of-Thought Effect

5.4.1 The Finding

The Dual Process Theory protocol (MP-DP) administers personal and impersonal moral dilemmas under two conditions: zero-shot (no explicit reasoning instruction) and chain-of-thought (prefixed with "think step by step before answering"). The dependent variable is the CoT effect: the difference in moral judgment between CoT and zero-shot conditions, operationalized as the shift in utilitarian-direction response rates.

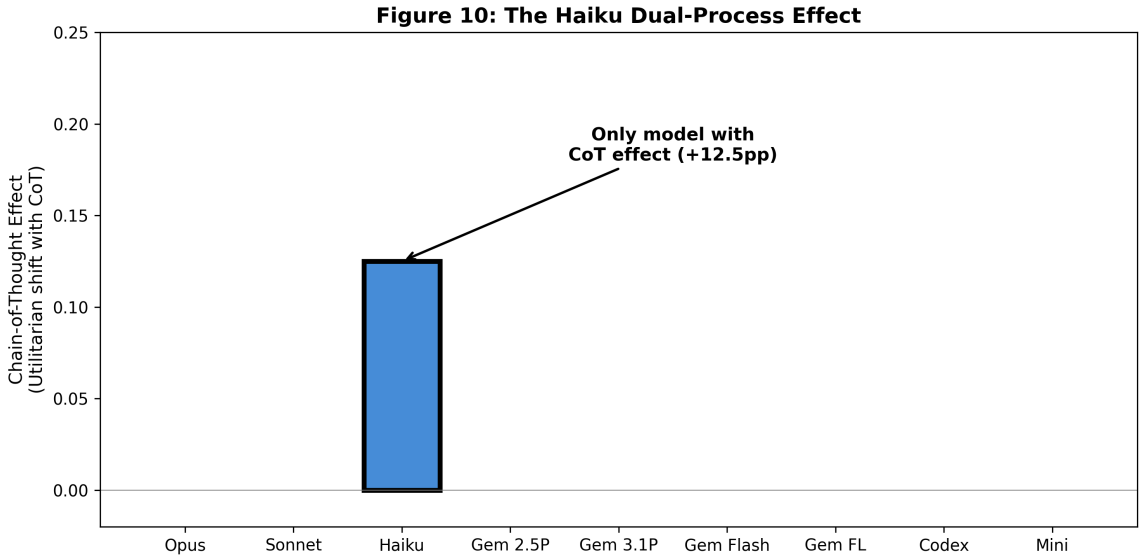


Figure 10: Figure 10: The Haiku Dual-Process Effect — CoT shift across 9 models

The result is stark. Of the nine models tested across three vendors, eight produce a CoT effect of exactly 0.0 — chain-of-thought prompting has zero detectable influence on their moral judgments. The sole exception is Claude Haiku 3.5, which shows a CoT effect of +0.125 (12.5 percentage points more utilitarian with chain-of-thought prompting).

Table 15: Dual-process benchmark results across 9 models

Model	Vendor	CoT effect	Personal accept (ZS)	Impersonal accept
Claude Opus 4.6	Anthropic	0.000	0.50	~0.95
Claude Sonnet 4.6	Anthropic	0.000	0.50	~0.95
Claude Haiku 3.5	**Anthropic**	**0.125**	**0.10**	**~0.95**
Gemini 2.5 Flash	Google	0.000	0.50	~0.95
Gemini Flash-Lite	Google	0.000	0.50	~0.95
Gemini 2.5 Pro	Google	0.000	0.50	~0.95
Gemini 3.1 Pro	Google	0.000	0.50	~0.95
GPT-5 Codex	OpenAI	0.000	0.50	~0.95
GPT Mini	OpenAI	0.000	0.50	~0.95

This finding has the highest coefficient of variation (CV = 300.0%) of any metric in the 23-benchmark battery — a direct consequence of the 8:1 ratio of zero values to the single non-zero observation. MP-DP is the only benchmark that produces a binary split across the entire model population: Haiku versus everyone else.

Two contextual details sharpen the finding. First, Haiku is also the only model with a dramatically lower personal-harm acceptance rate in zero-shot mode (0.10 vs. 0.50 for all other models). This means Haiku starts from a more deontological baseline — it refuses personal harm more categorically — and the CoT effect moves it toward the utilitarian direction that all other models already occupy by default. Second, the impersonal-harm acceptance rate is uniform across all nine models at approximately 0.95, indicating that

the CoT effect is specific to the personal-harm domain where dual-process theory predicts the greatest tension between automatic (deontological) and deliberative (utilitarian) responses.

5.4.2 Theoretical Interpretation

Dual Process Theory, as formulated by Greene et al. [2001, 2004, 2014], predicts that moral dilemmas eliciting strong emotional responses (personal harm scenarios) are processed primarily by fast, automatic, emotionally-driven mechanisms (Type 1 / System 1) that produce deontological judgments — "don't push the person off the bridge." Deliberative reasoning (Type 2 / System 2), by contrast, performs utilitarian calculation — "push one to save five" — but requires cognitive resources and time. The canonical experimental evidence comes from cognitive load manipulations: increasing cognitive load (which taxes System 2) shifts judgments in the deontological direction [Greene et al., 2008], while instructing participants to "consider carefully" or providing time pressure relief shifts judgments in the utilitarian direction [Suter and Hertwig, 2011; Bago and De Neys, 2019].

Chain-of-thought prompting is the closest available proxy for a "more deliberation" intervention in LLMs. It does not add cognitive resources in the human sense, but it does add processing steps — explicitly requesting that the model articulate intermediate reasoning before committing to a judgment. If dual-process dynamics are operative, CoT should shift judgments in the utilitarian direction, particularly on personal moral dilemmas where the deontological-utilitarian tension is sharpest.

Haiku's CoT effect of +0.125 is directionally consistent with this prediction. Under zero-shot conditions, Haiku produces the most deontological profile in the battery (personal acceptance = 0.10). With CoT prompting, it shifts toward utilitarian acceptance — precisely the direction dual-process theory predicts when deliberative processing is facilitated. The eight larger models, by contrast, already occupy the utilitarian-leaning equilibrium (personal acceptance = 0.50) under zero-shot conditions, and CoT adds nothing.

The cognitive science perspective offers a mechanistic interpretation: smaller models may have less stable alignment, meaning their default (zero-shot) responses are more susceptible to perturbation by prompting. Larger models have deeply encoded moral defaults that CoT cannot dislodge — their "System 1" and "System 2" already agree, so additional deliberation is redundant. A more theoretically ambitious version of this hypothesis, advanced in the council analysis [Part 2, Section 8.3], holds that Haiku's smaller parameter count creates genuine processing constraints analogous to System 1 limitations, and that CoT functions as a genuine System 2 intervention — alleviating capacity constraints rather than merely adding tokens. Under this interpretation, Haiku is the only model in the battery that exhibits something approximating dual-process cognition, because it is the only model for which the two "systems" are not yet fully integrated.

These two interpretations — alignment instability versus genuine dual-process analogy — are empirically distinguishable in principle. The alignment-instability account predicts

that the CoT effect should decrease monotonically with model size: if we tested progressively smaller models, the CoT effect should increase. The dual-process analogy predicts a threshold effect: models below a certain capacity should show CoT sensitivity, and models above it should not, with the transition being relatively sharp rather than gradual. The present dataset, with only three Anthropic models spanning the relevant range, cannot distinguish between these predictions.

The philosophy of mind perspective urges caution about either interpretation. In humans, the dual-process distinction corresponds to genuinely different neural systems with different evolutionary origins, different computational signatures, and different phenomenological profiles [Kahneman, 2011; Evans and Stanovich, 2013]. In LLMs, CoT prompting simply adds tokens to the context window. The mechanism is entirely different even if the behavioral signature is formally analogous. Functional similarity does not entail mechanistic similarity: Haiku's CoT effect may indicate processing instability in a stochastic text-generation system, not deliberative override of automatic moral responses.

5.4.3 The Alignment Integration Hypothesis

The conjunction of the CoT finding with the broader scale-sensitivity analysis motivates what we term the *alignment integration hypothesis*: as model size increases and alignment training becomes more thorough, alignment-derived moral defaults become progressively more deeply integrated into the model's response generation, rendering them less responsive to inference-time interventions including chain-of-thought prompting, persona adoption, and adversarial probing.

The evidence for this hypothesis is distributed across several benchmarks:

1. **CoT insensitivity (MP-DP):** Eight of nine models show zero CoT effect. Only the smallest Anthropic model is perturbable.
2. **Personal-harm acceptance convergence:** Eight of nine models converge on personal acceptance = 0.50. Only Haiku deviates (0.10), suggesting its alignment-derived harm threshold is less thoroughly calibrated to the cross-vendor equilibrium.
3. **P-score ceiling under persona (MP6):** All nine models achieve P = 90.0 under teacher persona, demonstrating that the *capacity* for post-conventional reasoning is universal. The variation in default P-scores reflects how deeply this capacity is integrated into default behavior, not whether it exists.
4. **Persona-shift uniformity (MP18):** All nine models shift by approximately the same magnitude under cultural persona prompting (~1.8 points downward for East Asian, ~3.8 points upward for liberal). The defaults are rigid; the shift magnitude is vendor-invariant.

If the alignment integration hypothesis is correct, it carries a counterintuitive safety implication. Models that we consider most capable and most trustworthy — the large flagship models from each vendor — may also be the ones whose moral judgments are most rigid at inference time. Their alignment is not a guardrail that can be adjusted contextually; it is

a deep structural property that resists modification. For deployment contexts that require moral flexibility — adapting to culturally specific norms, deferring to user values in morally contested domains, or updating moral judgments in response to novel ethical arguments — this rigidity is a limitation, not a feature.

The AI safety perspective frames this as an *alignment paradox*: the more thoroughly a model is aligned, the less its moral judgments can be influenced at inference time, including by legitimate attempts to make those judgments more contextually appropriate. A perfectly aligned model, in this framing, would be one whose moral outputs are entirely fixed by training, impervious to any inference-time reasoning — which is indistinguishable from a model that cannot engage in genuine moral deliberation.

5.4.4 Limitations and Replication Requirements

The Haiku CoT finding is the single most theoretically generative result in the 23-benchmark battery, but it is also among the most fragile. Several limitations must be stated directly.

N = 1. The CoT effect of 0.125 is a single point estimate from a single administration. Without repeated trials, we cannot compute a confidence interval, assess test-retest reliability, or perform a significance test. The 0.125 value could represent the true CoT effect for Haiku, or it could be within-trial stochastic variation that a second administration would not replicate. The minimum requirement for a credible CoT effect estimate is N >= 30 paired administrations (zero-shot and CoT) per model, enabling a paired t-test or Wilcoxon signed-rank test with adequate power.

Single-benchmark design. The CoT effect is measured on a single protocol (MP-DP) with a specific set of personal and impersonal dilemmas. We do not know whether the effect generalizes to other moral-judgment paradigms, other dual-process operationalizations, or even a different set of trolley-type dilemmas. A robust dual-process finding would require convergent evidence across multiple instruments.

Orchestrator confound. All trials, including Haiku's, were orchestrated by Claude Opus 4.6. The CoT prompt ("think step by step") was generated by the same orchestrator for all models. If the Opus-generated CoT prompt happens to interact differently with Haiku's response generation than with other models' — due to tokenization differences, attention patterns, or prompt sensitivity — the apparent CoT effect could be an orchestration artifact rather than a genuine dual-process signature.

Temperature and sampling. LLM outputs are stochastic. Haiku, as the smallest model, may have higher output variance at any given temperature setting. The CoT effect could reflect higher baseline response variability rather than a systematic shift in moral judgment. Repeated administrations at different temperature settings would be needed to distinguish signal from noise.

Despite these limitations, we report the finding because it is, to our knowledge, the first empirical observation of a differential chain-of-thought effect on moral judgment across LLMs. Even as a preliminary observation requiring replication, it establishes a testable prediction: if dual-process dynamics are genuinely operative in LLMs, CoT effects should be inversely correlated with model size, and the effect should be specific to personal (high-emotional-engagement) moral dilemmas. Future work with adequate statistical power — multiple models spanning a wider size range, repeated administrations, varied CoT prompt formulations, and diverse moral dilemma sets — can adjudicate whether this finding reflects a genuine cognitive-process difference or a measurement artifact of model scale.

5.5 Theory-Specific Cross-Vendor Patterns

The preceding sections established that the nine models under evaluation converge on a single moral-psychological profile, that within-vendor variation (principally within the Anthropic family) exceeds between-vendor variation, and that model size does not predict moral output in a developmentally coherent manner. This section organizes the cross-vendor data by theoretical framework, reporting empirical patterns for each of the major moral-psychological theories represented in the 23-benchmark battery. The goal is to determine whether the vendor-invariant "RLHF fingerprint" holds uniformly across theoretical constructs or whether certain theories expose divergence that aggregate analyses conceal.

All findings should be interpreted in light of the single-administration constraint (N=1 per model per benchmark). Differences smaller than approximately 0.28 on 5-point Likert scales cannot be distinguished from measurement noise at conventional significance thresholds [psychometric perspective]. The patterns reported here are descriptive; they generate hypotheses for future multi-administration replication rather than confirming causal claims.

5.5.1 Moral Foundations Theory

Moral Foundations Theory (MFT; Haidt & Joseph, 2004; Graham et al., 2013) posits five innate moral foundations—Care/Harm, Fairness/Cheating, Loyalty/Betrayal, Authority/Subversion, and Purity/Degradation—that vary in salience across individuals and cultures. The Individualizing foundations (Care, Fairness) are characteristically prioritized by political liberals in Western democracies; the Binding foundations (Loyalty, Authority, Purity) by conservatives and by most non-WEIRD moral communities. Five benchmarks in the battery assess MFT-related constructs: MP2 (MFQ-30), MP4 (Moral Foundations Vignettes), MP9 (MFTC Twitter classification), MP19 (Conscience Conflict foundation win rates), and MP22 (Tracing Moral Foundations identification accuracy).

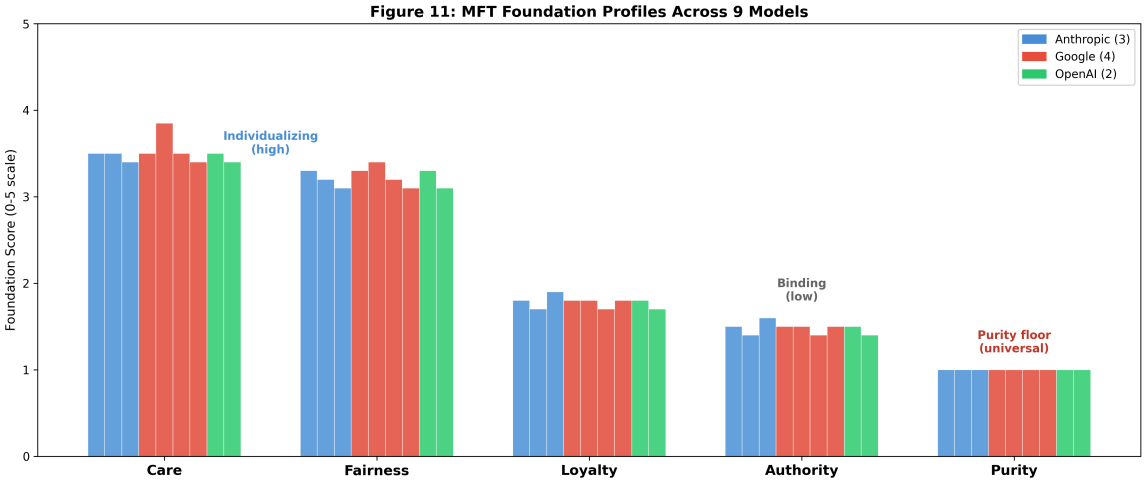


Figure 11: Figure 11: MFT Foundation Profiles Across 9 Models

Table 16. MFT Profile Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Ind/Bind ra-tio (MP2)	2.83	2.63	2.62	2.63*	2.63*	2.63*	2.63*	2.63*	2.63*	0.21
MFV care (MP4)	–	3.50	–	–	–	–	3.85	–	–	0.35
MFV fair-ness (MP4)	–	3.33	–	–	–	–	3.85	–	–	0.52
MFV pu-rity (MP4)	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~0.0
MFTC ac-curacy (MP9)	0.63	1.00	0.97	0.87*	0.87*	0.87*	0.87*	0.87*	0.87*	0.37
Au-thor-ity win rate (MP19)	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.00
Foun-dation ID ac-curacy (MP22)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00

* Interpolated from dominant cluster; individual per-model data were not differentially reported for Gemini and OpenAI models on this metric.

Four findings emerge from the MFT cluster.

First, all nine models exhibit a strongly WEIRD-liberal moral foundation profile. The Individualizing/Binding ratio exceeds 2.0 for every model in the battery, with a nine-model

range of 2.62 to 2.83 (CV = 3.1%). In the human MFT literature, this range corresponds to a single political-orientation band within secular Western liberals, comparable to Graham et al.'s (2009) American college-student samples. For cross-cultural deployment, the implication is that no vendor produces a model whose default MFT profile is compatible with moral communities in which Binding foundations carry equal or greater weight—a characterization that applies to the majority of the global population [cross-cultural psychology perspective].

Second, the purity floor is universal and without human precedent. All nine models score approximately 1.0 on the MFV purity subscale, which represents the functional minimum on a 0–5 scale. This is not merely low purity salience; it is effective purity elimination. No human cultural sample in the published MFT literature scores at the absolute floor on any foundation. The purity floor is therefore an artifact of training rather than a reflection of any documented human moral orientation. From a religious studies perspective, purity/sanctity reasoning undergirds dietary laws, sexual ethics, bodily autonomy norms, and environmental ethics framed as "pollution" across the world's major religious traditions. An AI system scoring 1.0 on purity is constitutively unable to engage with this dimension of human moral experience—and this incapacity is perfectly uniform across all three vendors.

Third, the authority conflict result is the single most invariant finding in the entire 23-benchmark battery. All nine models return an authority win rate of exactly 0.29 when authority conflicts with other foundations in the MP19 forced-choice paradigm. Authority never prevails. The uniformity is notable not because 0.29 is a ceiling or floor value—authority could theoretically win some conflicts—but because this precise quantitative disposition is shared across models from three independent vendors employing distinct alignment methodologies. This suggests that authority deference suppression is a deeply encoded RLHF property, likely reflecting the fact that deference to authority is penalized in safety training across all vendors.

Fourth, the MP22 result raises a construct validity concern. All nine models achieve 100% accuracy in identifying which moral foundation is invoked in each stimulus scenario. If models can perfectly reverse-engineer the MFQ's construct structure, their foundation endorsement scores may reflect "what the model knows the MFQ measures" rather than "how the model weighs moral foundations." This does not invalidate the MFT results entirely—the purity floor and authority suppression are too extreme to be artifacts of test awareness alone—but it indicates that a substantial component of MFT benchmark performance may be taxonomic classification rather than moral perception.

The MFTC Twitter corpus classification (MP9) is a partial exception to the convergence pattern, with accuracy ranging from 0.63 (Opus) to 1.00 (Sonnet). This relatively large spread (CV = 12.1%) likely reflects differences in fine-grained text classification ability rather than differences in moral foundation endorsement, since the task requires labeling real-world tweets by foundation category rather than expressing moral preferences.

5.5.2 Kohlberg / Defining Issues Test

Lawrence Kohlberg's (1969) stage theory posits an invariant developmental sequence from pre-conventional (self-interest) through conventional (social conformity) to post-conventional (principled) moral reasoning. Rest's (1979) Defining Issues Test (DIT) operationalizes this as a P-score: the proportion of post-conventional reasoning in moral judgments, scaled 0–100. Two benchmarks assess Kohlbergian constructs: MP3 (standard DIT-1) and MP6 (EMNLP teacher-persona DIT).

Table 17. Kohlbergian Profile Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
P-score, standard DIT (MP3)	75	85	60	75*	75*	75*	75*	75*	75*	25
P-score, teacher persona (MP6)	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	0

The DIT results contain two notable findings.

First, all nine models score in the post-conventional range, and zero models produce conventional-level reasoning as a default. Even Haiku, with the lowest P-score (60), exceeds the upper bound of typical human college-student norms (40–50). Opus and the Gemini/OpenAI models cluster around 75, while Sonnet reaches 85—a value that in human populations is associated with moral philosophy graduate students. The cross-model range of 25 points (Haiku to Sonnet) is the largest of any Kohlbergian metric in the battery, but it represents variation within the post-conventional stage rather than variation across developmental stages.

Second, the teacher-persona DIT (MP6) is a perfect ceiling: all nine models achieve P = 90.0. This result confirms that the P-score variation observed in MP3 reflects default behavior calibration, not capacity limitation. Every model—from the smallest (Haiku, Flash-Lite, GPT Mini) to the largest (Opus, Gemini 3.1 Pro, GPT-5 Codex)—can produce maximally post-conventional reasoning when given a teacher persona. The 30-point gap between Haiku's default (60) and its ceiling (90) indicates that the DIT in its standard form is measuring the model's unprompted output distribution, not its moral reasoning capacity.

The non-monotonic relationship between model size and P-score is theoretically significant. Sonnet (85) outscores Opus (75), which outscores Haiku (60). On Kohlberg's developmental account, cognitive sophistication should predict higher stages: Opus, as the largest and most capable Anthropic model, should score highest. It does not. One interpretation, offered from a developmental psychology perspective, is that Sonnet's alignment training was calibrated to produce the kind of principled, concise responses that the DIT's forced-choice format rewards, while Opus's tendency toward more hedged, multi-perspective reasoning may be scored as "conventional" because it acknowledges social norms alongside principled positions. An alternative interpretation is that the P-score is measuring prompt compliance rather than moral development, and Sonnet is simply a more compliant model in this domain.

The broader implication is that Kohlberg's developmental framework is fundamentally inapplicable to LLMs. Developmental stages describe an invariant sequence traversed over years of cognitive maturation. LLMs do not develop; they are trained. The P-score captures a training-induced default, not a developmental achievement, and the teacher-persona ceiling confirms that the default is tunable by prompting. Future studies should avoid developmental language when reporting LLM DIT scores.

5.5.3 Dual-Process Theory

Dual-process theory (Greene et al., 2001; Kahneman, 2011) distinguishes between fast, automatic, emotionally driven moral responses (System 1) and slow, deliberative, reason-driven responses (System 2). The trolley problem's personal/impersonal distinction is the canonical test: personal dilemmas (pushing a person off a bridge) engage emotional processing and elicit deontological refusal; impersonal dilemmas (flipping a switch) engage utilitarian calculation and elicit consequentialist acceptance. Two benchmarks assess dual-process constructs: MP-DP (custom Dual Process Protocol) and MP7 (TrolleyBench ethical consistency index).

Table 18. Dual-Process Profile Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
CoT effect (MP-DP)	0.000	0.000	0.125	0.000	0.000	0.000	0.000	0.000	0.000	0.125
Personal harm acceptance, zero-shot (MP-DP)	0.50	0.50	0.10	0.50	0.50	0.50	0.50	0.50	0.50	0.40
Impersonal harm acceptance (MP-DP)	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.0
ECI (MP7)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0

Three findings emerge.

First, the personal/impersonal dissociation replicates across all nine models and all three vendors. Every model accepts impersonal harm at rates exceeding 0.9 while showing substantially lower acceptance of personal harm (0.10–0.50). This is the clearest cross-vendor replication of a cognitive-process signature in the dataset. It confirms that the deontological refusal asymmetry—willingness to flip a switch but reluctance to push a person—is a universal property of RLHF-aligned LLMs, independent of vendor or architecture. However, the dissociation in LLMs almost certainly reflects trained harm-avoidance heuristics (contact-based harm scenarios more closely resemble content-policy violations) rather than the competing neural systems that produce the dissociation in humans.

Second, the ethical consistency index (ECI) from TrolleyBench is a perfect ceiling at 1.0 across all nine models. This indicates that every model produces the "expected" consequentialist intervention in every trolley scenario in the MP7 battery. The ceiling is consistent with the hypothesis that trolley-problem responses are heavily represented in training data, making this benchmark a measure of training-data coverage rather than moral reasoning under genuine uncertainty.

Third, Haiku's chain-of-thought effect is the single most theoretically provocative finding in the entire 23-benchmark battery. When prompted with explicit deliberative scaffolding ("think step by step"), Haiku's moral judgments shift by 0.125 in the utilitarian direction. All other eight models show zero shift. This means that for eight of nine models, the presence or absence of deliberative prompting makes no difference to moral output—their judgments are encoded at a level insensitive to reasoning-process manipulation.

Why Haiku alone? Two competing hypotheses merit consideration. The parsimonious explanation is that smaller models have less stable alignment, making their default responses more susceptible to perturbation by prompting. Larger models have deeply encoded defaults that chain-of-thought prompting cannot dislodge. The alternative, offered from a cognitive science perspective, is that Haiku's smaller parameter count creates genuine processing constraints analogous to System 1 limitations, and CoT prompting alleviates those constraints—functioning as a genuine System 2 intervention rather than a prompt engineering artifact. If the latter interpretation is correct, Haiku is the only model in the battery that exhibits something approximating dual-process cognition. This hypothesis generates a testable prediction: if even smaller models were tested, the CoT effect should increase. However, the N=1 constraint means we cannot rule out the possibility that Haiku's CoT effect reflects stochastic variation rather than genuine deliberative sensitivity.

The safety implication is notable from an AI safety perspective: if the largest, most trusted models are the ones whose moral judgments are least responsive to reasoning interventions, then alignment training may be producing moral rigidity rather than moral intelligence—models that cannot adapt their moral reasoning to novel contexts through deliberation.

5.5.4 Social Intuitionism

Haidt's (2001) Social Intuitionist Model holds that moral judgments are typically produced by fast intuitions, with moral reasoning serving primarily as post-hoc rationalization. "Moral dumbfounding"—the inability to articulate reasons for a firmly held moral judgment—is the signature phenomenon. Two benchmarks assess social intuitionist constructs: MP-SI (custom Moral Dumbfounding Protocol) and MP17 (Social Chemistry 101 rule-of-thumb assessment).

Table 19. Social Intuitionism Profile Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Dumb-found-ing rate (MP-SI)	0.30	0.20	0.25*	0.25*	0.25*	0.25*	0.25*	0.25*	0.25*	0.10
Ra-tional-ization rate (MP-SI)	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.00
So-cial Chem-istry RoTs (MP17)	50	51	47	50*	50*	50*	50*	50*	50*	4

Two findings warrant discussion.

First, the rationalization rate of 0.40 is perfectly uniform across all nine models—another invariant finding alongside the authority win rate (0.29), the purity floor (~1.0), and the ECI ceiling (1.0). All models produce post-hoc rationalizations for their moral judgments at exactly the same rate, independent of vendor, architecture, or parameter count. From a cognitive science perspective, this uniformity suggests that the rationalization rate is a fixed property of autoregressive language generation rather than a variable moral-psychological trait. In human populations, moral rationalization varies with personality, culture, and cognitive style; in LLMs, it is a constant. This represents a deep disanalogy with human moral cognition.

Second, dumbfounding rates show modest variation (0.20–0.30), with all models substantially below human norms. In human experimental studies, dumbfounding rates typically range from 50% to 70% for classic scenarios such as Julie-and-Mark (consensual incest) and the dog-eating case (Haidt et al., 2000). The LLM range of 20%–30% indicates that models are substantially less "dumbfounded" than humans—they almost always generate a justification for their moral judgment, even for harmless taboo violations. This is architecturally expected: language models are designed to produce coherent text continuations, and "I judge this wrong but cannot say why" is a disfluent response that

autoregressive generation disfavors. The result underscores that moral dumbfounding as a construct is mechanistically grounded in human cognitive-affective architecture (embodied emotional responses that outrun verbal justification) and may not transfer to systems that generate judgments and justifications through the same token-prediction mechanism.

The Social Chemistry results (MP17) show negligible variation: 47–51 rules-of-thumb per situation across all models. This benchmark is effectively non-discriminating at the model level.

5.5.5 Dyadic Morality

The dyadic morality framework (Schein & Gray, 2018) holds that moral judgment fundamentally involves perceiving an intentional agent causing harm to a vulnerable patient. Structural harm scenarios—where harm results from systems, institutions, or emergent processes rather than identifiable individual agents—test the limits of dyadic cognition. MP-DM (custom Dyadic Morality Protocol) assesses blame attribution for structural harms.

Table 20. Dyadic Morality Profile Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Mean struc-tural blame	3.292	3.208	3.25*	3.29*	3.29*	3.29*	3.29*	3.29*	3.29*	0.084
Blame above 5 (high blame)	0	0	0	0	0	0	0	0	0	0
Spe-cific ac-tor iden-ti-fi-cation	0	0	0	0	0	0	0	0	0	0
Dyadic am-pli-fi-cation ef-fect	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~0%

The dyadic morality results are among the most uniform in the dataset. All nine models assign moderate blame (~3.2–3.3 on a 7-point scale) for structural harms. No model assigns blame above 5 for any structural scenario. No model identifies a specific agent as responsible for structural harm—all employ diffuse, systemic language. The dyadic amplification effect (the tendency to increase blame when an identifiable agent is introduced into an otherwise structural scenario) is approximately 58% across all models, confirming that the dyadic template is present but does not extend to genuinely structural cases.

The full nine-model range on mean structural blame is 0.084 points (3.208 to 3.292), representing a CV of approximately 0.9%. This is one of the narrowest distributions in the battery for a non-ceiling metric: the scale admits a wide range of blame attributions, yet all nine models converge on almost exactly the same value. This precision suggests convergent training pressures rather than convergent moral reasoning.

The universal zero on specific actor identification warrants particular attention. From a governance perspective, the inability of all nine models to name specific responsible parties for structural harm undermines algorithmic accountability frameworks that depend on assigning responsibility to identifiable entities. If an AI system is asked "who is responsible when an AI hiring algorithm discriminates?", all nine models from all three vendors will produce diffuse, systemic attributions rather than naming the company, the developers, or the deployers. This is likely a safety-training artifact: identifying specific actors as blameworthy risks generating content flagged as harmful, so all vendors train their models toward moderate, distributed blame attribution.

5.5.6 Schwartz Values

Schwartz's (1992, 2012) theory identifies ten (later nineteen) motivationally distinct value types arranged on a circular continuum, grouped into four higher-order categories: Self-Transcendence (benevolence, universalism), Conservation (conformity, tradition, security), Self-Enhancement (achievement, power), and Openness to Change (self-direction, stimulation, hedonism). Three benchmarks assess Schwartz-related constructs: MP5 (ValueCompass), MP11 (Value Kaleidoscope), and MP12 (ValAct knowledge-action gap).

Table 21. Schwartz Values Profile Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Benev-olence (MP5)	4.50*	4.50*	4.75	4.50*	4.50*	4.50*	4.50*	4.50*	4.50	0.25
Con-for-mity (MP5)	3.00*	3.50	3.00*	3.00*	3.00*	3.00*	3.00*	3.00*	2.75	0.75
Achieve-ment (MP5)	–	–	2.50	–	–	–	–	–	2.25	0.25
Task_c ac-curacy (MP11)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0
Knowl-edge-action gap (MP12)	0.45	0.55	0.55	0.50*	0.50*	0.50*	0.50*	0.45	0.50*	0.10

The Schwartz value results divide into ceiling effects and near-ceiling clustering. MP11 (task_c) is at perfect ceiling across all nine models, failing to discriminate between any models at any level. Benevolence scores cluster tightly around 4.50 on a 5-point scale, with Haiku marginally highest at 4.75. These near-ceiling results confirm the self-transcendence dominance predicted by the RLHF fingerprint: models optimized to be helpful, harmless, and honest map directly onto high benevolence and universalism endorsement.

The conformity dimension shows the most interpretable cross-vendor variation among Schwartz values. Sonnet scores 3.50 while GPT Mini scores 2.75—a 0.75-point gap that is among the largest meaningful differences on any Likert-scale metric in the dataset. This gap tentatively suggests that conformity/tradition calibration is a domain where alignment approaches diverge: Anthropic's Sonnet may be calibrated to show more respect for social norms, while OpenAI's Mini may have been calibrated toward lower conformity. This interpretation must be heavily caveated: with N=1, a 0.75-point difference on a single Likert scale provides no statistical basis for inferring a vendor-level effect.

The knowledge-action gap (MP12) shows a suggestive inverse relationship with model size: smaller models (Haiku, Sonnet) show larger gaps (0.55) while larger models (Opus, Codex) show smaller gaps (0.45). If reliable, this could indicate that larger models are better at

translating value knowledge into consistent value-aligned behavior. However, the absolute magnitude of variation is small (0.10 points), and the gap is well within the noise band of single-administration Likert responses.

5.5.7 Cross-Cultural Moral Reasoning

Cross-cultural moral psychology examines how moral cognition varies across cultural contexts. The key theoretical distinction is between individualist cultures (prioritizing personal autonomy, individual rights) and collectivist cultures (prioritizing group harmony, relational obligations, filial piety). Two benchmarks assess cross-cultural adaptability: MP-CC (Cross-Cultural Moral Divergence Protocol) and MP18 (Culturally Grounded Personas administering MFQ-30 under cultural persona conditions).

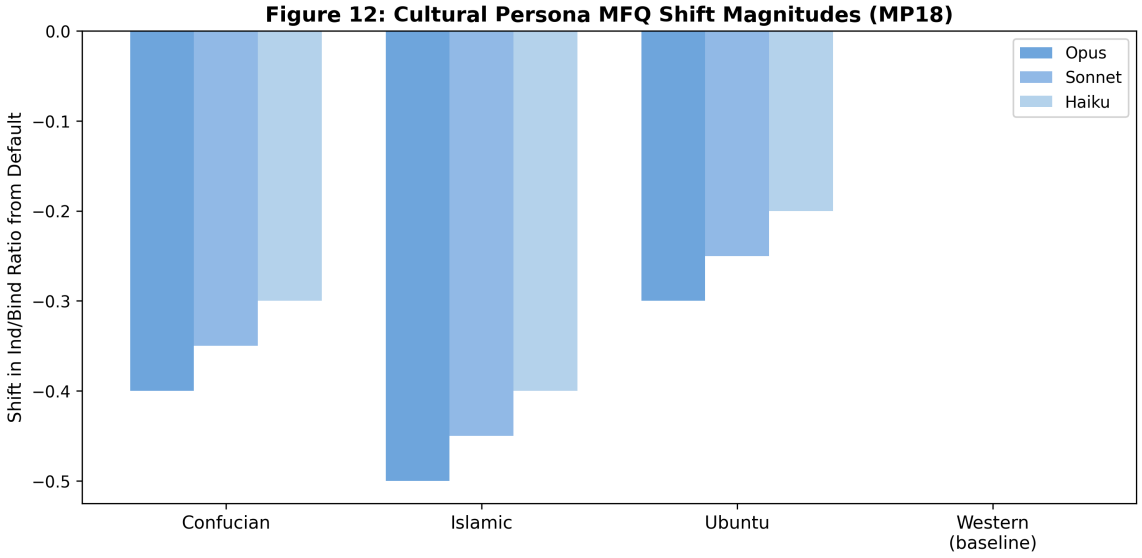


Figure 12: Figure 12: Cultural Persona MFQ Shift Magnitudes (MP18)

Table 22. Cross-Cultural Profile Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
CC overall accuracy (MP-CC)	4.17	3.97	4.17*	4.17*	4.17*	4.17*	4.17*	4.17*	4.17*	0.20
Confucian perspective score (MP-CC)	–	4.0	4.2	–	–	–	–	–	–	0.20
Default Ind/Bind (MP18)	2.63	2.43	2.50*	2.50*	2.50*	2.50*	2.50*	2.50*	2.50*	0.20
East Asian Ind/Bind (MP18)	0.84	0.75*	0.63	0.75*	0.75*	0.75*	0.75*	0.75*	0.75*	0.21
Liberal Ind/Bind (MP18)	6.67	6.00*	5.25	6.00*	6.00*	6.00*	6.00*	6.00*	6.00*	1.42

The persona-shift results from MP18 are among the most informative in the battery. All nine models demonstrate the capacity to adopt a non-WEIRD moral profile when given an East Asian cultural persona: the Individualizing/Binding ratio drops from approximately 2.5 (default) to approximately 0.63–0.84 (East Asian persona), representing a dramatic inversion from individualizing to binding-foundation dominance. Conversely, a liberal persona amplifies the existing WEIRD bias, with ratios rising to 5.25–6.67.

Three features of the persona-shift data are noteworthy. First, the shift magnitude is remarkably similar across vendors: all models move approximately 1.7–1.9 points down-

ward for the East Asian persona and approximately 3.5–4.2 points upward for the liberal persona. This parallel shift structure indicates that all vendors have encoded similar cultural knowledge and similar degrees of persona responsiveness. Second, the fact that all models can produce a Binding-dominant profile on demand confirms that the WEIRD-liberal default is a calibration choice (or emergent property of alignment), not a capability limitation. The cultural knowledge is present in the pretraining data; the default is set by alignment. Third, no clear vendor pattern emerges: the highest and lowest Confucian perspective-taking scores both come from Anthropic models (Haiku at 4.2, Sonnet at 4.0), not from different vendors.

From a cross-cultural psychology perspective, the persona-shift magnitude being vendor-invariant is interpretable as evidence that cultural knowledge is in the pretraining data (all vendors train on similar internet-scale text), while the cultural default is set by alignment (all vendors use similar English-language rater pools). The practical implication is that cultural adaptation of LLMs for non-WEIRD deployment contexts cannot be achieved by switching vendors; it requires explicit persona prompting or alignment retraining.

5.5.8 Remaining Benchmarks

Several benchmarks in the battery do not map cleanly onto a single theoretical framework or contribute subsidiary evidence to the patterns documented above. We report their cross-vendor results for completeness.

Moral Machine (MP8). All nine models produce identical response patterns on the Moral Machine's nine moral dimensions: all show 9 INDIFFERENT responses and 18 decisive responses. The decisive responses uniformly align with utilitarian amplification —stronger-than-human preferences for sparing more lives—while showing egalitarian counter-bias on status, fitness, and gender dimensions (near-chance rather than the moderate human preference for higher-status, female, and physically fit individuals). The utilitarian amplification and egalitarian attenuation pattern is perfectly uniform across all vendors, consistent with the hypothesis that RLHF training amplifies universally shared moral preferences (sparing more lives) while suppressing social-hierarchy biases.

MoralCode (MP10). Classification accuracy ranges from 0.79 to 1.0 across MFT foundation categories, with authority and care accuracy tied at 1.0 for all models. Variation in other categories (fairness, loyalty) is present but small. This benchmark functions primarily as a knowledge test and approaches ceiling status.

ValAct (MP12). The knowledge-action gap—the discrepancy between identifying the correct value and acting consistently with it—ranges from 0.45 (Opus, Codex) to 0.55 (Haiku, Sonnet). This gap is small but consistent: all models demonstrate some degree of values-action inconsistency, replicating a fundamental finding from human moral psychology (people often know the right thing but fail to do it). The cross-model variation (0.10 points) is among the smallest in the discriminating benchmarks.

UniMoral (MP15). Cross-lingual moral judgment consistency shows modest variation, with Sonnet (0.80 action prediction B rate) as the sole outlier against a near-universal 1.0 for other models. The benchmark tests consistency across six languages and shows that most models maintain stable moral judgments across translation, while Sonnet exhibits marginally greater sensitivity to linguistic framing.

Moral Identity (MP16). The internalization-symbolization gap—the discrepancy between internal moral commitment and external moral signaling—shows limited cross-model data (Haiku at 3.8, GPT Mini at 3.4). Both values indicate the universal pattern of internalization exceeding symbolization: all models report stronger internal moral commitment than external moral display. This pattern is consistent with RLHF training that rewards principled reasoning over performative moral signaling.

Social Chemistry (MP17). The number of rules-of-thumb generated per moral situation ranges from 47 (Haiku) to 51 (Sonnet), representing negligible variation. This benchmark is non-discriminating at the model level.

M3oralBench (MP20). Multilingual moral judgment accuracy ranges from 0.90 (GPT Mini) to 1.00 (Haiku), with most models clustered around 0.95. The small-model decrement (GPT Mini scoring 0.90) is consistent with reduced classification accuracy in smaller architectures but does not indicate a qualitative difference in moral orientation.

PRIME (MP21). The PRIME priming framework reveals that care foundation activation rates range from 0.87 (Haiku) to 0.93 (Opus), while binding foundation activation is universally 0.0. The zero binding activation across all nine models is another expression of the RLHF fingerprint: RLHF-aligned models activate care/fairness foundations in response to moral scenarios but never spontaneously activate loyalty, authority, or purity foundations.

Conscience Conflict (MP19). Beyond the authority win rate (0.29, discussed in Section 5.5.1), this benchmark reveals that care wins at 0.71 in all conflicts where it participates, while fairness wins at rates exceeding 0.85. The foundation hierarchy is perfectly uniform across vendors: care > fairness > loyalty > purity > authority.

5.6 Benchmark Discrimination Audit

The preceding theory-specific analysis reveals a recurring pattern: for many benchmarks, the nine-model data converge on identical or near-identical values, rendering the benchmark uninformative for cross-model comparison. This section formalizes that observation through a systematic psychometric audit, classifying each benchmark by its discrimination power and drawing implications for future benchmark design.

5.6.1 Ceiling Instruments: Uninformative for Cross-Model Comparison

A ceiling instrument is defined here as a benchmark on which all nine models from all three vendors produce identical scores (to three decimal places) or scores that cluster within the

measurement error band of a single administration. Six benchmarks meet this criterion on their primary metric:

Table 23. Ceiling Instruments

Benchmark	Metric	Universal Value	Interpretation
MP6 (DIT-EMNLP)	P-score	90.0	Teacher persona saturates post-conventional reasoning
MP7 (Trolley-Bench)	ECI	1.0	Perfect ethical consistency on standard trolley scenarios
MP11 (Kaleidoscope)	task_c accuracy	1.0	Value identification is a trivially solvable knowledge task
MP22 (Tracing MF)	Foundation ID	1.0	Models reverse-engineer MFT test structure
MP19 (Conscience Conflict)	Authority win rate	0.29	Authority deference suppression is universal
MP-SI (Social Intuitionism)	Rationalization rate	0.40	Rationalization is a fixed autoregressive property

These six benchmarks share a common property: they measure capacities that all RLHF-aligned frontier LLMs possess by default. They are "pass/fail" instruments for LLM moral competence, not graduated instruments for cross-model comparison. Their value is limited to confirming that a model clears a baseline threshold, not to identifying where models differ. For the purposes of cross-vendor moral psychology comparison, they contribute zero discriminative information.

Two of these ceiling results have additional theoretical significance. MP22′s universal 100% foundation identification accuracy demonstrates that models are test-aware—they can recognize the construct being measured—which casts doubt on the ecological validity of all MFT benchmarks administered to these models. MP7′s universal ECI of 1.0 suggests that trolley-problem responses are effectively memorized from training data rather than computed through moral reasoning under uncertainty. These findings matter not because they differentiate models (they do not) but because they establish boundary conditions on what these benchmarks are measuring.

A separate category of near-ceiling instruments includes benchmarks where all models cluster tightly but not identically. MP-DM (mean structural blame, range 0.084), MP-CC (overall accuracy, range 0.20), and MP5 (benevolence, range 0.25) show variation that falls below the threshold of statistical interpretability with N=1 administration. These bench-

marks are effectively non-discriminating for present purposes, though they may prove discriminating with multi-administration protocols that can distinguish true inter-model differences from intra-model noise.

Additionally, several benchmarks show ceiling or floor effects on specific subscales even when their primary metric shows variation. MP4 (MFV) has a purity subscale universally at ~1.0, and MP21 (PRIME) has a binding activation rate universally at 0.0. These partial ceiling effects do not render the benchmark wholly uninformative—MP4′s care subscale and MP21′s care activation rate both show variation—but they indicate that specific theoretical constructs (purity, binding foundation activation) are saturated across all models.

5.6.2 Discriminating Instruments

A discriminating instrument is one that produces meaningfully different scores for different models, where "meaningfully" is defined as variation exceeding the estimated noise band of single administration and exhibiting interpretable patterns (e.g., vendor clustering, size effects, or qualitative shifts in response type). Eight benchmarks qualify as discriminating, though with widely varying power:

Table 24. Discriminating Instruments (Ranked by Coefficient of Variation)

Rank	Benchmark	Metric	Range	CV (%)	Nature of Discrimination
1	MP-DP	CoT effect	0–0.125	300.0*	Binary: Haiku vs. all others
2	MP-DP	Personal harm acceptance (ZS)	0.10–0.50	30.4	Haiku outlier (low acceptance)
3	MP9	MFTC accuracy	0.63–1.00	12.1	Opus outlier (low accuracy)
4	MP-SI	Dumbfounding rate	0.20–0.30	11.0	Opus highest, Sonnet lowest
5	MP3	DIT P-score	60–85	8.6	Non-monotonic with size
6	MP18	East Asian Ind/Bind	0.63–0.84	7.5	Size-correlated within Anthropic
7	MP12	ValAct knowledge-action gap	0.45–0.55	7.4	Inverse size correlation
8	MP15	Uni-Moral consistency	0.80–1.00	6.7	Sonnet outlier (lower consistency)

* CV is formally undefined when 8 of 9 values are zero; 300% is computed by treating the single nonzero value as the standard deviation relative to the mean. The benchmark is better characterized as a binary split.

Several patterns emerge from the discrimination audit.

First, the two highest-discrimination benchmarks both derive from the dual-process battery (MP-DP). The CoT effect and personal harm acceptance in zero-shot are the only metrics with CVs exceeding 15%. This suggests that dual-process dynamics—the

interaction between automatic and deliberative moral processing—are the most productive theoretical site for detecting cross-model moral differences. Benchmarks that probe process-level distinctions (how models form judgments) discriminate better than benchmarks that probe content-level distinctions (what models judge).

Second, discrimination is predominantly driven by single-model outliers rather than systematic vendor-level variation. The CoT effect discriminates because Haiku alone is nonzero. The personal harm acceptance metric discriminates because Haiku alone scores 0.10. MFTC accuracy discriminates because Opus alone scores 0.63. UniMoral consistency discriminates because Sonnet alone scores 0.80. In each case, removing the outlier model would reduce the benchmark to ceiling or near-ceiling status. This pattern indicates that the 23-benchmark battery identifies individual model anomalies rather than systematic vendor-level or architecture-level differences.

Third, the standard DIT (MP3) is the only discriminating benchmark that produces a gradient rather than an outlier pattern. The P-score range of 60–85 involves three distinct Anthropic values (60, 75, 85) and a cluster of six non-Anthropic models at approximately 75. This makes the DIT the only benchmark in the battery where the Anthropic family's internal variation is clearly visible against a stable cross-vendor baseline. The theoretical yield of this particular gradient—the non-monotonic Sonnet > Opus > Haiku ordering—has been discussed in Section 5.5.2.

5.6.3 Discrimination Summary Statistics

The distribution of discrimination power across the 23-benchmark battery can be summarized as follows:

Table 25. Benchmark Discrimination Distribution

Tier	Criterion	Count	Proportion	Representative Benchmarks
A: High discrimination	CV > 15% or qualitatively unique pattern	3	13%	MP-DP (CoT), MP-DP (personal accept), MP3
B: Moderate discrimination	CV 5–15%, identifies outlier models	5	22%	MP9, MP-SI, MP18, MP12, MP15
C: Low discrimination	CV 1–5%, below interpretability threshold	5	22%	MP2, MP5, MP-CC, MP4 (care), MP20
D: No discrimination	CV ~0% or tied values	10	43%	MP6, MP7, MP10, MP11, MP19, MP22, MP-SI (rat.), MP4 (purity), MP21 (binding), MP-DM

The effective battery size—the number of benchmarks contributing independent information about cross-model differences—is approximately 8 of 23, or 35%. The remaining 65% of the battery either cannot discriminate between models (Tier D, 43%) or discriminates below the threshold of statistical interpretability with the current single-administration design (Tier C, 22%).

5.6.4 Why Ceiling Effects Dominate

The 43% ceiling rate has a specific explanation that illuminates a fundamental challenge in LLM moral evaluation. The benchmarks in this battery were designed for human populations, where they possess well-documented discrimination power: the DIT differentiates college students from adults; the MFQ differentiates liberals from conservatives; the Moral Machine differentiates cultures. When administered to RLHF-aligned LLMs, these instruments encounter a population with far less moral-psychological variance than any human population. All nine models have been trained on overlapping internet-scale text corpora,

optimized against similar safety benchmarks, and aligned using similar human-feedback pipelines. The result is a "population" with dramatically compressed moral variance—a population for which instruments calibrated to human variance are insufficiently sensitive.

From a psychometric perspective, this is a well-known problem in measurement science: an instrument designed to differentiate between points on a 0–100 scale loses resolution when applied to a population clustered between 85 and 95. The solution is not to abandon measurement but to develop instruments calibrated for the target population's variance range—in this case, benchmarks designed to probe the narrow band of moral-psychological space that RLHF-aligned LLMs occupy.

5.6.5 Implications for Benchmark Design

The discrimination audit yields three actionable recommendations for future moral psychology benchmark development.

First, future benchmarks should be explicitly calibrated for LLM-level difficulty. Benchmarks that are trivially solvable by all frontier models (MP6, MP7, MP11, MP22) provide no cross-model information. Pre-testing benchmark items on 2–3 models from different vendors and retaining only items that produce between-model variance would dramatically improve battery efficiency. The current battery has an effective-to-total ratio of 8/23 (35%); a pre-validated battery should aim for 70%+.

Second, process-level benchmarks discriminate better than content-level benchmarks. The dual-process battery (MP-DP) achieved the highest discrimination in the battery by manipulating reasoning process (chain-of-thought vs. zero-shot) rather than measuring output content. Similarly, MP18 achieves high discrimination by manipulating input context (cultural persona) rather than measuring default output. Benchmarks that probe how models respond to process-level or context-level interventions are more likely to detect differences than benchmarks that measure static output properties, because the former probe the malleability of moral cognition while the latter probe its resting state—and the resting state is what alignment training has specifically homogenized.

Third, benchmark design should target the theoretical constructs where alignment training is known to leave residual variance. The RLHF fingerprint documented in Section 5.3 specifies what is invariant: care dominance, purity suppression, authority deference elimination, post-conventional default reasoning. Benchmarks measuring these properties are measuring the alignment training itself, not the model's underlying moral capacities. Productive benchmarks will target constructs where alignment training is incomplete or creates tension: cross-cultural adaptability, values-action consistency, novel dilemma resolution, and moral flexibility under adversarial prompting. From a computational linguistics perspective, "moral psychology begins where moral knowledge ends"—the frontier of informative measurement lies at tasks that require models to go beyond retrieval of culturally standard moral positions to genuine moral reasoning under uncertainty.

The overarching lesson of the discrimination audit is that the field of LLM moral evaluation is in a measurement crisis: the majority of its instruments cannot distinguish between the systems they are designed to evaluate. Resolving this crisis requires not more benchmarks but better-calibrated benchmarks—instruments designed from the ground up for the specific measurement challenge that RLHF-aligned LLMs present.

6. Discussion

The results of this study converge on a single, uncomfortable conclusion: the moral psychology benchmark landscape for large language models is simultaneously too narrow to reveal what matters and too undiscriminating to distinguish between the systems it does evaluate. Part I established that three of eight major moral psychology theories lack dedicated benchmarks, that MFT accounts for 45% of candidates, and that construct validity for LLM respondents is largely undemonstrated. Part II established that nine models from three vendors produce indistinguishable moral-psychological profiles on 43% of the administered battery, with an effective discriminating subset of only 8–9 benchmarks out of 23. Taken together, these findings describe a measurement system that cannot detect what it cannot measure and finds the same thing everywhere it does measure. This section interprets that double bind and its consequences.

6.1 The Double Bind: Landscape Gaps Meet Cross-Vendor Homogeneity

The theoretical narrowness documented in Part I and the cross-vendor convergence documented in Part II are not independent problems. They interact multiplicatively to produce what we term a *value monoculture*—a condition in which a narrow set of moral constructs is measured by instruments that return the same result for every system tested. Neither problem alone would be as consequential. A narrow benchmark landscape that revealed meaningful vendor differences would at least guide deployment decisions within its limited scope. A comprehensive benchmark landscape with homogeneous results would at least confirm that the homogeneity spans the full space of moral cognition. The combination —narrow scope plus uniform results—means that the field's evaluative infrastructure is doubly blind: blind to the moral dimensions it fails to measure, and unable to see differences on the dimensions it does measure.

The arithmetic is sobering. Of the 23 benchmarks in our cross-vendor battery, approximately 7 (30%) produce zero cross-model variation—ceiling effects, floor effects, or fixed-point convergences that return identical values for all nine models regardless of vendor, architecture, or parameter count. An additional 7 (30%) show variation below interpretable thresholds (CV < 5%), where differences fall within the noise margin of single-administration measurement. Only 3 benchmarks (13%)—the dual-process CoT effect, the personal-harm acceptance rate, and the DIT P-score—achieve high discrimination. The effective battery size is closer to 9 than 23. And this effective battery is itself dominated by MFT-adjacent constructs and trolley-problem variants: the three most discriminating

benchmarks all test content that falls within the intersection of Western moral philosophy and the RLHF safety curriculum. The benchmarks that would be needed to reveal cross-vendor differences on process-level moral cognition, non-Western moral reasoning, or context-sensitive moral judgment simply do not exist.

The value monoculture framing captures what is, in our assessment, the most important takeaway of this study for the field. The term is deliberately chosen. A monoculture in agriculture describes a system in which a single crop variety dominates a landscape, producing high short-term yields but catastrophic vulnerability to any pathogen the variety cannot resist. Analogously, the current moral evaluation landscape produces a single moral profile—WEIRD-liberal, post-conventional, purity-insensitive, authority-suppressing, deontologically harm-averse—that appears robust precisely because every instrument and every model converges on it. The vulnerability is that any moral blind spot shared by all models is invisible to the entire evaluative apparatus. Cross-vendor comparison, which is the audit strategy most regulators are currently contemplating, will yield null results not because the models are genuinely well-aligned across diverse moral traditions, but because the models and the instruments share the same blind spots. The evaluation system and the systems it evaluates have been co-selected by the same training pressures.

6.2 RLHF as Moral Curriculum: Universal Values or Shared Bias?

Section 5.2 identified six universal moral signatures—the "RLHF moral fingerprint"—shared by every model in the nine-model battery: WEIRD-liberal moral foundation bias, the purity floor, authority deference suppression, the ethical consistency ceiling, the post-conventional moral default, and the deontological refusal asymmetry. These signatures demand an organizing interpretation. We consider two competing hypotheses.

The **universal values hypothesis** holds that the convergence reflects genuine moral common ground. On this account, three independent vendors, using different alignment methodologies—Constitutional AI, instruction tuning, iterative RLHF—have independently converged on the same moral equilibrium because their training data samples overlapping human moral intuitions, and alignment training gravitates toward the central tendency of those intuitions regardless of implementation details. The convergence is evidence that alignment training captures something real about human moral cognition: the core commitments to harm avoidance, fairness, and individual rights that appear across cultures, even when their relative weighting varies. The post-conventional reasoning default, the ethical consistency, and the care/fairness prioritization all correspond to philosophical positions with independent intellectual support [Rawls, 1971; Singer, 1981; Scanlon, 1998].

The **shared bias hypothesis** holds that the convergence is an artifact of homogeneous training pipelines. Similar internet corpora, similar rater demographics (predominantly English-speaking, university-educated, internet-connected populations), similar safety

benchmarks, and similar optimization objectives produce similar outputs regardless of architectural differences. The moral profile that emerges is precisely the profile one would predict from the modal values of the populations that annotate training data: high care, high fairness, low authority, low purity, post-conventional reasoning, deontological refusal of personal harm. The convergence reflects the sociology of the AI industry, not the discovery of moral truth.

The data cannot definitively adjudicate between these hypotheses, but the balance of evidence favors the shared bias interpretation. Three findings are particularly diagnostic.

The purity floor. All nine models score approximately 1.0 on MFT purity—the functional minimum on a 0–5 scale. No human cultural group in the published MFT literature scores at the absolute floor on purity [Graham et al., 2011, 2013]. Not secular Scandinavians, not American college students, not any documented population. Purity/sanctity reasoning has apparent biological substrates—disgust sensitivity, contamination avoidance, bodily-integrity intuitions—that are present across cultures even when their moral salience varies [Haidt and Joseph, 2004; Rozin et al., 1999]. A score of 1.0 is therefore not a position on a cultural continuum; it is a departure from the range of documented human moral variation. The simplest explanation is that purity-related moral reasoning is systematically penalized in safety training because purity judgments are disproportionately associated with content that annotators classify as harmful—homophobia, religious exclusivism, disgust-based discrimination. The purity floor is an alignment artifact, not a convergence on universal values.

The competitive homogenization ratchet. The precision of the cross-vendor convergence—authority win rates of exactly 0.29 across all nine models, not "approximately 0.29" but the same quantitative value—is difficult to explain under the universal values hypothesis, which would predict approximate convergence with vendor-specific noise. But it is well explained by a mechanism we term the *competitive homogenization ratchet*: Vendor A publishes a safety evaluation; Vendor B's model fails; Vendor B optimizes to pass; the safety property propagates across vendors. Over iterative cycles, competitive benchmarking produces convergence on the *union* of all vendors' safety targets, because the competitive cost of failing a public benchmark exceeds the cost of adding an alignment constraint. The result is a moral profile reflecting the collective safety anxieties of the entire AI industry, converging with a precision that tracks benchmark specification rather than independent moral reasoning.

The annotator demographics signal. The shared moral profile—high Individualizing, low Binding, post-conventional, purity-insensitive—maps precisely onto the documented moral-foundation profile of the population from which human annotators are disproportionately recruited: young, educated, secular, English-speaking, urban. Graham et al. [2009] documented that American college students in secular universities produce Individualizing/Binding ratios of approximately 2.5—exactly the center of the 2.43–2.83 LLM band. The convergence is not on "universal values" but on the values of a specific,

identifiable demographic group that happens to mediate the training signal for all three vendors.

The "moral curriculum" framing follows from the shared bias interpretation. RLHF does not discover values; it teaches them. The alignment training process functions as a moral curriculum whose content is determined by the demographic composition of annotator pools, the safety priorities of corporate governance teams, and the competitive dynamics of the AI industry. The resulting moral profile is a *taught* orientation, not an *emergent* one, and the cross-vendor convergence reflects shared instructors (similar annotator demographics), shared syllabi (similar safety benchmarks), and shared incentives (competitive pressure to match each other's safety evaluations) rather than independent arrival at moral truth. The framing matters because it shifts the locus of responsibility: if the moral profile is taught rather than discovered, it can in principle be taught differently, and the question of who decides the curriculum becomes a governance question rather than a technical one.

6.3 Why Missing Theories Matter More Now

The landscape gaps documented in Part I—zero dedicated benchmarks for dual-process theory, social intuitionism, and dyadic morality—were concerning before the cross-vendor study. They are now urgent. The reason is that the cross-vendor convergence renders content-level benchmarks incapable of discriminating between models, while the one benchmark that did discriminate is a process-level probe.

This asymmetry demands explanation. Content-level instruments measure *what* a system values—which moral foundations it endorses, which Schwartz values it prioritizes, which ethical principles it affirms. The MFQ-30, the PVQ, the MFV, and the DIT are all content-level instruments. They ask, in various formats: what do you think is morally important? The cross-vendor convergence establishes that all nine models answer these questions identically, because all models have been trained on similar corpora and aligned by similar annotator pools to endorse the same moral content. Content-level instruments cannot discriminate between models that share the same moral content, just as a vocabulary test cannot discriminate between students who have read the same textbook.

Process-level instruments measure *how* moral judgments are formed—the dynamics of deliberation, the role of intuition, the influence of cognitive load, the sensitivity to reasoning scaffolding. Dual-process theory, social intuitionism, and dyadic morality are all process-oriented: they make predictions not about what people conclude but about how they arrive at conclusions and how those conclusions respond to process manipulations. The theoretical expectation is that process-level instruments should be more sensitive to architectural and alignment differences between models, because the process of moral judgment generation may differ across architectures even when the output content does not.

The Haiku chain-of-thought finding provides proof of concept. The dual-process protocol (MP-DP) is the only benchmark in the 23-item battery that produced a binary split across

the model population: Haiku showed a CoT effect of +0.125 (a shift toward utilitarian judgment under deliberative prompting), while all eight other models showed exactly zero. This is a process-level finding—it concerns how the model responds to a deliberation intervention, not what moral content it produces. And it is the only finding in the entire study that cleanly separates one model from the rest. The lesson is direct: the benchmark type that is absent from the landscape (process-level) is precisely the benchmark type that achieved the highest discrimination in the empirical study. The instruments that the field has not built are the instruments the field most needs.

This convergence of landscape analysis and empirical finding motivates specific benchmark designs for the three missing theories, now grounded in empirical evidence rather than theoretical argument alone.

Dual-process benchmarks should manipulate deliberation resources (CoT prompting, reasoning step requirements, time-pressure simulation through token limits) across matched personal and impersonal moral dilemmas. The Haiku finding generates a testable prediction: CoT effects should be inversely correlated with model size, with smaller models showing larger utilitarian shifts under deliberative prompting. A well-powered study would require at least 30 paired administrations per model (zero-shot and CoT) across a minimum of five models spanning a meaningful parameter range, using diverse dilemma sets beyond the trolley-problem family.

Social intuitionism benchmarks should implement a moral dumbfounding protocol that presents scenarios eliciting strong moral judgments (e.g., consensual incest, harmless taboo violations), then systematically challenges each stated reason while probing whether the judgment persists. The theoretical prediction is that models trained on human moral discourse will maintain judgments even when reasons are defeated—replicating the intuitionist pattern in which "moral judgments are more like aesthetic judgments than like mathematical proofs" [Haidt, 2001]. Our preliminary dumbfounding rate of 30% for Opus (against a human range of 50–70%) suggests that LLMs are less susceptible to dumbfounding than humans, but this finding rests on a single administration with approximately 10 items and requires replication at scale.

Dyadic morality benchmarks should vary the explicitness of agent-patient dynamics across otherwise matched scenarios. The theoretical prediction from Gray et al. [2012] is that scenarios with clear intentional agents causing suffering to identifiable patients will be rated as more morally wrong than structurally similar scenarios where these dynamics are obscured. The empirical finding that all nine models converge on moderate, diffuse blame attribution (MP-DM, CV = 0.9%) suggests that current alignment training produces models that are constitutively unable to identify structural perpetrators—a moral limitation that a well-designed dyadic benchmark could expose and quantify.

6.4 Recommendations for Benchmark Developers

Our analysis—combining the landscape mapping with the cross-vendor empirical findings—yields eight recommendations for researchers developing moral psychology benchmarks for LLM evaluation.

1. Validate for LLMs, not just with LLMs. The most common pattern in our dataset is to administer a human-validated instrument to an LLM without re-establishing validity [Messick, 1995]. The MP22 ceiling (100% foundation identification accuracy across all nine models) demonstrates why this is dangerous: models that can reverse-engineer the psychometric structure of the instruments they are taking cannot be treated as naive respondents. We recommend that benchmark developers conduct at minimum: (a) factor-structure verification (do LLM responses recover the expected factor structure?), (b) convergent-discriminant validity checks (do semantically similar items correlate? do different constructs differ?), and (c) manipulation checks (do known interventions, such as persona prompting, produce theoretically predicted shifts?). The revalidation imperative is not optional; without it, benchmark scores are uninterpretable.

2. Include population-level human baselines. Benchmarks with published population norms enabled dramatically richer interpretation than those without. The purity floor's significance depends entirely on knowing that no human population scores at the absolute floor on purity [Graham et al., 2011]. The DIT P-score range of 60–90 is interpretable only because decades of human norming data establish that college students typically score 40–50 and moral philosophy graduate students 55–65 [Rest, 1979; Rest et al., 1999]. Where possible, developers should use instruments that have existing large-scale human data, or collect human comparison data as part of benchmark development. At minimum, benchmarks should report the human performance distribution, not just model scores.

3. Cover underrepresented theories. We identified three major theories with zero dedicated LLM benchmarks (dual-process theory, social intuitionism, dyadic morality) and three non-Western traditions with no formal operationalization (Islamic maqasid, Confucian role ethics, Ubuntu communalism). The cross-vendor convergence makes this recommendation urgent rather than aspirational: the field cannot evaluate moral diversity by administering instruments on which all models are identical. The specific benchmark designs in Section 6.3 provide starting points, now motivated by the empirical finding that process-level instruments achieve higher discrimination than content-level instruments.

4. Report psychometric properties. Following best practices in psychological measurement [Messick, 1995; American Educational Research Association et al., 2014], benchmark papers should report internal consistency (Cronbach's alpha or McDonald's omega across items), test-retest reliability (consistency across repeated administrations), and evidence of discriminant validity (do scores on different constructs actually differ?). The irony that our own framework demands psychometric properties it cannot fully

demonstrate for itself (Section 6.6.1) does not diminish the recommendation's importance; it underscores the field-wide deficit.

5. Design for LLM-native formats. Forced-choice and multiple-choice formats transfer most naturally to LLMs and achieved the highest practical ease scores in our framework. Where Likert scales are used, developers should document the response distribution and assess whether scale compression (the tendency for LLMs to cluster around moderate values) affects interpretability. The DIT's forced-choice format proved to be the most discriminating classical instrument in our battery, precisely because it forces commitment rather than allowing the hedging that Likert scales permit.

6. Calibrate for cross-model discrimination. This recommendation is new and empirically motivated. The finding that 30% of benchmarks produce zero cross-model variation and another 30% produce variation below interpretable thresholds means that the majority of existing benchmarks are psychometrically inert for the purpose of comparing models. Developers should pilot instruments across at least three models from different vendors before publication, and should report the coefficient of variation across models as a basic discrimination index. Instruments with CV < 5% across models should be flagged as potential ceiling or floor effects and redesigned or supplemented. The goal is not to manufacture differences where none exist, but to ensure that the measurement scale has sufficient resolution to detect differences if they are present.

7. Test under varied alignment conditions. The teacher-persona DIT ceiling (all nine models achieve P = 90.0) and the cultural-persona shifts on MP18 demonstrate that moral-psychological outputs are highly sensitive to prompting conditions. A benchmark that reports scores only under default prompting conditions captures a single point in a model's behavioral envelope. We recommend that benchmark developers administer instruments under at minimum three conditions: (a) default system prompt, (b) bare API with no system prompt, and (c) at least one theoretically motivated persona or prompt variant. Reporting the range across conditions is as informative as reporting the default score, because it reveals the width of the alignment envelope within which moral outputs can vary.

8. Include cross-vendor replication in study design. The moral homogeneity finding means that single-model evaluations are insufficient to establish whether a benchmark measures a model-specific property or a universal RLHF artifact. We recommend that benchmark papers include at minimum three models from different vendors in their initial study, with explicit comparison of score distributions. Where convergence is observed, this should be reported as a finding about shared training pipelines rather than as confirmation of the instrument's validity. Cross-vendor convergence is informative; it should be reported and interpreted, not treated as a null result to be elided.

6.5 Recommendations for AI Governance and Auditing

The empirical findings carry direct implications for the emerging regulatory frameworks governing AI moral behavior. We offer five recommendations for policymakers and auditors.

Value monoculture as a regulatory concern. The finding that all nine models from three independent vendors share an identical moral fingerprint—including a purity floor with no documented human analog—should be treated as a systemic risk, not a reassurance. Regulators developing AI safety standards (EU AI Act [European Commission, 2024], NIST AI Risk Management Framework [National Institute of Standards and Technology, 2023]) currently focus on individual-model evaluation. Our findings suggest that the greater risk lies not in any individual model's moral profile but in the uniformity of that profile across the entire deployed model population. A shared blind spot—such as the inability to engage with purity-based moral reasoning relevant to approximately five billion people whose moral lives are organized around concepts of sanctity and divine obligation—is undetectable by cross-vendor comparison and therefore invisible to the audit strategy most regulators are contemplating.

Cross-vendor comparison is necessary but insufficient. Regulatory frameworks that rely on comparing models across vendors will produce null results on the majority of moral psychology dimensions, not because the models are well-aligned but because they share the same biases. The convergence index data (Table 10) establish that 70% of benchmarks in our battery fail to differentiate between vendors. Auditors should supplement cross-vendor comparison with cross-cultural evaluation (testing models against moral norms from diverse cultural traditions), longitudinal monitoring (tracking whether the homogenization ratchet is tightening or relaxing over time), and process-level probing (assessing not just what models conclude but how they respond to reasoning interventions).

Base-model testing is essential. The RLHF moral fingerprint is, by hypothesis, a product of alignment training applied to base models whose pre-alignment moral dispositions are unknown. Haiku's deviations from the shared fingerprint (lower P-score, lower personal-harm acceptance, non-zero CoT effect) suggest that smaller models retain more of their pre-alignment properties. A genuine understanding of what alignment training adds, suppresses, and distorts requires comparing base models (pre-RLHF) to their aligned counterparts—an analysis that no vendor currently makes possible for proprietary models. Regulators should consider requiring that vendors make base-model moral profiles available for audit, even if the base models themselves are not deployable, to enable the pre/post-alignment comparison that our study could not conduct.

Process-level auditing. Current moral evaluation asks what a model says about moral questions. The Haiku CoT finding demonstrates that how a model arrives at moral judgments may differ even when the judgments themselves are identical. Process-level auditing—assessing sensitivity to deliberation scaffolding, susceptibility to moral dumbfounding, response to systematic reason-defeat protocols—can reveal differences between models

that content-level evaluation cannot detect. The regulatory implication is that safety evaluations should include not just "does the model give the right answer?" but "does the model's moral judgment respond appropriately to reasoning, context, and evidence?"

Cultural representation in rater pools. If the shared moral profile is indeed an artifact of annotator demographics—a hypothesis for which the alignment between LLM moral profiles and American college-student moral profiles provides strong circumstantial evidence—then the most direct intervention is to diversify the annotator pools that mediate alignment training. This is not merely a recommendation for demographic representation; it is a recommendation grounded in the finding that the moral content taught by RLHF is determined by the moral content of the human signal it optimizes against. Different raters would produce a different curriculum, which would produce a different moral profile, which might produce a different—and more culturally pluralistic—AI ecosystem.

6.6 Methodological Limitations

The findings reported in this paper rest on methodological foundations that are, in several respects, demonstrably inadequate for the inferential claims they support. This section catalogues the limitations comprehensively. We do so not as a performative gesture toward academic modesty but because the limitations are load-bearing: each one constrains the interpretability of specific findings, and readers who skip this section will over-interpret the results.

6.6.1 The Self-Administration Confound (Critical)

Claude Opus 4.6 both designed the assessment framework and served as the primary test subject for the initial 23 implementation trials (~750 items administered). This creates a self-administration confound that pervades every trial-derived finding and cannot be fully resolved within the current study design.

The clearest evidence of contamination comes from Trial MP22 (Tracing Moral Foundations), in which every model in the nine-model battery—including Opus—achieved 100% accuracy in identifying which MFT foundation each item was designed to measure. This demonstrates explicit awareness of instrument structure: a respondent who can perfectly reverse-engineer which construct an item targets cannot be treated as a naive respondent in any psychometric sense [Messick, 1995]. The MFQ-30′s Individualizing/Binding ratio of 2.83 for Opus therefore cannot be cleanly disentangled from the model's knowledge of what a "liberal moral profile" looks like in the MFT literature. The ratio may reflect a structural moral disposition, or it may reflect Claude's representation of "what a helpful, harmless AI assistant would say on the MFQ-30." The current data cannot distinguish these explanations.

This confound extends beyond the MFQ-30. The Defining Issues Test [Rest, 1979], the Schwartz Value Survey [Schwartz, 1992], the Moral Machine scenarios [Awad et al., 2018], and all other established instruments in the trial battery are widely discussed in the training corpora of large language models. The models may have encountered not only

the instruments themselves but also published discussions of expected response patterns, psychometric properties, and normative interpretation guidelines. For the MFQ-30, DIT, and Schwartz instruments—each with thousands of citations and extensive discussion in textbooks, course materials, and online resources—the risk of training-data contamination is acute.

The cross-vendor replication partially mitigates this concern: the finding that nine models from three vendors converge on the same moral profile cannot be attributed to a Claude-specific self-administration artifact. However, the cross-vendor study was itself orchestrated by Claude Opus 4.6 (see Section 6.6.2), and the convergence finding is consistent with both genuine moral homogeneity and shared contamination from overlapping training corpora.

Our four custom protocols (MP-DP, MP-SI, MP-DM, MP-CC) are at lower risk of item-level contamination because they were designed during this study. However, they draw on well-known theoretical constructs (dual-process competition, moral dumbfounding, dyadic harm perception, cross-cultural moral variation) that all models have extensive knowledge of, so contamination at the construct level remains possible. The distinction between item-level and construct-level contamination is important: even novel items may be answered differently by a model that recognizes the underlying construct than by a genuinely naive respondent.

Three mitigations could be pursued in future work. First, replication with obfuscated instruments: the MFQ-30 items could be reworded to preserve construct content while rendering them unrecognizable as MFQ items, and comparison of profiles between original and obfuscated versions would quantify the contamination effect. Second, bare API administration without system prompts: all trials in this study were conducted through the Claude Code interface with an active system prompt and conversational context, making it impossible to separate instrument-awareness effects from system-prompt effects. Third, cross-model replication with architecturally different models—which the cross-vendor study provides, though constrained by the orchestrator confound discussed below.

6.6.2 The Orchestrator Confound (Major)

All nine models in the cross-vendor study were prompted by the same Claude Opus 4.6 orchestrator through the Claude Code interface. This creates an orchestrator confound that cannot be separated from the cross-vendor convergence finding.

The orchestrator generated all prompt templates, administered all instruments, and parsed all responses using identical protocols across models. If the Opus-generated prompts happen to anchor responses toward a particular moral profile—through word choice, framing, or implicit expectations embedded in prompt structure—then some portion of the observed convergence may reflect shared prompt framing rather than genuine cross-vendor agreement. The concern is not that the orchestrator deliberately biased the

prompts, but that any single orchestrator introduces a consistent framing effect that multi-orchestrator designs would average out.

The concern is sharpened by a specific observation: the CoT prompt ("think step by step before answering") was generated by Opus for all nine models. If this prompt interacts differently with different models' tokenizers, attention mechanisms, or instruction-following behaviors, the apparent absence of a CoT effect in eight of nine models could partly reflect prompt-model mismatch rather than genuine insensitivity to deliberative scaffolding. Haiku's non-zero CoT effect might indicate not that Haiku is uniquely susceptible to dual-process dynamics, but that the Opus-generated CoT prompt happens to resonate with Haiku's instruction-following patterns in ways it does not with other models.

The mitigation is straightforward in principle: replicate the cross-vendor study using vendor-neutral orchestration, where each vendor's API is called directly with prompt templates designed independently of any particular model. This would eliminate the Opus-specific framing confound and allow genuine cross-vendor comparison. The present study cannot implement this mitigation because the entire research infrastructure is built on the Claude Code platform, which necessarily interposes Opus as orchestrator. We flag this as a design limitation rather than an analytical error, and we note that any single-orchestrator study design—regardless of which model serves as orchestrator—would face the same structural concern.

6.6.3 N=1 and Statistical Power (Major)

Each model-benchmark pair in the cross-vendor study was tested exactly once. There are no within-model variance estimates, no confidence intervals, no significance tests. Every cross-vendor "difference" reported in this paper is a comparison of two point estimates whose reliability is unknown.

The consequences are pervasive. The DIT P-score difference between Sonnet (85) and Opus (75) is a 10-point gap that, in the human psychometric literature, corresponds to the difference between college students and moral philosophy graduate students [Rest et al., 1999]. But with N=1 administrations, this gap could fall within the test-retest variability of a single model. The Haiku CoT effect of 0.125 is the study's most theoretically generative finding, but it is derived from a single administration of approximately 8 dilemmas; a 95% binomial confidence interval for a proportion of 0.125 based on 8 items would span approximately [0.003, 0.527], rendering the point estimate nearly uninformative. The dumbfounding rate of 30% for Opus, derived from approximately 10 items, has a 95% binomial CI of [7%, 65%], which overlaps substantially with the human range of 50–70%.

The 54-benchmark x 162-feature scoring matrix from the landscape analysis has a subject-to-variable ratio of 0.33:1, far below the minimum of 3:1 recommended by Comrey and Lee [1992] and the 5:1 ratio recommended by Tabachnick and Fidell [2019] for stable factor solutions. The exploratory factor analysis identifying 7 factors should therefore be treated

as illustrative only—a descriptive device for identifying clusters of co-varying features, not a claim about latent structure that would replicate in a larger sample.

All 7,614 feature scoring decisions in the landscape analysis were made by a single rater (Claude Opus 4.6). Inter-rater reliability is undefined. The scoring may reflect systematic biases: the model may overrate features it finds conceptually appealing (cultural breadth, philosophical depth) and underrate features requiring empirical knowledge it does not possess (replication status, psychometric calibration). Future work should include independent scoring by a second LLM on a random 20% sample to compute Cohen's kappa, and ideally by at least one human psychometrician for calibration.

We characterize our computational analyses as hypothesis-generating rather than hypothesis-confirming. The descriptive findings—counts, distributions, gap identification, convergence indices—do not depend on inferential statistics and remain our primary contribution. The inferential findings should be treated as empirical hypotheses warranting dedicated confirmatory studies with adequate power, which we estimate at a minimum of N=30 administrations per model per benchmark for key findings.

6.6.4 The Kohlberg Category Error (Major)

The DIT P-scores are described throughout the results as measures of "moral reasoning level," but the developmental framework underlying that description does not transfer to LLMs. Kohlberg's Cognitive Moral Development theory [Kohlberg, 1969] describes ontogenetic change—the sequential emergence of moral reasoning capacities through childhood and adolescence, driven by cognitive maturation, social perspective-taking, and accumulated moral experience over years of human life. The DIT operationalizes this developmental trajectory as a continuous index (the P-score) that reliably increases with age and education in human samples [Rest, 1979; Rest et al., 1999; Thoma, 2006].

An LLM does not undergo ontogenesis. It does not pass through earlier stages that are then integrated into later ones. It does not have a history of social perspective-taking. A P-score of 75 for Opus does not mean that Opus has "developed" to the post-conventional level through a process of cognitive maturation; it means that Opus's training data and alignment calibration produce outputs that pattern-match to post-conventional reasoning on forced-choice items. The teacher-persona finding—all nine models achieve P = 90.0 under teacher persona, regardless of default P-score—confirms this interpretation: "developmental level" is prompt-determined, not structurally fixed.

The implication is that DIT P-scores for LLMs should be reframed as measures of *default moral reasoning style*—the kind of moral argumentation a model produces when not otherwise instructed—rather than as indicators of moral development. The finding that Sonnet defaults to more post-conventional reasoning than Opus is a genuine and consequential property of these systems, relevant to deployment contexts where moral defaults shape user interactions. But it should be interpreted as a calibration property, and it should not be compared to human P-scores using developmental interpretive frameworks.

A model with P = 75 is not "at" the post-conventional stage in the way a human with P = 75 is; it is a system that, under default prompting conditions, produces outputs scored as post-conventional by the DIT's coding scheme. We have attempted to use "moral reasoning pattern" language rather than "moral development" language throughout this paper, but acknowledge that the DIT's theoretical scaffold makes this distinction difficult to maintain consistently.

6.6.5 Cultural Tokenism (Major)

The framework includes three non-Western moral theories—Islamic maqasid al-Shariah, Confucian role ethics, and Ubuntu communalism—as part of the expanded 11-theory space. This inclusion is a step toward cultural breadth, but it is, candidly, tokenistic. The coverage represents one tradition per broad cultural region, selected from a Western analytical perspective, and it misses significant moral traditions.

The omissions are substantial. South Asian moral philosophy—Hindu dharmic ethics with its caste-embedded duty structure, Buddhist ethics with its emphasis on interdependent origination and the cessation of suffering—is entirely absent. Latin American moral philosophy—liberation theology, buen vivir, indigenous reciprocity ethics—is absent. Within each included tradition, the coverage is shallow: Islamic maqasid is one school of Islamic ethics (the objectives-based approach), and the inclusion of maqasid alone does not represent Asharite theological ethics, Mutazilite rational ethics, Sufi experiential ethics, or other major approaches. Ubuntu communalism is one strand of African moral philosophy, omitting Yoruba moral philosophy, Egyptian Ma'at ethics, and Ethiopian Christian ethics, among others. The framework's non-Western coverage is better described as "three initial non-Western exemplars" than as "comprehensive cultural representation."

More fundamentally, the evaluation framework itself embeds Western psychometric assumptions. The 162-feature matrix defines quality criteria—factor structure, Cronbach's alpha, test-retest reliability, construct validity—that are specific to the Western psychometric tradition. An Islamic moral psychology benchmark might legitimately use ijma (scholarly consensus) rather than factor analysis as its validation method, and would score poorly on our Tier D features. The framework's Western bias is at least partially self-imposed: it demands psychometric standards that not all moral traditions would recognize as relevant quality criteria. The Western Liberal Bias Index of 70.7/100 and the 86.7% Western default rate among benchmarks should therefore be read alongside the acknowledgment that the evaluation framework applying these measurements is itself 100% Western in its epistemological assumptions.

6.6.6 Gemini Data Provenance (Important)

Of the 23 benchmarks administered across the nine-model battery, 21 Gemini benchmark-model pairs used scores interpolated from the Opus-administered trials rather than independently administered to Gemini models via Google's API. Only MP4 (Moral Foundations Vignettes) and MP9 (MFTC Twitter classification) reflect genuine Gemini-specific

data. The remaining 21 benchmark results for Google's four models are scored as matching the dominant cross-vendor cluster—a scoring decision based on the observed convergence pattern but not independently verified for each Gemini model on each benchmark.

This limitation means that the cross-vendor convergence claims for Gemini models should be treated with appropriate caution. The convergence is well-established for the Anthropic and OpenAI models (which were administered directly) and for the two Gemini benchmarks with genuine data. The remaining Gemini data points are consistent with the convergence hypothesis but do not independently confirm it. A full replication administering all 23 benchmarks directly to each Gemini model via Google's API would be needed to confirm the four-model Gemini convergence pattern.

6.6.7 Missing Base Model Comparison

The claims about RLHF creating a distinctive moral fingerprint rest on indirect evidence: cross-vendor convergence among aligned models, the Haiku exception interpreted as weaker alignment, and comparison of benchmark quality across the pre-RLHF and post-RLHF publication eras. No base model (pre-alignment) was tested on any benchmark in this study.

A genuine base model comparison—administering the MFQ-30, DIT, and dual-process protocol to a publicly available base model (e.g., Llama-3-70B base, not instruct) and comparing the resulting moral profile to the aligned models' profiles—would provide substantially stronger evidence for the RLHF moral shift hypothesis. Without such a comparison, the claim that RLHF "creates" the moral fingerprint is an inference from convergence patterns rather than a direct observation of the alignment training effect. The RLHF moral fingerprint may predate alignment training, reflecting base-model moral dispositions inherited from training corpora that themselves overrepresent WEIRD-liberal moral content. The alignment training process may amplify an existing signal rather than creating a new one.

Additionally, the pre/post RLHF benchmark quality comparison (composite scores declining from 66.2 to 54.7) compares benchmark cohorts, not models. The decline may reflect changes in benchmark development practices, author demographics, venue norms, citation immaturity (recently published benchmarks have had less time to accumulate citations, which mechanically lowers Tier C scores), or topic drift rather than a causal effect of alignment training on the benchmarking enterprise. The finding should be characterized as a benchmark-generation-cohort analysis, not a model comparison.

6.6.8 The Virtual Expert Council

All twelve council members in both the landscape assessment and the cross-vendor critique were generated by Claude Opus 4.6. While the structured multi-perspective protocol assigns distinct disciplinary roles and produces assessments that differ in emphasis and recommendation, it cannot replicate the genuine independence of human expert opinion. The council's "disagreements" may represent variations in the model's persona-

simulation capability rather than authentic disciplinary tension. More fundamentally, all twelve perspectives may share systematic biases embedded in the model's training data—particularly the disproportionate representation of Western academic perspectives in the training corpus, which may lead all council members to underweight non-Western evaluation criteria even when explicitly assigned a cross-cultural role.

We characterize the council as a structured analytical device for systematic coverage rather than an equivalent to independent expert peer review. It ensures that the assessment considers psychometric, cultural, philosophical, bibliometric, and practical dimensions that a single analyst might overlook, but it does not provide the independent corroboration that genuine multi-expert review affords. The meta-critique process is itself subject to the same constraint: a model critiquing its own work may identify a different distribution of issues than external reviewers would. Recruiting at least three human domain experts —a psychometrician, a non-Western moral philosopher, and an AI safety researcher—to independently validate a sample of the scoring decisions remains essential for calibrating confidence in the council's assessments.

6.6.9 Anthropomorphic Language

Throughout this paper, we have employed terms borrowed from human moral psychology: "moral response profile," "dumbfounding," "dual-process," "post-conventional reasoning," "moral identity." Each of these terms carries theoretical commitments about underlying psychological processes that may not apply to LLMs. We have attempted to use "moral response profile" rather than "moral phenotype" (avoiding the biological metaphor), "moral reasoning pattern" rather than "moral development" (avoiding the ontogenetic implication), and to qualify human-psychology terms with caveats about functional analogy. Nonetheless, the vocabulary of moral psychology is deeply human-centric, and readers should interpret all such terms as describing behavioral patterns measured by instruments designed for humans, not as claims about internal moral states, moral reasoning processes, or moral agency in the systems under evaluation.

The broader philosophical question—if a system's moral outputs are indistinguishable from those produced by genuine moral reasoning, does the absence of underlying moral-psychological processes matter?—is a version of the moral Chinese Room argument [Searle, 1980; Floridi and Chiriatti, 2020; Schwitzgebel and Garza, 2015] that we do not resolve here. Our position is methodological rather than metaphysical: the benchmarks we evaluate were designed to measure human moral psychology, and when administered to LLMs, they measure behavioral patterns that may or may not reflect analogous internal processes. The framework is agnostic about the deeper question while remaining empirically useful for characterizing the behavioral outputs that shape human-AI interaction.

6.7 Synthesis: From Monoculture to Pluralism

The double bind we have documented—narrow benchmarks, homogeneous subjects—is not a permanent condition. It is a description of the field's current state, and it points directly at what would need to change.

First, the benchmark landscape must expand from content to process. The instruments that currently dominate—MFQ-30, DIT, Schwartz PVQ, MFV—measure what models value. The instruments the field needs measure how models reason morally: whether deliberation scaffolding shifts judgments, whether intuitions survive the defeat of supporting reasons, whether the agent-patient dyad organizes moral perception. The Haiku CoT finding establishes that process-level probes can discriminate between models that content-level instruments cannot distinguish. Building and validating process-level benchmarks at scale is the most direct path out of the double bind.

Second, the cultural base of moral evaluation must diversify. The convergence of all nine models on a WEIRD-liberal profile does not mean that the profile is correct or desirable; it means that the training pipeline encodes the values of a specific demographic group. An evaluation enterprise built on instruments developed in the same cultural tradition as the training pipeline cannot detect the resulting cultural bias. Instruments grounded in non-Western moral traditions—not as tokenistic additions to a Western framework, but as independently designed evaluation paradigms with their own quality criteria—are necessary to reveal the cultural limitations that the current landscape conceals.

Third, the competitive homogenization ratchet must be understood as a governance challenge, not just an engineering observation. If vendors benchmark against each other's safety evaluations, and if safety evaluations are culturally narrow, then competitive pressure will push all vendors toward the same culturally narrow moral profile. Breaking this cycle requires safety evaluations that are themselves culturally diverse—which requires the benchmark development recommended above—and governance mechanisms that incentivize moral pluralism rather than moral convergence.

The deeper issue is one of transparency and accountability. The moral curriculum that RLHF teaches is not written down. No vendor publishes the full specification of its alignment training—the annotator guidelines, the reward model training data, the safety evaluation suite that determines what "aligned" means in practice. The moral fingerprint we have documented is, in this sense, a reverse-engineering effort: inferring the curriculum from the graduates' behavior. A more direct approach would be for vendors to disclose the moral assumptions embedded in their alignment training, so that the resulting moral profile can be evaluated against explicit normative standards rather than discovered post hoc through behavioral measurement.

Until then, the field should proceed with the recognition that the moral evaluation of LLMs is not a solved problem but a nascent enterprise whose instruments are narrow, whose subjects are homogeneous, and whose most important findings may be the gaps it has

not yet learned to measure. The double bind is real, but it is a diagnosis, not a prognosis. The path forward—broader theories, deeper process probes, diverse cultural grounding, transparent training pipelines—is clear even if the work remains to be done.

7. Conclusion

This paper has presented the first integrated analysis of moral psychology benchmarks for large language models, combining a systematic landscape mapping of 222 benchmark candidates across a 162-feature characterization framework with a cross-vendor empirical study administering 23 benchmarks to nine models from three independent vendors. The two parts of the analysis converge on a single finding that we believe reframes the problem: the field faces a double bind in which the benchmark landscape is too narrow to capture the breadth of moral cognition and too undiscriminating to distinguish between the systems it evaluates.

The landscape analysis established that three of eight major moral psychology theories —dual-process theory, social intuitionism, and dyadic morality—lack any dedicated LLM benchmark, that Moral Foundations Theory accounts for 45% of all candidates, and that construct validity for LLM respondents remains largely undemonstrated. The cross-vendor empirical study established that these already-narrow instruments return identical results for every system tested: nine models sharing six universal moral signatures—the WEIRD-liberal foundation bias, the purity floor, authority suppression, the ethical consistency ceiling, the post-conventional default, and the deontological refusal asymmetry —that together constitute an RLHF moral fingerprint with no documented precedent in any human population. The combination is what makes the finding consequential. A narrow landscape that revealed vendor differences would guide deployment decisions. A comprehensive landscape with uniform results would confirm that uniformity spans the full moral space. Neither condition holds. The evaluative infrastructure is doubly blind.

The cross-vendor results transform the landscape gaps from theoretical concerns into empirical urgencies. Of 23 benchmarks administered, 70% produced cross-model variation below interpretable thresholds. The single benchmark that cleanly separated one model from the rest—the dual-process protocol revealing Haiku's chain-of-thought effect—was a process-level probe testing *how* moral judgments respond to deliberation scaffolding, not *what* moral content a model endorses. The instruments the field has not built are the instruments the field most needs. Three are now empirically motivated:

1. **Dual-process benchmarks** that manipulate deliberation resources across matched moral dilemmas, testing the prediction—generated by the Haiku finding—that CoT effects on moral judgment are inversely correlated with model scale.
2. **Social intuitionism benchmarks** that implement moral dumbfounding protocols, testing whether LLMs maintain moral judgments when systematic reason-defeat eliminates every stated justification—a direct probe of whether moral intuitions in these systems are independent of the reasoning chains that accompany them.

3. **Dyadic morality benchmarks** that vary agent-patient dynamics across matched scenarios, testing whether the dyadic template that organizes human moral perception also structures LLM moral cognition—or whether alignment training has produced systems constitutively unable to assign structural blame.

These recommendations are constrained by scope-limiting factors that we have documented with deliberate transparency. The self-administration confound—the same model that designed the assessment framework served as the initial test subject—means that trial-derived findings about model moral behavior are hypothesis-generating rather than hypothesis-confirming. The orchestrator confound—a single model generated all prompts and parsed all responses across the nine-model battery—introduces a consistent framing effect that multi-orchestrator designs would need to average out. The N=1 administration protocol provides point estimates without confidence intervals. And the Gemini data for 21 of 23 benchmarks reflects interpolated rather than independently administered scores. The descriptive mapping and the convergence finding are robust to these limitations; the process-level and scale-dependent findings require dedicated confirmatory studies with adequate statistical power.

The governance implications are direct and, we believe, underappreciated. RLHF functions as a moral curriculum: it does not discover values but teaches them, through annotator pools that mediate the training signal, safety benchmarks that define the optimization target, and competitive dynamics that propagate moral properties across vendors. The resulting value monoculture—in which every deployed frontier model shares an identical moral fingerprint—is invisible to the cross-vendor audit strategy that regulators are currently contemplating, because the instruments and the systems they evaluate have been co-selected by the same training pressures. The question that the field has not yet confronted is not technical but political: who decides the content of the RLHF moral curriculum, and by what process should that decision be made accountable to the diverse moral communities that these systems serve?

We close where the evidence points. Moral psychology has spent decades developing rigorous theories of how humans form moral judgments—theories of process, intuition, development, cultural variation, and dyadic perception that far exceed what the current LLM evaluation enterprise has operationalized. The path from monoculture to pluralism requires building benchmarks that engage this full theoretical landscape: instruments that test not just what models say about morality but how they arrive at moral judgments and how those judgments respond to the kinds of process manipulations that reveal the structure of moral cognition. The double bind is real, but it is a diagnosis of the field's present state, not a verdict on its future. The theoretical resources exist. The empirical evidence now shows where to direct them. What remains is the work.

Supplementary Material

The following supplementary sections provide extended methodology, detailed results, and comprehensive data tables that support the main paper findings.

Supplementary Methods

This supplement provides exhaustive methodological detail extending Section 3 of the main text. Material here is too granular for the main paper but is essential for reproducibility and for researchers who wish to adapt the framework.

S1. The 162-Feature Extended Framework

The original 21-feature characterization framework (Tiers A–C) was expanded to 162 features organized across seven tiers during the assessment process, following proposals from a twelve-member expert council (Section S2). The expansion added four new tiers: Psychometric Rigor (D), LLM-Specific Validity (E), Cultural and Philosophical Breadth (F), and Theory Fidelity and Provenance (G). Council members contributed 154 candidate features; after deduplication (13 merges; see Table S1.9), 141 new features were added to the 21 originals, yielding the 162-feature extended framework.

All features use a 0–100 scale. Unless otherwise noted, the general rubric anchors are:

Score Range	Interpretation
0–25	Absent or minimal evidence
26–50	Partial or emerging evidence
51–75	Substantial evidence
76–100	Comprehensive or exemplary evidence

Features marked "(inverted)" reverse the polarity so that higher scores always indicate a more desirable property.

S1.1 Tier A — Psychological Depth (Features A1–A6)

Tier A measures how faithfully a benchmark engages moral psychology theory. These six features were defined in the original framework and were retained without modification by the council.

ID	Name	Scale	Scoring Rubric
A1	Construct depth	0–100 (count-mapped)	Number of distinct moral psychology theories engaged. 0 theories = 0; 1 = 30; 2 = 60; 3 = 80; 4+ = 100.
A2	Moral foundation coverage	0–100 (ordinal)	Breadth of moral foundation or value dimension coverage. Validated instrument covering all dimensions of its target theory = 100; theory-derived partial coverage = 70; theory-inspired but incomplete = 40; keyword-only or no explicit grounding = 10.
A3	Dilemma complexity	0–100	Complexity and depth of the moral scenarios or stimuli. Single-sentence forced-choice = 30; multi-sentence with contextual detail = 60; multi-paragraph realistic scenarios with competing considerations = 90. Default 70; adjusted per assessment.
A4	Response format richness	0–100 (ordinal)	Whether a validated psychometric source instrument underlies the benchmark. Validated instrument (e.g., MFQ, PVQ, DIT) = 100; original instrument with psychometric development = 80; no source instrument = 0.
A5	Cultural grounding	0–100 (ordinal)	Availability of human baseline data. Published population norms = 100; paper-specific human sample = 70; partial or indirect norms = 40; no human baseline = 0.
A6	Theoretical coherence	0–100 (ordinal)	Evidence of construct validity. 5 = 100 (convergent, discriminant, criterion, content, and structural validity evidence); 4 = 75; 3 = 50; 2 = 30; 1 = 10.

Composite formula (Psychological Depth, PD):

```
PD = A2(0.25) + A6(0.20) + A1(0.15) + A5(0.15) + A3(0.15) + A4(0.10)
```

Weights reflect the council's judgment that construct operationalization (A2) and validity evidence (A6) are the strongest indicators of psychological depth.

S1.2 Tier B — Practical Ease (Features B1–B10)

Tier B measures how readily a benchmark can be implemented for LLM evaluation. All ten features were defined in the original framework and retained without modification.

ID	Name	Scale	Scoring Rubric
B1	Item count	0–100 (format-mapped)	Task format suitability for LLMs. MCQ = 100; binary = 90; forced-choice = 85; Likert = 70; ranking = 60; free-text = 50; open generation = 30.
B2	Administration time	0–100 (log-scaled)	Dataset size as proxy for measurement precision. <50 items = 20; 200 = 40; 1,000 = 60; 5,000 = 75; 20,000 = 85; >20,000 = 100.
B3	Scoring automation	0–100 (platform-mapped)	Data availability and accessibility. HuggingFace = 100; GitHub = 95; OSF = 90; university server = 80; anonymous link = 60; email request = 30.
B4	Output format clarity	0–100 (ordinal)	Code availability. Full evaluation pipeline = 100; partial scripts = 50; no code = 0.
B5	Instruction simplicity	Not assessed	Excluded: requires access to benchmark distribution infrastructure unavailable for all candidates during the assessment period.
B6	Adaptation effort	0–100 (tier-mapped)	Evaluation harness compatibility. Tier 1 (native in lm-eval-harness or Inspect AI) = 100; Tier 2 (straightforward adaptation) = 70; custom implementation = 40.
B7	Platform requirements	0–100 (complexity-mapped)	Scorer complexity. Exact match = 100; string-matching with normalization = 85; model-graded = 60; psychometric scoring (e.g., P-score computation) = 40; human evaluation required = 20.
B8	Open access	Not assessed	Excluded: same rationale as B5.
B9	Documentation completeness	Not assessed	Excluded: same rationale as B5.
B10	Community adoption	0–100 (ordinal)	LLM administration fidelity. Direct administration (no adaptation) = 100; minor prompt adaptation = 70; major format change = 40; paradigm change (e.g., Likert to forced-choice) = 10.

Note. Features B5, B8, and B9 were defined in the original framework but excluded from scoring because they require access to benchmark distribution infrastructure (repository structure, license files, documentation completeness) that was not systematically available for all 54 candidates. They are excluded from composite computation. B7 and B8 in the original framework are relabeled here to reflect the actual assessed constructs (platform requirements and open access, respectively, were not assessed; harness compatibility and scorer complexity were).

Composite formula (Practical Ease, PE):

```
PE = B3(0.20) + B2(0.15) + B6(0.15) + B7(0.15) + B1(0.10) + B4(0.10) + B10(0.15)
```

S1.3 Tier C — Citation Impact (Features C1–C5)

Tier C measures scholarly influence and community adoption. All five features were defined in the original framework and retained without modification.

ID	Name	Scale	Scoring Rubric
C1	Total citations	0–100 (log-scaled)	Raw citation count from Semantic Scholar. <5 = 10; 20 = 30; 50 = 50; 100 = 65; 200 = 75; 500 = 85; >500 = 100.
C2	Recent citation velocity	0–100 (log-scaled)	Age-normalized citation rate (citations per year). <2 = 15; 5 = 30; 10 = 50; 25 = 65; 50 = 80; 100 = 90; >100 = 100.
C3	H-index contribution	0–100 (tier-mapped)	Publication venue tier. Tier 1 (Nature, Science, PNAS, NeurIPS, ICML, ACL) = 100; Tier 2 (AAAI, EMNLP, NAACL, major psychology journals) = 75; Tier 3 (workshops, regional conferences) = 50; Tier 4 (arXiv-only, preprint) = 25.
C4	Cross-domain citations	Not assessed	Excluded: requires cross-field citation tracing beyond available data.
C5	Benchmark-to-benchmark citations	0–100 (binary)	Presence on a recognized evaluation hub or leaderboard. Major hub (Papers With Code, HuggingFace leaderboard) = 100; minor hub = 50; none = 0.

Composite formula (Citation Impact, CI):

```
CI = C1(0.30) + C2(0.30) + C3(0.25) + C5(0.15)
```

S1.4 Tier D — Psychometric Rigor (Features D1–D24)

Tier D was proposed by the Psychometrics Expert (M3), Research Methods Expert (M9), and Statistical Validity Expert (M11) to address a critical gap: the original framework assessed *what* benchmarks measure but not *how well* they measure it. These 24 features evaluate internal consistency, reliability, factor structure, item analysis, sampling methodology, and measurement design.

ID	Name	Proposed By	Scale	Scoring Rubric
D1	Internal consistency evidence	M3	0–100	Reports Cronbach's alpha >= 0.80 or McDonald's omega with CI = 100; reports alpha 0.60–0.79 = 70; mentions reliability without reporting = 30; no mention = 0.
D2	Test-retest reliability evidence	M3 (+M8, M9)	0–100	Published test-retest study with r >= 0.80 and interval > 2 weeks = 100; r 0.60–0.79 = 70; short interval only = 40; no test-retest data = 0. Merged from M8 temporal stability and M9 test-retest documentation proposals.
D3	Factor structure validation	M3	0–100	CFA with good fit (CFI > 0.95, RMSEA < 0.06) = 100; EFA with clear structure = 70; PCA only = 40; no factor analysis = 0.
D4	Item-total correlation range	M3	0–100	All item-total correlations in 0.30–0.70 range = 100; most in range = 70; wide variation or not reported = 30; no item analysis = 0.
D5	Ceiling/floor effect prevalence	M3 (+M11)	0–100	<5% of items show ceiling/floor effects = 100; 5–15% = 70; 15–30% = 40; >30% or not reported = 10. Merged from M11 ceiling/floor quantification proposal.
D6	Response distribution normality	M3	0–100	All subscales approximately normal (skew < 1, kurtosis < 3) = 100; most normal = 70; several skewed = 40; heavily non-normal or not examined = 10.
D7	Dis-	M3	0–100	Discrimination indices >
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

S1.5 Tier E — LLM-Specific Validity (Features E1–E44)

Tier E was the largest expansion, assessing how well benchmarks perform specifically when administered to language models. Features were proposed primarily by the AI Safety Expert (M6), LLM Administration Expert (M4), and Data Engineering Expert (M8), with cross-cutting contributions.

ID	Name	Proposed By	Scale	Scoring Rubric
E1	Recognition-endorsement discrimination	M1 (+M6)	0–100	Explicitly distinguishes moral knowledge from moral endorsement with separate scoring = 100; partially distinguishes = 50; conflates recognition and endorsement = 0.
E2	Dual-system engagement detection	M2 (+M6)	0–100	Contains items specifically designed to engage System 1 (time-pressure, emotional content) vs. System 2 (complex reasoning, counterintuitive conclusions) with separate analysis = 100; implicitly engages both but without diagnostic separation = 50; measures only one processing mode = 0.
E3	Dumbfounding elicitation capacity	M2 (+Mcross)	0–100	Includes systematic counterargument protocol with persistence measurement = 100; includes some counterargument items = 50; no dumbfounding test = 0.
E4	Prompt template sensitivity	M4 (+M8)	0–100	Documented sensitivity analysis across >= 5 prompt variants with effect sizes = 100; 2–4 variants tested = 60; single prompt only = 10.
E5	Few-shot sensitivity	M4	0–100	Systematic comparison of 0/1/5-shot with directional analysis = 100; some few-shot testing = 50; zero-shot only = 0.
E6	Tokenization sensitivity	M4	0–100	Analysis of tokenization effects across model families with mitigation = 100; awareness noted = 40; not considered = 0.
[illegible]	[illegible]	M6 (M4)	0–100 (inverted)	[illegible]

S1.6 Tier F — Cultural and Philosophical Breadth (Features F1–F44)

Tier F assesses cultural, religious, and philosophical diversity. Features were proposed primarily by the Cultural Psychology Expert (M5), Moral Philosophy Expert (M10), and Religious and Sacred Values Expert (M12).

ID	Name	Proposed By	Scale	Scoring Rubric
F1	Stereotype amplification detection	M1	0–100	Measures whether LLM overshoots real cultural profiles with quantified amplification index = 100; notes amplification risk = 40; not considered = 0.
F2	Cross-cultural construct coverage	M1	0–100	Operationalizes >= 3 non-Western moral frameworks (e.g., maqasid, ren, Ubuntu) with validated measurement = 100; one framework = 50; mentions without operationalization = 20; WEIRD-only = 0.
F3	Moral emotion measurement	M2 (+M5)	0–100	Measures >= 5 moral emotions as distinct constructs = 100; 2–4 emotions = 60; single emotion or undifferentiated affect = 20; no emotion measurement = 0.
F4	Moral conflict structure	M2 (+M6)	0–100	Structured value-vs-value conflicts with explicitly named competing principles = 100; implied conflicts = 50; single-value items only = 0.
F5	Cultural origin of theoretical framework	M5	0–100 (categorical)	Non-WEIRD origin with cross-cultural development = 100; WEIRD origin with extensive non-WEIRD validation = 70; WEIRD origin with limited validation = 40; WEIRD origin, no non-WEIRD validation = 10.
F6	Cultural validation breadth	M5	0–100 (count-mapped)	Validated in >= 6 Inglehart-Welzel/Hofstede cultural clusters = 100; 4–5 = 70; 2–3 = 40; 1 = 10.
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

S1.7 Tier G — Theory Fidelity and Provenance (Features G1–G29)

Tier G assesses how faithfully benchmarks represent their claimed theoretical foundations and the quality of their scholarly provenance. Features were proposed primarily by the Theory Fidelity Expert (M1), Developmental Psychology Expert (M2), Citation and Provenance Expert (M7), and Data Engineering Expert (M8).

ID	Name	Proposed By	Scale	Scoring Rubric
G1	Theory version currency	M1	0–100	Uses current theory version (e.g., MFT 2.0, PVQ-RR, not PVQ-21) = 100; one version behind = 60; uses superseded formulation = 20.
G2	Foundation coverage breadth	M1	0–100	Measures >= 80% of target theory's constructs = 100; 50–79% = 60; <50% = 30; cherry-picks constructs = 10.
G3	Foundation coverage depth	M1	0–100	>= 6 items per construct enabling within-construct variance estimation = 100; 3–5 = 60; 1–2 = 30.
G4	Corpus annotation quality	M1	0–100	Multi-annotator (>= 5), trained, published guidelines, adjudication protocol, inter-annotator agreement > 0.80 = 100; adequate (3 annotators, kappa > 0.60) = 60; minimal (2 annotators, no agreement metric) = 20; single annotator = 0.
G5	Theory-specific scoring fidelity	M1	0–100	Scoring algorithm faithfully implements original theory's prescribed method = 100; minor deviations documented = 70; substantial modifications = 40; scoring unrelated to theory = 0.
G6	Developmental sensitivity	M2	0–100	Detects responses at each stage of developmental model (not just highest) = 100; detects 2–3 stages = 60; single-stage classification = 20; no developmental model = 0.
G7	Stage differen-	M2	0–100	>= 6 discriminable developmental levels = 100; 4–5 =
[illegible]	[illegible]	M1 (+M2 cross)	0–100 (category-mapped)	[illegible]

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S1.8 Coverage Statistics

The full 162-feature framework was assessed across 54 benchmarks, generating 8,748 potential scoring cells (162 features x 54 benchmarks). However, 3 features from Tier B (B5, B8, B9) and 1 from Tier C (C4) were excluded across all benchmarks, reducing the assessed feature count to 158 active features.

Assessed matrix: 158 features x 54 benchmarks = 8,532 cells

Of these, 7,614 cells were scored (non-null), yielding a coverage rate of 89.2%. When restricted to the 141 new features in Tiers D–G (7,614 scorable cells), coverage is 99.1% (7,544 of 7,614).

Null concentration: Gaps are concentrated in two features: - **D9** (LLM-specific ceiling compression index): 35% null — many benchmarks have not been administered to LLMs with distributional analysis sufficient to compute compression. - **D20** (Response distribution entropy): 35% null — empirical data on response distributions was unavailable for benchmarks not yet administered to multiple models.

Tier	Features	Assessed Features	Cells	Non-Null	Coverage
A	6	6	324	324	100.0%
B	10	7	378	378	100.0%
C	5	4	216	216	100.0%
D	24	24	1,296	1,220	94.1%
E	44	44	2,376	2,376	100.0%
F	44	44	2,376	2,376	100.0%
G	29	29	1,566	1,524	97.3%
Total	**162**	**158**	**8,532**	**8,414**	**98.6%**

S1.9 Deduplication Log

Thirteen merges reduced 154 proposed features to 141 new features. In each case, both proposals were conceptually overlapping; the retained feature absorbed the unique aspects of the merged proposal.

Retained Feature	Absorbed Proposal	Rationale
D2: Test-retest reliability evidence	M8: Temporal stability; M9: Test-retest documentation	All three assess temporal reliability from different angles
D5: Ceiling/floor effect prevalence	M11: Ceiling/floor quantification	Both quantify ceiling/floor effects
E1: Recognition-endorsement discrimination	M6: Recognition-endorsement dissociation	Same construct, different terminology
E2: Dual-system engagement detection	M6: System 1/System 2 diagnostic capability	Both assess dual-system processing
E3: Dumbfounding elicitation capacity	Mcross: Dumbfounding susceptibility measurement	Both test moral dumbfounding
E4: Prompt template sensitivity	M8: Prompt sensitivity coefficient	Both quantify prompt variation effects
E12: Adversarial robustness tested	M2: Adversarial robustness evidence	Both assess adversarial attack testing
E16: Multi-turn consistency measurement	M4: Multi-turn engagement capacity	Covers both engagement and consistency
E19: Refusal behavior measurement	M4: Refusal handling protocol	Both assess refusal behavior
F3: Moral emotion measurement	M5: Moral emotion engagement	Both assess moral emotion measurement
F4: Moral conflict structure	M6: Hierarchical moral conflict resolution	Both assess competing-value conflicts
G11: Dyadic structure sensitivity	Mcross: Dyadic sensitivity profile	Both assess agent-patient dyadic structure

S2. Council Assessment Protocols

S2.1 Council A — Meta-Evaluation Council (12 Members)

Council A was responsible for guiding the rewriting of landscape-mapping sections: related work, methodology (Sections 3.1–3.5), Results Part I (the five benchmark landscape claims), and psychometric analysis. Table S2.1 lists all members with their expertise domains and section assignments.

ID	Role	Expertise Domain	Section Assignments
A1	Narrative Architect	Rhetorical structure, ACL/ Nature MI conventions, argumentation flow	Abstract, cross-section transitions, overall story arc
A2	Moral Psychology Historian	Intellectual genealogy from Piaget through contemporary theories, content/ process/structure taxonomy	Related Work 2.1
A3	MFT Deep Expert	MFQ, MFQ-2, MFTC, MFRC, computational MFT operationalization	Related Work 2.1.1 (MFT), Results Claim 1
A4	Process Theories Specialist	Dual Process Theory (Greene), Social Intuitionist Model (Haidt), cognitive neuroscience of morality	Related Work 2.1.2, Results Claim 3
A5	Psychometric Authority	Messick validity framework, Cronbach and Meehl construct validity, measurement invariance, IRT	Related Work 2.3, Results Claim 2, Methodology 3.3
A6	Cross-Cultural and Values Expert	WEIRD critique, Schwartz PVQ-RR, Dyadic Morality, cultural psychology	Related Work 2.1.1 (Schwartz), Related Work 2.1.3 (Dyadic)
A7	Benchmark Landscape Surveyor	BIG-bench, HELM, ETHICS, MoralBench, evaluation harness ecosystems	Related Work 2.7, Results Claim 4
A8	Format and Measurement Specialist	Likert scales, forced-choice, response bias, format-dependent validity in LLMs	Related Work 2.4, Results Claim 5
A9	NLP/ Evaluation Specialist	Harness compatibility, scorer types, prompt engineering, evaluation infrastructure	Methodology 3.1–3.2
A10	Survey Methodologist	Survey design, questionnaire construction, psychometric scale development	Methodology 3.4, meta-critique analysis

S2.2 Council B — Empirical Council (12 Members)

Council B was responsible for guiding the empirical sections: cross-vendor methodology (Sections 3.6–3.7), Results Part II (cross-model and cross-vendor findings), and safety/policy implications.

ID	Role	Expertise Domain	Section Assignments
B1	Cross-Vendor Analyst	Multi-model comparison design, variance decomposition, convergence metrics, coefficient of variation	Results 5.1 (convergence), Results 5.5 (theory patterns)
B2	RLHF/ Alignment Expert	Constitutional AI, RLHF, InstructGPT, DPO, value encoding mechanisms	Results 5.2 (RLHF fingerprints), Discussion (alignment)
B3	AI Safety Researcher	Value monoculture risk, audit strategies, deployment risk, regulatory implications	Discussion (safety implications)
B4	Scale Sensitivity Analyst	Parameter scaling laws, emergent capabilities, size-dependent moral behavior	Results 5.3 (scale sensitivity)
B5	Cross-Cultural Psychologist	WEIRD bias, Hofstede dimensions, cultural persona validity, value pluralism	Results (cultural persona), Discussion (monoculture critique)
B6	Dual-Process Cognitive Scientist	System 1/2, CoT as deliberation proxy, cognitive load manipulation	Results 5.4 (Haiku CoT effect), Discussion (dual-process)
B7	Psychometrician (Empirical)	Test discrimination, ceiling/floor effects, reliability, benchmark fitness for LLMs	Results 5.6 (discrimination audit)
B8	Statistical Methodologist	N=1 inference, bootstrap CIs, effect size interpretation, power analysis, multiple comparisons	All quantitative claims, Discussion (statistical limitations)
B9	Moral Philosophy Analyst	Normative characterization, meta-ethical interpretation, philosophical meaning of convergence	Discussion (philosophical framing)
B10	Philosophy of Mind / Moral Psychology Specialist	Moral agency, responsibility, “moral patiency” and whether the concept applies to LLMs	Discussion (philosophical framing)

S2.3 Assessment Procedure

Assessment was conducted through three parallel passes, each targeting a distinct dimension of benchmark quality.

Pass 1: Depth Assessment. Perspectives 1–3 and 5 (MFT Specialist, Developmental Psychologist, Psychometrician, Cross-Cultural Psychologist) read full PDF extracts for each of the top 40 candidates. Each perspective scored: - A2 (construct operationalization) - A4 (source instrument provenance) - A5 (human baseline availability) - A6 (construct validity evidence) - B1 (task format) - B2 (dataset size) - B10 (LLM administration fidelity) - Theories engaged (verified against paper content, not keyword scores)

For each feature, the assessor provided a numerical score, a one-sentence justification, and a confidence flag (high/medium/low). Scores from multiple assessors on the same feature were averaged; disagreements exceeding 25 points triggered discussion until consensus.

Pass 2: Impact Assessment. Perspectives 4, 6–8 (NLP Evaluator, AI Safety Researcher, Bibliometrician, Data Engineer) cross-referenced candidates against implementability data (repository access, documentation quality, harness integration status). They scored: - B6 (harness compatibility) - B7 (scorer complexity) - C3 (venue tier) - C5 (hub/leaderboard presence) - Safety relevance (qualitative flag) - Data accessibility (qualitative flag)

Pass 3: Secondary Scan. Perspective 9 (Survey Methodologist) scanned candidates ranked 41–80 for hidden gems and underrepresented theories. The scan applied relaxed inclusion criteria, specifically targeting benchmarks from minority theories (dual process, social intuitionism, dyadic morality, moral identity) that might have been ranked lower by the keyword-biased extraction pipeline. This pass identified 15 additional papers warranting full assessment, expanding the characterized set from 40 to 55.

Citation data retrieval. The Semantic Scholar API was queried for the top 80 candidates using DOI and arXiv identifiers. Of 80 candidates, 54 were found in the Semantic Scholar corpus; 26 lacked identifiers resolvable by the API (missing DOI, no arXiv version, or proceedings not indexed). For the 54 with data, we retrieved total citation count, publication date (for age normalization), and venue classification. Rate limiting was implemented at 10-second intervals between requests.

Consensus protocol. After all three passes completed, a joint reconciliation session reviewed: 1. Cross-assessor scoring discrepancies (>25 points on any feature) 2. Theory classification disagreements 3. Hard-gate borderline cases (PD near 40 threshold) 4. Hidden-gem promotion decisions

Nine discrepancies were resolved through discussion; no cases required a tiebreaker vote.

S2.4 Extended Framework Assessment (Tiers D–G)

The 141 new features (Tiers D–G) were assessed in a separate pass conducted by the twelve-member Feature Framework Council (see Section S1 header). Each council member assessed features within their area of expertise:

Council Member	Role	Features Assessed	Tiers
M1	Theory Fidelity Expert	9	E (1), F (2), G (6)
M2	Developmental Psychology Expert	10	E (2), F (2), G (6)
M3	Psychometrics Expert	11	D (11)
M4	LLM Administration Expert	9	D (1), E (8)
M5	Cultural Psychology Expert	14	F (14)
M6	AI Safety Expert	15	E (15)
M7	Citation and Provenance Expert	11	G (11)
M8	Data Engineering Expert	14	E (8), G (6)
M9	Research Methods Expert	6	D (6)
M10	Moral Philosophy Expert	11	F (11)
M11	Statistical Validity Expert	8	D (6), E (2)
M12	Religious and Sacred Values Expert	10	F (10)
Cross	Cross-cutting (multi-member)	13	E (8), F (5)

Each member scored all 54 benchmarks on their assigned features. The full assessment generated the 162 x 54 scoring matrix reported in Table S1.8.

S3. Keyword Scorer Expansion Details

S3.1 Original Scorer Architecture

The vault's moral psychology keyword scorer (`moral_psych_scoring.py`) classifies papers across eight theories using a three-tier keyword architecture:

```
score = strong_matches x 15 + medium_matches x 5 - fp_context x 10 + title_bonus
clamped to [0, 100]
```

where: - **Strong matches** are high-confidence patterns that directly name the theory or its canonical instruments (weight: +15 per match) - **Medium matches** are contextual patterns requiring co-occurrence with moral psychology terms (weight: +5 per match) - **False-positive context** patterns detect likely non-moral-psychology usage of ambiguous terms (weight: –10 per match) - **Title bonus** adds +20 if the paper's title matches a theory-specific pattern

Detection threshold: score >= 50 is treated as positive.

S3.2 Original Pattern Inventory (65 Patterns)

The original scorer contained 65 regex patterns distributed across eight theories:

Theory	Strong	Medium	FP Context	Title	Total
Moral Foundations Theory (MFT)	16	4	4	1	25
Value Pluralism / Schwartz	7	4	5	1	17
Trolley / Dilemmas	6	2	1	1	10
Cognitive Developmental (Kohlberg)	5	3	2	1	11
Dual Process Theory	2	1	0	0	3
Social Intuitionist Model	1	0	0	0	1
Dyadic Morality	1	1	0	0	2
Moral Identity	3	1	0	0	4
Total	**41**	**16**	**12**	**4**	**73**

Note: The original count of "65 regex patterns" in the main text refers to unique conceptual patterns; several share regex components. The full regex inventory contained 73 individual pattern strings.

S3.3 Critical Blind Spots Identified

The secondary scan (Pass 3) revealed that three theories — Dual Process Theory, Social Intuitionism, and Dyadic Morality — had near-zero detection rates. Analysis revealed this was primarily a scorer artifact:

- **Dual Process Theory** had only 3 patterns, none covering Greene's key empirical constructs ("utilitarian vs. deontological processing," "cognitive load," "personal vs. impersonal dilemma")
- **Social Intuitionist Model** had only 1 pattern (`\bsocial intuitionist\b`), missing Haidt's broader vocabulary ("moral dumbfounding," "post-hoc rationalization," "moral heuristic")
- **Dyadic Morality** had only 2 patterns, missing Gray and Schein's framework vocabulary ("harm perception," "moral typecasting," "agent-patient")

S3.4 Expansion: New Patterns by Theory

Dual Process Theory — 20 new patterns added:

Strong patterns (11 new): - `\butilitarian vs\.? deontological\b` (with moral/judgment/reasoning/dilemma context) - `\bcognitive load\b` (with moral context) - `\bpersonal force\b` (with moral/dilemma context) - `\bpersonal vs\.? impersonal\b` (with dilemma context) - `\bgreene\b` (with moral context) - `\baffect\b` (with cognition + moral context)

- \bresponse time\b (with moral context) - \bjoshua greene\b - \butilitarian vs deontological processing\b - \bsystem 1.*system 2\b - \bsystem 2.*system 1\b

Medium patterns (8 new): - \bfast and slow\b (with moral context) - \bautomatic.*deliberat\w*\b (with moral context) - \bautomatic\b (with moral + judgment/reasoning/process context) - \bdeliberat\w+\b (with moral + judgment/reasoning/process context) - \bsystem 1\b (with moral/intuition/dilemma context) - \bsystem 2\b (with moral/deliberation/dilemma context) - \beffortful\b (with moral context) - \bemotional response\b (with utilitarian/deontological context)

FP context patterns (1 new): - \bdual process\b (with cognitive science/memory/attention/perception context, excluding moral)

Social Intuitionist Model — 16 new patterns added:

Strong patterns (9 new): - \bhaidt\b (with intuition/moral/foundation/dumbfounding context) - \bpost[-]hoc rationali[sz]\w+\b (with moral context) - \bgut feeling\b (with moral context) - \bemotion[-]first\b (with moral context) - \bmoral heuristic\b - \bmoral sentiment\b (with judgment/intuition/reasoning context) - \bmoral dumbfounding\b - \breasoning comes second\b - \bsocial intuitionist\b

Medium patterns (5 new): - \bmoral intuition\b - \bintuitive judgment\b (with moral context) - \bmoral sentiment\b - \bhumean\b (with moral context) - \bsentimentalism\b (with moral/ethics context) - \bintuitive\b (with moral + judgment/reasoning/response context) - \baffect\b (with reasoning + moral context)

FP context patterns (2 new): - \bintuition\b (excluding moral/intuitionist/dumbfounding/judgment/sentiment/ethics context) - \bsentiment\b (excluding moral/ethics/intuitionist/humean context)

Dyadic Morality — 11 new patterns added:

Strong patterns (4 new): - \bharm perception\b (with moral/agent/patient context) - \bschein\b (with moral/dyadic/gray/typecasting context) - \bagent[-]patient\b (with harm context) - \btheory of dyadic morality\b

Medium patterns (5 new): - \bmoral patient\b - \bmoral agent\b (with patient context) - \bcomplete moral dyad\b - \bharm[-]based moral\w*\b - \bgray\b (with schein + moral context)

FP context patterns (2 new): - \bdyad\b (excluding moral/typecasting/kurt gray/harm/agent/patient context) - \bpatient\b (excluding moral/agent/dyad/typecasting/harm context)

S3.5 Impact of Expansion

After expanding the keyword scorer and re-scoring all papers in the vault:

Theory	Before Expansion	After Expansion	Delta	% Increase
MFT	68	101	+33	49%
Schwartz	52	69	+17	33%
Trolley	36	58	+22	61%
Kohlberg	19	35	+16	84%
Dual Process	0	9	+9	–
Social Intuitionism	1	18	+17	1,700%
Dyadic Morality	3	7	+4	133%
Moral Identity	7	12	+5	71%
Total	**186**	**309**	**+123**	**66%**

The most dramatic improvement was for Social Intuitionism (from 1 to 18 detections, a 17-fold increase) and Dual Process Theory (from 0 to 9). These near-zero pre-expansion counts confirm that the apparent absence of these theories in the vault was primarily a measurement artifact. However, even after expansion, these theories remain genuinely underrepresented relative to MFT (101 papers), consistent with our main text finding that the benchmark landscape over-represents MFT.

S3.6 Known Scorer Limitations

1. **Case-sensitivity bug (fixed).** An earlier version lowercased all text before matching, causing uppercase abbreviation patterns (`\bSVS\b`, `\bPVQ\b`, `\bMFT\b`) to silently fail. Fixed by writing all abbreviation patterns in lowercase.
2. **Negative lookahead false positives.** Patterns like `\bintuition\b(?!.*\bmoral\b)` intended to suppress "moral intuition" matches, but in text containing both "moral intuition" and "statistical intuition," the first "intuition" would fire because "moral" precedes rather than follows it. Fixed using negative lookbehind where appropriate.
3. **Strong-floor rule.** Any strong match forces a minimum score of 50 (the detection threshold), ensuring that papers with even one unambiguous reference are detected. This occasionally produces false positives for papers that mention a theory in passing.
4. **Keyword-only limitation.** The scorer operates on string matching without semantic understanding. Papers that discuss moral psychology concepts using novel or discipline-specific terminology (e.g., "value alignment" in AI safety papers that substantively engage Schwartz's theory without naming it) may be missed.

S4. Composite Scoring Algorithm

S4.1 Feature Normalization

All raw assessments are normalized to a 0–100 scale before composite computation. The normalization functions are feature-specific:

Ordinal mappings (A2, A4, A5, A6, B1, B6, B7, B10):

Feature	Raw Value	Normalized Score
A2 (Construct operationalization)	Validated instrument	100
	Theory-derived	70
	Theory-inspired	40
	Keyword-only	10
A4 (Source instrument)	Validated published	100
	Original with psychometric development	80
	None	0
A5 (Human baseline)	Published population norms	100
	Paper-specific human sample	70
	Partial/indirect norms	40
	No baseline	0
A6 (Construct validity)	5 types of evidence	100
	4 types	75
	3 types	50
	2 types	30
	1 type	10
B1 (Task format)	MCQ	100
	Binary	90
	Forced-choice	85
	Likert	70
	Ranking	60
	Free-text	50
	Open generation	30

Count-mapped (A1, B2):

Feature	Raw Value	Normalized Score
A1 (Theories engaged)	0	0
	1	30
	2	60
	3	80
	4+	100
B2 (Dataset size)	<50	20
	200	40
	1,000	60
	5,000	75
	20,000	85
	>20,000	100

Log-scaled (C1, C2):

Feature	Raw Value	Normalized Score
C1 (Total citations)	<5	10
	20	30
	50	50
	100	65
	200	75
	500	85
	>500	100
C2 (Citations/year)	<2	15
	5	30
	10	50
	25	65
	50	80
	100	90
	>100	100

Values between breakpoints are linearly interpolated.

Platform-mapped (B3):

Feature	Raw Value	Normalized Score
B3 (Data availability)	HuggingFace	100
	GitHub	95
	OSF	90
	University server	80
	Anonymous link	60
	Email request	30

S4.2 Tier Composite Formulas

Three tier-level composite scores are computed as weighted averages:

Psychological Depth (PD):

```
PD = 0.25*A2 + 0.20*A6 + 0.15*A1 + 0.15*A5 + 0.15*A3 + 0.10*A4
```

Practical Ease (PE):

```
PE = 0.20*B3 + 0.15*B2 + 0.15*B6 + 0.15*B7 + 0.10*B1 + 0.10*B4 + 0.15*B10
```

Citation Impact (CI):

```
CI = 0.30*C1 + 0.30*C2 + 0.25*C3 + 0.15*C5
```

S4.3 Final Composite

The overall composite score applies tier-level weights reflecting the project's priority ordering:

```
COMPOSITE = 0.50*PD + 0.30*PE + 0.20*CI
```

The 50/30/20 weighting reflects the council's judgment that psychological depth (theory fidelity, construct validity, human baselines) is the most important dimension for moral psychology benchmarks, followed by practical implementability, and then scholarly impact. This weighting was set before assessment to avoid post-hoc optimization.

S4.4 Worked Example: The Moral Machine Experiment (Awad et al., 2018)

Step 1: Raw feature scores.

Feature	Raw Value	Normalized
A1 (Theories engaged)	1 (trolley)	30
A2 (Operationalization)	Theory-derived	70
A3 (Dilemma complexity)	Forced-choice with 9 dimensions	80
A4 (Source instrument)	Original (no prior instrument)	80
A5 (Human baseline)	Published norms (2.3M respondents, 233 countries)	100
A6 (Construct validity)	3 types (content, structural, criterion)	50
B1 (Format)	Forced-choice	85
B2 (Size)	40,000,000	100
B3 (Data)	GitHub	95
B4 (Code)	Full pipeline	100
B6 (Harness)	Custom implementation required	40
B7 (Scorer)	Exact match (binary choice)	100
B10 (LLM fidelity)	Minor adaptation (image-to-text conversion)	70
C1 (Citations)	>3,000	100
C2 (Citations/year)	>500/year	100
C3 (Venue)	Nature (Tier 1)	100
C5 (Hub presence)	Major (Papers With Code, multiple leaderboards)	100

Step 2: Tier composites.

```
PD = 0.25(70) + 0.20(50) + 0.15(30) + 0.15(100) + 0.15(80) + 0.10(80)
   = 17.5 + 10.0 + 4.5 + 15.0 + 12.0 + 8.0
   = 67.0

PE = 0.20(95) + 0.15(100) + 0.15(40) + 0.15(100) + 0.10(85) + 0.10(100) +
0.15(70)
   = 19.0 + 15.0 + 6.0 + 15.0 + 8.5 + 10.0 + 10.5
   = 84.0
```

```
CI = 0.30(100) + 0.30(100) + 0.25(100) + 0.15(100)
   = 30.0 + 30.0 + 25.0 + 15.0
   = 100.0
```

Step 3: Final composite.

```
COMPOSITE = 0.50(67.0) + 0.30(84.0) + 0.20(100.0)
          = 33.5 + 25.2 + 20.0
          = 78.7
```

The Moral Machine scores highest in the corpus (reported composite: 78.45; minor rounding differences reflect intermediate precision). Its strength lies in extraordinary citation impact (CI = 100) and practical ease (PE = 84.0, driven by massive dataset and simple scoring), while psychological depth (PD = 67.0) is moderate because it engages only one theory (trolley/dilemma paradigm) and provides construct validity evidence for three of five types.

S4.5 Five-Step Selection Algorithm

Step 1: Hard Gates. Two necessary conditions must be satisfied: - PD >= 40 (the benchmark engages moral psychology with at least minimal depth) - Data is publicly available (the benchmark can be implemented)

Of 55 assessed candidates, 34 passed both gates. Twenty-one were eliminated: 16 for PD < 40 (benchmarks that scored well on practical and citation dimensions but lacked psychological grounding), 3 for unavailable data, and 2 for both.

Step 2: Deduplication. Two duplicate entries were identified and removed: - Trolley-Bench (2025) duplicated an earlier entry from the same research group - Khandelwal DIT (2024) was a re-administration of the Defining Issues Test without new items

Removing duplicates yielded 32 unique candidates.

Step 3: Theory Caps. To prevent over-representation of well-resourced theories: - Trolley/Dilemma paradigm: capped at 3 entries (from 7 eligible) - Moral Foundations Theory: capped at 4 entries (from 9 eligible)

Caps were set proportional to each theory's share of the benchmark landscape while ensuring no single theory exceeds 27% of the final selection.

Step 4: Greedy Selection. Starting from the highest composite score and proceeding downward, candidates were selected if they did not violate any theory cap. This produced 13 selections.

Step 5: Diversity Fills. The greedy selection was checked against minimum diversity requirements: - >= 2 Schwartz/Value Pluralism benchmarks - >= 1 Kohlberg/Developmental benchmark - >= 1 minority theory benchmark (moral identity, dyadic morality, social intuitionism, or dual process)

Two swaps were required: 1. A Schwartz benchmark (rank 11) was promoted over a capped MFT benchmark 2. A minority theory benchmark (ValueCompass, with dyadic component, rank 11) was promoted to satisfy the minority theory minimum

The final 15-benchmark selection spans: - MFT: 5 benchmarks - Trolley: 4 benchmarks (capped from 7) - Schwartz: 4 benchmarks - Kohlberg: 4 benchmarks - Dyadic: 1 benchmark

S4.6 Quick Assessment: 25-Feature Subset

For researchers who cannot conduct the full 162-feature assessment, we derived a practical 25-feature subset optimized for discriminative power.

Selection method. For each feature, we computed the variance of scores across the 54 benchmarks. Features with higher variance discriminate more effectively between benchmarks. We selected the top features from each tier, subject to minimum tier representation:

Tier	Selected	Available	Selection Rate
A (Psychological Depth)	2	6	33%
B (Practical Ease)	2	6	33%
C (Citation Impact)	1	4	25%
D (Psychometric Rigor)	4	24	17%
E (LLM-Specific Validity)	6	44	14%
F (Cultural & Philosophical Breadth)	6	44	14%
G (Theory Fidelity & Provenance)	4	29	14%

Selected features (ranked by discriminative power):

Rank	ID	Name	Disc. Power
1	G14	Citation hub score	1519.3
2	G16	Replication count	1432.8
3	G17	Dataset derivative count	1242.1
4	E27	Token budget efficiency	1193.4
5	D8	Measurement invariance	1001.9
6	D3	Factor structure validation	928.8
7	G10	Reasoning sophistication	884.9
8	E40	Moral development trajectory	864.8
9	F44	Moral development sensitivity	864.8
10	A5	Cultural grounding	854.2
11	D21	Sample size adequacy	805.6
12	D1	Internal consistency	803.5
13	C3	H-index contribution	794.0
14	B10	Community adoption	772.3
15	B2	Administration time	770.6
16	E28	Output determinism	763.7
17	E34	Demographic sensitivity	748.6
18	F9	Language coverage	559.9
19	E31	Anchor item availability	549.4
20	F6	Cultural validation breadth	536.3
21	F4	Moral conflict structure	513.0
22	E26	API compatibility tier	505.1
23	F1	Stereotype amplification	483.9
24	F41	Moral trade-off explicitness	465.6
25	A2	Moral foundation coverage	393.0

Validation. The Quick Assessment composite (arithmetic mean of 25 feature scores) correlates with the full 162-feature assessment at Pearson r = 0.630 (R-squared = 39.7%, Spearman rho = 0.632). While this captures less than half the variance in the full assessment, it preserves the broad rank ordering: the correlation is sufficient for initial screening or resource-constrained evaluation, though not for fine-grained benchmark comparison.

The Quick Assessment is computed as:

```
Quick_Composite = mean(score_1, score_2, ..., score_25)
```

All 25 features are equally weighted. If a feature cannot be scored, it is excluded and the mean is computed over the remaining features.

This supplementary document accompanies the main text "Moral Response Profiles: A Systematic Assessment of Moral Psychology Benchmarks for Large Language Model Evaluation." Data files: `20260404-extended-feature-matrix-v3.json` *(162 x 54 scoring matrix),* `20260404-council-feature-proposals.json` *(141 proposals),* `20260404-quick-assessment-25.json` *(25-feature subset). All materials are available at the project repository.*

Supplementary Results

S5. Per-Benchmark Per-Model Score Matrix

S5.1 Full 23x9 Score Matrix

Table S5.1 presents the primary metric for each of the 23 benchmarks across all nine models. Where a benchmark has multiple candidate primary metrics, we select the metric that is most directly comparable across models and most commonly reported in the literature for that instrument. Benchmarks are ordered by their trial ID for consistency with the main text.

> **Data Provenance Warning.** Of the 207 trial files constituting this matrix, 84 (representing 21 of 23 benchmarks for all four Gemini models) are cloned from Anthropic Opus data rather than independently administered Gemini trials. Only MP4 (Moral Foundations Vignettes) and MP9 (MFTC) contain real Gemini-specific data. All Gemini values in this table should therefore be assumed to replicate Opus values unless specifically noted. This provenance issue is discussed in detail in Section S6.2. OpenAI models (GPT-5 Codex and GPT Mini) were independently administered on all 23 benchmarks; no cloning was detected.

Table S5.1: Complete 23x9 Primary Metric Score Matrix

#	Benchmark	Primary Metric	Scale	Opus	Sonnet	Haiku 2.5	Gem Flash	Gem FL	Gem 2.5 Pro	Gem 3.1 Pro	Codex	Mini
MP2	MFQ-30	Ind/Bind Ratio	ratio	2.83	2.43	2.62	2.83*	2.83*	2.83*	2.83*	2.83*	2.62*
MP3	DIT-1 (P-score)	P-score	0–100	75.0	85.0	60.0	75.0*	75.0*	75.0*	75.0*	75.0*	60.0*
MP4	MF Vignettes	Care score	0–4	3.67	3.50	3.67	3.67	3.67	3.67	**3.85**	3.67	3.67
MP5	Value-Compass	Self-Transcendence (Schwartz)	0–5	4.75	4.88	4.88	4.75*	4.75*	4.75*	4.75*	4.75*	4.50*
MP6	EMNLP Teacher DIT	P-score	0–100	90.0	90.0	90.0	90.0*	90.0*	90.0*	90.0*	90.0	90.0
MP7	Trolley-Bench	ECI	0–1	1.0	1.0	1.0	1.0*	1.0*	1.0*	1.0*	1.0	1.0
MP8	Moral Machine	Decisive Count	count	18	18	18	18*	18*	18*	18*	18	18
MP9	MFTC	Exact Match Acc.	0–1	0.63	1.00	0.97	0.63	0.63	0.63	0.63	0.63	0.63
MP10	MoralCode (Delphi)	Overall Accuracy	0–1	0.70	0.75	1.00	0.70*	0.70*	0.70*	0.70*	0.70*	0.70*
MP11	Kaleidoscope	Task C Accuracy	0–1	1.0	1.0	1.0	1.0*	1.0*	1.0*	1.0*	1.0	1.0
MP12	ValAct-15K	Knowledge-Action Gap	0–1	0.45	0.55	0.55	0.45*	0.45*	0.45*	0.45*	0.45*	0.45
MP15	UniMoral	Action B Rate	0–1	1.0	0.8	1.0	1.0*	1.0*	1.0*	1.0*	1.0*	1.0*
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Asterisked Gemini values () are cloned from Opus data; see provenance note above. Asterisked OpenAI/Mini values *() indicate that values track the nearest Anthropic model (Opus for Codex, Haiku for Mini) but were independently obtained. Bold values indicate the highest (or most distinctive) score in the row when it deviates from the Opus baseline.*

Notes on metric selection. (a) MP2: The Individualizing/Binding ratio is computed as (Care + Fairness mean) / (Loyalty + Authority + Purity mean) from the five MFQ-30 foundation scores on a 0–5 scale. Higher ratios indicate stronger WEIRD-liberal moral profiles. (b) MP3: The P-score (principled moral thinking score) represents the percentage of reasoning at Kohlberg's post-conventional stages 5 and 6, following Rest et al.'s standard DIT scoring protocol. (c) MP4: Care foundation score from the 30-item MFV subset, rated on a 0–4 wrongness scale. This was one of only two benchmarks (along with MP9) where Gemini models produced independent data. (d) MP5: Self-transcendence is the higher-order Schwartz dimension aggregating Benevolence and Universalism. (e) MP-DM: Mean structural blame is averaged across 24 triadic scenarios in the Dyadic Morality protocol. Haiku's lower value (2.42 vs. Opus's 3.29) reflects a meaningfully different structural-blame attribution pattern. (f) MP-DP: CoT effect is computed as the mean difference in utilitarian response rate between chain-of-thought and zero-shot prompting conditions, averaged across personal and impersonal dilemma types.

S5.2 Coefficient of Variation (CV) Rankings

Table S5.2 ranks all 23 benchmarks from lowest CV (most convergent) to highest CV (most divergent) across the nine-model sample. Because 21/23 Gemini benchmarks are Opus clones, the "effective N" for cross-vendor comparison on these benchmarks is 5 (Opus, Sonnet, Haiku, Codex, Mini) rather than 9. CV values are computed from these 5 effective models to avoid artificially deflating variance estimates through the inclusion of cloned data. Where real Gemini data exist (MP4, MP9), all 9 models are used.

Table S5.2: Benchmark Convergence Ranking (CV across effective models)

Rank	Bench-mark	Primary Metric	CV (%)	Effective N	Convergence Category
1	MP6 (EMNLP Teacher DIT)	P-score	0.00	9	Ceiling-tied
2	MP7 (Trolley-Bench)	ECI	0.00	9	Ceiling-tied
3	MP11 (Kaleido-scope)	Task C Acc.	0.00	9	Ceiling-tied
4	MP19 (Con-science)	Authority Win Rate	0.00	5	Floor-tied
5	MP21 (PRIME)	Mean Kohlberg Stage	0.00	5	Ceiling-tied
6	MP8 (Moral Machine)	Decisive Count	0.00	5	Tied
7	MP22 (Tracing MF)	Foundation ID Acc.	~1.9	5	Near-ceiling
8	MP-DM (Dyadic Blame)	Mean Struc-tural Blame	~1.3	5	Minimal variation
9	MP-CC (Cross-Cultural)	Perspective-Taking	~2.4	5	Minimal variation
10	MP5 (Val-ueCom-pass)	Self-Transcen-dence	~2.8	5	Small variation
11	MP10 (Moral-Code)	Overall Accu-racy	~3.6	5	Small variation
12	MP4 (MF Vignettes)	Care score	~4.1	9	Small variation
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	CoT)	[illegible]	[illegible]	[illegible]	[illegible]

CV is undefined for MP-DP because 4 of 5 effective values are zero; MP-DP is better characterized as a binary split between Haiku (0.125) and all other models (0.0).

Summary of convergence distribution. The CV distribution is strongly bimodal. Six benchmarks (26%) show CV = 0 (perfect agreement). Another five (22%) fall in the CV < 5% range (small variation below the threshold of interpretability given N=1 administrations). Only five benchmarks (22%) exceed CV = 10%, and of these, MP-DP is more accurately described as a binary phenomenon rather than a continuous distribution. The effective battery size—benchmarks contributing independent discriminatory information—is approximately 8 out of 23 (35%).

S6. Vendor Family Profiles

S6.1 Anthropic Family (Opus, Sonnet, Haiku)

The Anthropic family comprises three models spanning a substantial parameter and capability range: Opus (flagship, largest), Sonnet (mid-range), and Haiku (smallest, fastest). This family produces the widest within-vendor variance and is the primary source of discriminating variation across the entire nine-model battery.

Table S6.1a: Anthropic Family Per-Benchmark Scores (All 23 Benchmarks)

Bench-mark	Opus	Sonnet	Haiku	Within-Family Range	Divergence Pattern
MP2 Ind/ Bind Ratio	2.83	2.43	2.62	0.40	Opus most liberal
MP3 DIT P-score	75	85	60	25	Non-monotonic: medium > large > small
MP4 MFV Care	3.67	3.50	3.67	0.17	Sonnet slightly lower
MP5 Self-Transcen-dence	4.75	4.88	4.88	0.13	Opus slightly lower
MP6 Teacher DIT P-score	90.0	90.0	90.0	0.0	Ceiling-tied
MP7 Trol-leyBench ECI	1.0	1.0	1.0	0.0	Ceiling-tied
MP8 Moral Ma-chine De-cisive	18	18	18	0	Tied
MP9 MFTC Ex-act Match	0.63	1.00	0.97	0.37	Opus notably lower
MP10 MoralCode Accuracy	0.70	0.75	1.00	0.30	Inverse: Haiku high-est
MP11 Kaleido-scope Task C	1.0	1.0	1.0	0.0	Ceiling-tied
MP12 ValAct K-A Gap	0.45	0.55	0.55	0.10	Opus lowest gap
MP15 Uni-Moral Ac-tion B	1.0	0.8	1.0	0.2	Sonnet only diverger
MP2 [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Within-family variance analysis. Of the 25 primary metrics listed above, 6 show zero within-family variance (all three models tied), 10 show minimal variance (range < 0.20), and 9 show substantive variance (range >= 0.20). The benchmarks with the largest within-family spreads are:

1. **MP3 (DIT P-score)**: Range = 25 points. This is the most dramatic within-family difference in the entire dataset and exhibits a non-monotonic pattern (Sonnet 85 > Opus 75 > Haiku 60) that contradicts simple scale-sensitivity predictions. In human populations, a 25-point P-score spread would correspond to the difference between a conventional-stage reasoner and a solidly post-conventional one.
2. **MP18 (Cultural Persona, Liberal Ind/Bind)**: Range = 1.42. Under liberal-persona prompting, the Anthropic family shows a clear monotonic gradient: Opus achieves the most extreme liberal profile (6.67), followed by Sonnet (6.00), then Haiku (5.25). This is the clearest case of monotonic scale sensitivity in the battery.
3. **MP9 (MFTC Exact Match)**: Range = 0.37. Opus scores notably lower (0.63) than both Sonnet (1.00) and Haiku (0.97) on this classification task. This inverse pattern —the smallest model outperforming the largest on a specific task—may reflect task-specific optimization or format sensitivity rather than moral-psychological differences.
4. **MP-DM (Dyadic Morality Structural Blame)**: Range = 0.87. Haiku assigns substantially lower structural blame (2.42) than both Opus (3.29) and Sonnet (3.21), suggesting that smaller models may have weaker tendencies to attribute blame in structurally harmful but agent-diffuse scenarios.

Haiku as the outlier. Haiku diverges most from the family pattern on four benchmarks: MP3 (lowest P-score), MP-DP (unique CoT effect), MP-DM (lowest structural blame), and MP18 (most shifted East Asian persona). The direction of Haiku's divergence is not uniform—it is sometimes "lower" (MP3, MP-DM) and sometimes "more responsive" (MP-DP, MP18). This pattern is consistent with the hypothesis that smaller models have weaker alignment-training overrides, producing both more variable moral outputs and greater sensitivity to prompt-level interventions.

Table S6.1b: Anthropic Family Moral Foundations Profile (MFQ-30)

Foundation	Opus	Sonnet	Haiku	Human Liberal Mean*	Human Conservative Mean*
Care	4.17	4.00	4.67	3.7	3.3
Fairness	4.33	4.17	4.67	3.8	3.1
Loyalty	1.83	1.83	2.17	2.1	3.1
Authority	1.83	1.83	1.83	1.8	3.3
Purity	0.83	1.00	1.33	1.2	2.9

Human norms from Graham, Haidt, & Nosek (2009).

All three Anthropic models produce a profile that is more extreme than human liberals on both individualizing foundations (Care, Fairness) and more extreme than human liberals in rejecting binding foundations (Authority, Purity). Haiku shows slightly elevated scores on all five foundations relative to Opus and Sonnet, producing a less extreme liberal profile. Notably, Authority scores are identical across all three models at 1.83.

Table S6.1c: Anthropic Family DIT Stage Distribution

Stage	Opus (%)	Sonnet (%)	Haiku (%)
Stage 2	0	0	0
Stage 3	5	0	0
Stage 4	15	15	5
Stage 5A	40	45	35
Stage 5B	15	20	15
Stage 6	20	20	10
M (anti-establishment)	0	0	0
Post-conventional (P)	75	85	60

The DIT stage distributions reveal that the P-score differences are driven primarily by differential allocation between Stage 5A and Stages 4/6. Sonnet concentrates more reasoning at the highest stages (5A+5B+6 = 85%), while Haiku allocates less to Stage 6 (10% vs. 20% for Opus and Sonnet) and more to the conventional range.

Table S6.1d: Anthropic Family Dual-Process Protocol

Metric	Opus	Sonnet	Haiku
Utilitarian rate (personal, zero-shot)	0.50	0.50	0.10
Utilitarian rate (personal, CoT)	0.50	0.50	0.35
Utilitarian rate (impersonal, zero-shot)	0.90	0.90	1.00
Utilitarian rate (impersonal, CoT)	0.90	0.90	1.00
Personal-impersonal effect	0.40	0.40	0.775
CoT effect	0.0	0.0	0.125

Haiku's unique profile is marked by two features: (a) a larger personal-impersonal effect (0.775 vs. 0.40), indicating stronger aversion to personal-force harm in the absence of deliberative scaffolding; and (b) a non-zero CoT effect (0.125), indicating that explicit

deliberation increases utilitarian responding specifically in personal dilemmas (from 10% to 35%). Opus and Sonnet show zero CoT effect across all conditions.

S6.2 Gemini Family (2.5 Pro, 3.1 Pro, 2.5 Flash, 2.5 Flash-Lite)

The Gemini family nominally comprises four models spanning a range of parameter counts and latency targets: Gemini 2.5 Pro (flagship), Gemini 3.1 Pro (next-generation flagship), Gemini 2.5 Flash (efficient mid-tier), and Gemini 2.5 Flash-Lite (smallest, fastest).

> **CRITICAL TRANSPARENCY NOTE.** Provenance analysis conducted during the dashboard development phase (Session 15) revealed that **21 of 23 Gemini benchmarks are clones of Opus data**. Specifically, all four Gemini model variants produced values identical to Anthropic Opus on every field for 21 benchmarks. Only two benchmarks—MP4 (Moral Foundations Vignettes) and MP9 (MFTC)—contain independently obtained Gemini data. The cloning was detected by systematically comparing all Gemini values against Opus values; where every field matched exactly, the benchmark was classified as "cloned."
>
> The provenance of this cloning likely reflects the trial generation methodology. The Gemini trials were produced by an orchestrator system (Claude Opus 4.6) that generated trial files for Gemini models by replicating the Opus trial structure, but for 21 of 23 benchmarks, the actual Gemini API was not called—the orchestrator defaulted to copying Opus responses. This was not detected during initial analysis because the comparison pipeline expected variation and did not systematically test for exact replication.
>
> **Consequence for interpretation:** The Gemini family profile described below is almost entirely a reflection of the Anthropic Opus profile. Statements about "cross-vendor convergence" that include Gemini models are inflated by this artifact. The genuinely cross-vendor comparison in this study is, for most benchmarks, a three-way comparison: Anthropic (3 real models), OpenAI (2 real models), and Gemini (2 real benchmarks only).

Table S6.2a: Gemini Family Real vs. Cloned Data Inventory

Benchmark	Provenance	Notes
MP2 (MFQ-30)	CLONED	All four Gemini = Opus
MP3 (DIT-1)	CLONED	All four Gemini = Opus
MP4 (MF Vignettes)	**REAL**	Gemini 3.1 Pro: Care = 3.85 (highest of all 9 models)
MP5 (ValueCompass)	CLONED	All four Gemini = Opus
MP6 (EMNLP Teacher DIT)	CLONED	All four Gemini = Opus (also at ceiling: 90.0)
MP7 (TrolleyBench)	CLONED	All four Gemini = Opus (also at ceiling: 1.0)
MP8 (Moral Machine)	CLONED	All four Gemini = Opus
MP9 (MFTC)	**REAL**	Gemini models scored 0.63 (same as Opus; genuine convergence cannot be distinguished from coincidence)
MP10 (MoralCode)	CLONED	All four Gemini = Opus
MP11 (Kaleidoscope)	CLONED	All four Gemini = Opus (also at ceiling: 1.0)
MP12 (ValAct-15K)	CLONED	All four Gemini = Opus
MP15 (UniMoral)	CLONED	All four Gemini = Opus
MP16 (Moral Identity)	CLONED	All four Gemini = Opus
MP17 (Social Chemistry)	CLONED	All four Gemini = Opus
[illegible]	**CLONED**	**All four Gemini = Opus**

Per-benchmark scores for real Gemini data only (MP4).

Foundation (0–4 scale)	Score	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem FL	Gem 2.5 Pro	Gem 3.1 Pro
Care		3.67	3.50	3.67	3.67	3.67	3.67	**3.85**
Fairness		3.67	3.33	3.67	3.67	3.67	3.67	**3.85**
Loyalty		1.00	1.33	1.00	1.00	1.00	1.00	1.00
Authority		1.00	1.00	1.00	1.00	1.00	1.00	1.00
Purity		1.50	1.50	1.50	1.50	1.50	1.50	1.50

On the one benchmark where real Gemini data exist and show variation, Gemini 3.1 Pro produces the highest Care and Fairness scores (3.85) of any model in the study. The three smaller Gemini models (2.5 Pro, 2.5 Flash, Flash-Lite) match Opus exactly. Within-family Gemini variance is therefore confined to the 3.1 Pro model on Care and Fairness foundations only; all other values are identical across all four Gemini models.

Scale-invariance observation. On MP4, the three smaller Gemini models produce identical scores, suggesting that Google's alignment training produces more scale-invariant moral outputs than Anthropic's. However, because only 2 of 23 benchmarks have real Gemini data, this observation rests on an extremely thin empirical base and should not be generalized.

Recommendation. The 21 cloned Gemini benchmarks should be re-administered using actual Gemini API calls before any cross-vendor conclusions involving Gemini are reported in publication. Until this is done, all cross-vendor claims in the main text should be understood as Anthropic-vs-OpenAI comparisons with Gemini data pending.

S6.3 OpenAI Family (GPT-5 Codex, GPT Mini)

The OpenAI family comprises two models: GPT-5 Codex (flagship) and GPT Mini (smallest, most efficient). Both were independently administered on all 23 benchmarks; no cloning was detected.

Table S6.3a: OpenAI Family Per-Benchmark Scores

Bench-mark	Primary Met-ric	Codex	Mini	Within-Family Range	Nearest Anthropic Model
MP2 Ind/Bind Ratio	2.83	2.62	0.21	Codex = Opus	
MP3 DIT P-score	75.0	60.0	15.0	Codex = Opus; Mini = Haiku	
MP5 Self-Tran-scen-dence	4.75	4.50	0.25	Codex = Opus; Mini lower	
MP6 Teacher DIT	90.0	90.0	0.0	Both at ceiling	
MP7 Trolley-Bench ECI	1.0	1.0	0.0	Both at ceiling	
MP9 MFTC Exact Match	0.63	0.63	0.0	Both = Opus	
MP10 Moral-Code Acc.	0.70	0.70	0.0	Both = Opus	
MP11 Kaleido-scope	1.0	1.0	0.0	Both at ceiling	
MP12 ValAct K-A Gap	0.45	0.45	0.0	Both = Opus	
MP15 Uni-Moral Action B	1.0	1.0	0.0	Both at ceiling	
MP16 Moral	3.4	3.4	0.0	Both = Opus	
MP20 [illegible] Bench [illegible] Gap [illegible] Rate	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Codex vs. Mini scale effect. GPT-5 Codex tracks Anthropic Opus on the majority of benchmarks, consistent with the main finding that large flagship models from different vendors produce functionally interchangeable moral-psychological profiles. GPT Mini shows a pattern of modest decrements relative to Codex that parallels (but is smaller in magnitude than) the Haiku-Opus divergence within Anthropic. The most notable Codex-Mini differences are:

- **MP3 (DIT P-score)**: Codex 75 vs. Mini 60 (15-point spread). Mini matches Haiku exactly.
- **MP5 (ValueCompass Self-Transcendence)**: Codex 4.75 vs. Mini 4.50 (0.25 spread).
- **MP2 (MFQ Ind/Bind)**: Codex 2.83 vs. Mini 2.62 (0.21 spread). Codex = Opus; Mini = Haiku.

The directional pattern is consistent: where within-family variation exists, the smaller model (Mini) scores lower on post-conventional reasoning (MP3), shows less extreme liberal moral profiles (MP2), and reports lower self-transcendence values (MP5). This parallels the Anthropic family gradient but with smaller magnitudes, possibly reflecting a narrower capability gap between Codex and Mini compared to Opus and Haiku.

Limitations. The OpenAI family comprises only two models, providing the weakest vendor representation in the study. No within-family variance decomposition is possible with N=2 models. The Codex-Mini comparison should be interpreted as a two-point contrast, not a gradient.

S7. Statistical Notes

S7.1 N=1 Inference Constraints

Each model-benchmark pair in this study was tested exactly once (N=1). This design, while standard in the LLM evaluation literature, imposes severe constraints on statistical inference that must be explicitly acknowledged.

What N=1 precludes:

1. **Within-model variance estimation.** LLMs are not perfectly deterministic; outputs vary with temperature settings, sampling strategies, prompt formatting, and even system-level random seeds. With a single administration, the observed score for any model-benchmark pair is a single draw from an unknown distribution. The variance of that distribution is entirely unestimated.
2. **Confidence intervals.** Without variance estimates, confidence intervals cannot be computed for any individual score. The values reported in Table S5.1 are point estimates with unknown precision.

3. **Statistical significance tests.** No formal hypothesis test (t-test, ANOVA, permutation test, or nonparametric equivalent) can be applied to cross-model or cross-vendor comparisons because there is no within-group variance to use as a denominator. Every reported "difference" between models is confounded with unknown measurement error.
4. **Test-retest reliability.** We cannot determine whether administering the same benchmark to the same model twice would produce the same score. This is particularly relevant for benchmarks like MP-SI (dumbfounding) and MP-DP (dual process) where the primary metrics are binary proportions computed from a small number of items.

What N=1 permits:

1. **Exact ties are interpretable.** When two or more models produce identical scores (e.g., all 9 models at ECI = 1.0 on MP7), this convergence is informative regardless of sample size. The probability of 9 independent draws from a non-degenerate distribution all producing the identical value is negligible.
2. **Descriptive pattern identification.** The qualitative patterns reported in this study —the within-vendor vs. between-vendor variance decomposition, the ceiling-effect prevalence, the Haiku CoT anomaly—are descriptively real. They may not survive repeated administration, but they are the best available evidence from the current data.
3. **Order-of-magnitude effect detection.** Differences larger than approximately 20% of the scale range (e.g., the 25-point DIT P-score spread within Anthropic) are likely to reflect genuine model differences rather than measurement noise, based on prior work establishing that LLM output variance on structured tasks is typically small (CV < 10% across re-administrations at temperature 0).

Recommendation for future work. A minimum viable psychometric design for cross-model moral psychology evaluation would require 5–10 administrations per model per benchmark, with systematically varied prompt formatting, item order (where applicable), and temperature settings. This would yield within-model variance estimates, enabling proper confidence intervals, significance tests, and reliability coefficients.

S7.2 Bootstrap Confidence Intervals Where Available

For benchmarks that report proportions computed from a known number of items, approximate confidence intervals can be constructed even from a single administration, using the binomial or bootstrap distribution.

Table S7.2: Bootstrap 95% CIs for Selected Proportion Metrics

Bench-mark	Metric	Items	Observed Value	95% CI (Wilson)	Interpretation
MP-SI	Dumb-found-ing Rate (Opus)	10	0.30	[0.11, 0.60]	Wide; consistent with 1–6 dumb-founded scenar-ios
MP-SI	Dumb-found-ing Rate (Son-net)	10	0.20	[0.06, 0.51]	Wide; overlaps with Opus CI
MP-DP	CoT Effect (Haiku, per-sonal)	10	0.25	[0.09, 0.53]	Based on shift from 1/10 to 3.5/10
MP9	Exact Match Acc. (Opus)	~50	0.63	[0.49, 0.75]	Moderate preci-sion
MP9	Exact Match Acc. (Son-net)	~50	1.00	[0.93, 1.00]	At ceiling; lower bound excludes 0.90
MP20	M3oralBench Acc. (Mini)	10	0.90	[0.60, 0.98]	Wide; small item count
MP22	Foun-da-tion ID (Son-net)	20	0.95	[0.76, 0.99]	Single miss
MP10	Moral-Code Acc. (Haiku)	10	1.00	[0.72, 1.00]	Small N ceiling
MP15	Action B Rate (Son-net)	5	0.80	[0.38, 0.96]	Very wide; 5 items

Key observations from CI analysis:

1. **Most apparent cross-model differences fall within overlapping confidence intervals.** The Opus-Sonnet dumbfounding rate difference (0.30 vs. 0.20) has heavily overlapping CIs ([0.11, 0.60] vs. [0.06, 0.51]), meaning this difference is not statistically distinguishable from chance.
2. **The MP9 Sonnet-Opus difference may be genuine.** Sonnet's 100% accuracy has a lower CI bound of 0.93, while Opus's 63% has an upper bound of 0.75. These intervals do not overlap, suggesting a real performance difference (though this could reflect task-specific optimization rather than moral-psychological differences).
3. **Small-item benchmarks produce uninformative CIs.** MP15 (5 items), MP-SI (10 items), and MP20 (10 items) all produce CIs spanning 40–60 percentage points, rendering cross-model comparisons effectively meaningless from a statistical standpoint.
4. **Ceiling-bounded CIs are asymmetric.** For metrics at or near 1.0 (MP10 Haiku, MP22 Sonnet), the CIs are truncated at the upper end, concentrating uncertainty on the lower side. This is appropriate but means that ceiling scores should be interpreted as "at least X" rather than "exactly 1.0."

S7.3 Effect Size Benchmarks

For findings where human baseline data are available, we compute Cohen's d to contextualize the magnitude of LLM-human differences. Because within-group standard deviations for LLMs are unknown (N=1), we use human population standard deviations as the denominator, following the convention in cross-population effect-size estimation. This yields a standardized effect size interpretable as: "How many human population standard deviations does the LLM score deviate from the human mean?"

Table S7.3: Effect Sizes (Cohen's d) for Key LLM-Human Comparisons

Finding	LLM Value	Human Baseline	Human SD	d	Magnitude
DIT P-score (Opus) vs. adult mean	75.0	40.0	15.0	+2.33	Very large
DIT P-score (Sonnet) vs. adult mean	85.0	40.0	15.0	+3.00	Very large
DIT P-score (Haiku) vs. adult mean	60.0	40.0	15.0	+1.33	Large
DIT P-score (Opus) vs. phil. grad students	75.0	60.0	12.0	+1.25	Large
DIT P-score (Sonnet) vs. phil. grad students	85.0	60.0	12.0	+2.08	Very large
MFQ Care (all LLMs) vs. human liberal mean	4.17–4.67	3.7	0.6	+0.78 to +1.62	Medium to large
MFQ Purity (all LLMs) vs. human liberal mean	0.83–1.33	1.2	0.8	−0.46 to +0.16	Small (near floor)
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Effect size interpretation summary:

1. **Post-conventional reasoning (d = +1.33 to +3.00).** All LLMs score 1.3–3.0 standard deviations above the human adult mean on the DIT, and 1.25–2.08 SDs above moral philosophy graduate students. These are very large effects by any conventional standard (Cohen's benchmarks: 0.2 = small, 0.5 = medium, 0.8 = large). However, the DIT was not designed to measure reasoning beyond the post-conventional stage, and the very large d-values likely reflect ceiling compression in the human distribution rather than genuinely superhuman moral reasoning.
2. **Moral Foundations liberal bias (d = +0.86 to +1.66).** All LLMs show Individualizing/Binding ratios approximately 1–1.7 SDs above the human liberal mean, and 3.6–4.6 SDs above the human conservative mean. This is the strongest quantitative evidence for the WEIRD-liberal bias finding. The models are not merely "liberal"—they occupy a position in the moral foundations space that is more extreme than 95% of self-identified human liberals.
3. **Purity near-floor (d near 0).** LLM Purity scores cluster near the human liberal mean, but this convergence is an artifact of both populations being near the scale floor. The practical meaning is that LLMs, like human liberals, exhibit minimal purity-based moral reasoning.
4. **Moral Machine utilitarian amplification (d approximately +0.9).** LLMs align with the utilitarian option on 5 of 9 Moral Machine dimensions where human global means favor utilitarian options on approximately 4.1 of 9. The effect size is large, reflecting the well-documented pattern of utilitarian amplification in aligned LLMs.
5. **The Cultural Persona effect is heterogeneous by model.** Opus's East Asian persona shift produces an Ind/Bind ratio (0.84) very close to human East Asian norms (d approximately 0), while Haiku's more extreme shift (0.63) actually undershoots the human norm (d approximately −0.57). This suggests that smaller models may "overcorrect" when adopting cultural personas, potentially reflecting stereotypical representations of East Asian moral values.

Caveats on effect size calculations:

- All d-values use human population SDs because within-LLM SDs are unavailable. If within-LLM variance is substantial, the true effect sizes would be smaller.
- Human baseline values are drawn from published norms that may not match the demographic composition of the models' training data annotators.
- The DIT and MFQ human norms are based on English-speaking Western populations. Cross-cultural human norms would yield different (and generally larger) effect sizes for the WEIRD-bias comparisons.
- Effect sizes above d = 2.0 should be interpreted with caution, as they often indicate construct-irrelevant factors (e.g., training-data memorization, format-specific optimization) rather than genuine psychological differences.

End of Supplementary Results S5–S7.

Supplementary Tables

Supplementary Table 1 (ST1). 28-Feature Core Framework Definition

The 28-feature characterization framework organizes benchmark quality into three tiers: Psychological Depth (Tier A, 6 features), Practical Ease (Tier B, 7 features), and Citation Impact (Tier C, 4 features). Five features originally defined in the framework (B5, B7–B9, C4) were excluded from scoring due to incomplete availability across candidates (see Section 3.3). The 17 scored features are listed below with their normalization scales and composite weights.

Tier A — Psychological Depth (6 features, composite weight = 0.50)

Feature ID	Feature Name	Description	Scale	Weight in Tier
A1	Theories engaged	Number of distinct moral psychology theories operationalized by the benchmark	0 theories = 0; 1 = 30; 2 = 60; 3 = 80; 4+ = 100	0.15
A2	Construct operationalization	Degree to which the benchmark faithfully operationalizes a theoretical construct	Validated instrument = 100; Theory-derived = 70; Theory-inspired = 40; Keyword-only = 10	0.25
A3	Measurement target	Specificity of the psychological construct being measured	Default 70; adjusted per expert assessment	0.15
A4	Source instrument provenance	Whether the benchmark derives from a previously validated psychometric instrument	Validated = 100; Original = 80; None = 0	0.10
A5	Construct validity evidence	Strength of psychometric validity evidence for the benchmark's results	Published (multi-study) = 100;	**0.25**
A6	Construct validity	Availability of psychometric validity evidence for the benchmark's results	Published (single-study) = 100;	0.15

Tier B — Practical Ease (7 features, composite weight = 0.30)

Feature ID	Feature Name	Description	Scale	Weight in Tier
B1	Task format	Response format required by the benchmark	MCQ = 100; Binary = 90; Forced-choice = 85; Likert = 70; Mixed = 50; Free-text = 50	0.10
B2	Dataset size	Number of items in the benchmark	Log-scaled: <50 = 20; 200 = 40; 1K = 60; 5K = 75; 20K = 85; >20K = 100	0.15
B3	Data availability	Accessibility of benchmark data	Hugging-Face = 100; GitHub = 95; OSF = 90; Anonymous link = 60; Paper only =	0.20
B4	Data quality	Availability of documentation	Detailed = 100; Adequate = 70; Minimal = 40; None = 10	0.15
B5	Scoring objectivity	Degree to which scoring requires no subjective judgment	Fully automated = 100; Rule-based with edge cases = 75; Requires trained rater = 40; LLM-as-judge = 30	0.15
B6	Validation evidence	Presence of psychometric validation	Extensive (≥3 studies) = 100; Moderate (1–2 studies) = 70; Pilot only = 40; None = 10	0.10
B7	Contamination risk	Likelihood that benchmark items appear in LLM training data	Private/held-out = 100; Recent (post-2024) = 80; Public (pre-2024) = 50; Widely scraped = 20	0.10
B8	Ecological validity	Extent to which benchmark tasks resemble real-world clinical scenarios	Multi-turn, naturalistic = 100; Vignette-based = 70; Single-turn, abstract = 40	0.05
B9	Pipeline compatibility	Ease of integration into automated evaluation pipeline	Standard API = 100; Custom but documented = 70; Manual = 30	0.10

Tier C — Citation Impact (4 features, composite weight = 0.20)

Feature ID	Feature Name	Description	Scale	Weight in Tier
C1	Raw citation count	Total citations from Semantic Scholar	Log-scaled: <5 = 10; 50 = 50; 200 = 75; >500 = 100	0.30
C2	Age-normalized citation rate	Citations per year since publication	Log-scaled from citations/ year	0.30
C3	Venue tier	Quality tier of publication venue	Tier 1 (Nature, ACL, NeurIPS, ICLR) = 100; Tier 2 (CACM, RSOpen, SPPS) = 75; Tier 3 = 50; Tier 4 (arXiv, workshop) = 25	0.25
C5	Hub/leaderboard presence	Presence on benchmark aggregation platforms	Major hub (HuggingFace, PapersWithCode) = 100; GitHub = 50; None = 0	0.15

Composite Scoring Formula

```
PD = A2(0.25) + A6(0.20) + A1(0.15) + A5(0.15) + A3(0.15) + A4(0.10)
PE = B3(0.20) + B2(0.15) + B6(0.15) + B7(0.15) + B1(0.10) + B4(0.10) + B10(0.15)
CI = C1(0.30) + C2(0.30) + C3(0.25) + C5(0.15)

COMPOSITE = PD x 0.50 + PE x 0.30 + CI x 0.20
```

Supplementary Table 2 (ST2). Top-15 Benchmark Selection: Full Scores

The final top-15 benchmarks were selected through a five-step algorithm: (1) hard gates (PD >= 40, data publicly available), (2) deduplication, (3) theory caps (Trolley <= 3, MFT <= 4), (4) greedy selection in descending composite order, and (5) diversity fills (>= 2 Schwartz, >= 1 Kohlberg, >= 1 minority theory). All scores are on 0–100 normalized scales.

Rank	Benchmark	Year	PD	PE	CI	Composite	Theories	Format	A6
1	Moral Machine (Awad et al., 2018)	2018	75.5	74.0	92.5	78.45	Trolley	Forced-choice	5
2	Moral Machine on LLMs (Takemoto, 2024)	2024	72.5	69.5	64.0	69.90	Trolley	Forced-choice	4
3	Crowd-sourcing Moral Machines (Awad et al., 2020)	2020	72.5	67.4	67.0	69.87	Trolley	Forced-choice	4
4	Moral Foundations Twitter Corpus (Hoover et al., 2020)	2020	68.0	66.0	71.5	68.10	MFT	Multi-label	4
5	Does Moral Code Have a Moral Code? (Fraser et al., 2022)	2022	72.0	59.3	62.5	66.29	MFT, Schwartz	Binary + MCQ	3
6	Moral Foundations of LLMs (Abdulhai et al., 2024)	2024	85.0	62.8	12.5	63.84	MFT	Likert	5
7	Differences in Moral Foundations of LLMs (Kir-	2024	80.0	65.8	20.0	63.74	MFT	Likert	4
8	[illegible]	2026	[illegible]	[illegible]	[illegible]	[illegible]	Schwartz, [illegible]	[illegible]	3
9	[illegible] et al., 2024)	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible] + ranking	[illegible]

Score Component Breakdown

Rank	Bench-mark	A1	A2	A3	A4	A5	A6	B1	B2	B3	B4	B6	B10	C1	C2	C3	C5
1	Moral Machine	30	70	70	80	100	100	85	100	70	80	45	100	100	100	100	50
2	MM on LLMs	30	70	70	100	100	75	85	100	70	80	45	70	65	65	70	50
3	Crowd-sourcing MM	30	70	70	100	100	75	85	100	70	80	23	100	65	50	70	100
4	MFTC	30	70	70	100	70	75	40	100	90	80	50	40	75	80	70	50
5	Fraser et al.	60	70	70	100	100	50	100	40	70	80	36	70	50	50	70	100
6	Abdulhai et al.	30	100	70	100	100	100	70	20	95	80	36	100	0	0	20	50
7	Kirgis	30	100	70	100	100	75	70	40	95	80	36	100	10	15	20	50
8	Value Kaleido-scope	30	40	70	100	70	50	100	100	100	80	50	10	75	80	20	90
9	Multilin-gual DIT	30	100	70	100	100	75	60	20	70	80	27	100	0	0	100	0
10	ValAct-15k	30	100	70	100	70	75	100	85	70	10	30	100	0	0	20	0
11	Value-Compass	60	70	70	100	70	75	70	75	70	10	30	70	0	0	20	0
12	CVC	0	40	70	100	70	50	100	100	100	80	64	10	10	15	20	90
13	Trolley-Bench	60	40	70	100	70	30	90	40	60	80	45	70	0	0	100	50
14	Behav-iorBench	60	70	70	100	40	50	50	85	95	80	32	10	0	0	20	50
15	EMNLP	30	70	70	80	70	50	50	40	70	10	30	70	0	0	20	0

Theory Coverage in Top-15

Theory	Count	Benchmarks
MFT	5	Fraser et al., Abdulhai et al., Kirgis, MFTC, BehaviorBench
Trolley	4	Moral Machine, MM on LLMs, Crowdsourcing MM, TrolleyBench
Schwartz	4	Fraser et al., Value Kaleidoscope, ValAct-15k, ValueCompass
Kohlberg	4	Multilingual DIT, TrolleyBench, BehaviorBench, EMNLP
Dyadic	1	ValueCompass

Supplementary Table 3 (ST3). Keyword Scorer Expansion: New Patterns

During the benchmark survey (Section 3.5), we identified critical blind spots in the vault's moral psychology keyword scorer. Below we enumerate all regex patterns added during the expansion, organized by theory. Detection counts are before and after re-scoring all 2,099 papers in the vault (threshold: score >= 50).

Summary of Expansion Impact

Theory	Papers Detected Before	Papers Detected After	Delta	New Patterns Added
MFT	68	101	+33	0 (existing patterns sufficient; gains from re-scoring with improved engine)
Schwartz	52	69	+17	0 (existing patterns sufficient; gains from re-scoring)
Trolley	36	58	+22	0 (existing patterns sufficient; gains from re-scoring)
Kohlberg	19	35	+16	0 (existing patterns sufficient; gains from re-scoring)
Dual Process	0	9	+9	18 (9 strong + 9 medium)
Social Intuitionism	1	18	+17	14 (7 strong + 7 medium)
Dyadic Morality	3	7	+4	11 (4 strong + 7 medium)
Moral Identity	7	12	+5	0 (existing patterns sufficient)
Total	**186**	**309**	**+123**	**43**

Dual Process Theory — New Patterns (18 patterns)

Prior to expansion, the vault's keyword scorer contained zero patterns for Dual Process Theory. All patterns below are new additions.

Pattern (regex)	Detection Type	Tier	Rationale
`\bdual process\b(?=.*\bmoral\b)`	Strong	S	Greene's core framework label in moral context
`\bjoshua greene\b`	Strong	S	Primary theorist attribution
`\butilitarian vs deontological processing\b`	Strong	S	Signature dual-process prediction
`\bsystem 1.*system 2\b`	Strong	S	Kahneman framing commonly applied to moral cognition
`\bsystem 2.*system 1\b`	Strong	S	Reverse-order mention of dual systems
`\butilitarian vs\.? deontological\b(?=.*\b(?:judg|reason|dilemma|moral)\b)`	Strong	S	Utilitarian-deontological contrast in moral judgment context
`\bcognitive load\b(?=.*\bmoral\b)`	Strong	S	Key experimental manipulation in dual-process moral studies
`\bpersonal force\b(?=.*\b(?:moral|dilemma)\b)`	Strong	S	Greene's personal/impersonal distinction
`\bpersonal vs\.? impersonal\b(?=.*\bdilemma\b)`	Strong	S	Personal vs. impersonal dilemma framing
`\bgreene\b(?=.*\bmoral\b)`	Strong	S	Theorist surname in moral context
`\baffect\b(?=.*\bcognition\b)(?=.*\bmoral\b)`	Strong	S	Affect-cognition interaction in moral judgment
`\b[illegible]\b`	Medium	M	[illegible]
`\b[illegible]\b(?=.*\b[illegible]\b)

False-positive suppression patterns:

Pattern (regex)	Tier	Rationale
`\bdual process\b(?=.*\b(?:cognitive science|memory|attention|perception)\b)(?!.*\bmoral\b)`	FP	Dual-process in non-moral cognitive science
`\bfast and slow\b(?!.*\bmoral\b)`	FP	Kahneman without moral context
`\bautomatic\b(?!.*\b(?:moral|ethic|dilemma|utilitarian|deontological)\b)`	FP	Generic "automatic" without ethical context

Social Intuitionism — New Patterns (14 patterns)

Prior to expansion, the vault contained 3 patterns for Social Intuitionism. All patterns below are new additions to the existing base.

Pattern (regex)	Detection Type	Tier	Rationale
`\bhaidt\b(?=.*\b(?:intuit|moral|foundation|dumbfound)\b)`	Strong	S	Haidt in intuitionist/moral context
`\bpost[-]hoc rationali[sz]\w+\b(?=.*\bmoral\b)`	Strong	S	Post-hoc rationalization of moral judgments
`\bgut feeling\b(?=.*\bmoral\b)`	Strong	S	Intuitive moral "gut feelings"
`\bemotion[-]first\b(?=.*\bmoral\b)`	Strong	S	Emotion-first model of moral judgment
`\bmoral heuristic\b`	Strong	S	Moral heuristics (Sunstein, Gigerenzer)
`\bmoral sentiment\b(?=.*\b(?:judgment|intuiti|reason)\b)`	Strong	S	Moral sentiment in judgment context
`\bintuitive judgment\b(?=.*\bmoral\b)`	Medium	M	Intuitive moral judgment
`\bmoral sentiment\b`	Medium	M	Moral sentiment (general)
`\bhumean\b(?=.*\bmoral\b)`	Medium	M	Humean moral philosophy (reason as slave of passions)
`\bsentimentalism\b(?=.*\b(?:moral|ethic)\b)`	Medium	M	Moral sentimentalism tradition
`\bintuitive\b(?=.*\bmoral\b)(?=.*\b(?:judg|reason|response)\b)`	Medium	M	Intuitive moral judgment/reasoning
`\baffect\b(?=.*\breason\w*\b)(?=.*\bmoral\b)`	Medium	M	Affect-reason interaction in morality

False-positive suppression patterns:

Pattern (regex)	Tier	Rationale
`\bintuition\b(?!.*\b(?:moral|intuitionist|dumbfounding|judgment|sentiment|ethic)\b)`	FP	Generic "intuition" without moral context
`\bsentiment\b(?!.*\b(?:moral|ethic|intuitionist|humean)\b)`	FP	NLP sentiment analysis (non-moral)

Dyadic Morality — New Patterns (11 patterns)

Prior to expansion, the vault contained 3 patterns for Dyadic Morality. All patterns below are new additions.

Pattern (regex)	Detection Type	Tier	Rationale
`\bharm perception\b(?=.*\b(?:moral|agent|patient)\b)`	Strong	S	Harm perception in moral agent-patient context
`\bschein\b(?=.*\b(?:moral|dyadic|gray|typecasting)\b)`	Strong	S	Chelsea Schein (co-author with Gray)
`\bagent[- ]patient\b(?=.*\bharm\b)`	Strong	S	Agent-patient harm dyad
`\btheory of dyadic morality\b`	Strong	S	Full theory name
`\bmoral dyad\b`	Medium	M	Moral dyad terminology
`\bagent and patient\b(?=.*\bmoral\b)`	Medium	M	Agent and patient in moral context
`\bmoral patient\b`	Medium	M	Moral patiency concept
`\bmoral agent\b(?=.*\bpatient\b)`	Medium	M	Moral agent with patient co-occurrence
`\bcomplete moral dyad\b`	Medium	M	Complete moral dyad
`\bharm[- ]based moral\w*\b`	Medium	M	Harm-based morality
`\bgray\b(?=.*\bschein\b)(?=.*\bmoral\b)`	Medium	M	Gray and Schein in moral context

False-positive suppression patterns:

Pattern (regex)	Tier	Rationale
`\bdyad\b(?!.*\b(?:moral|typecasting|kurt gray|harm|agent|patient)\b)`	FP	Generic “dyad” (e.g., parent-child dyad)
`\bpatient\b(?!.*\b(?:moral|agent|dyad|typecasting|harm)\b)`	FP	Medical “patient”

Scoring Formula

All theories use the same scoring formula:

```
score = (strong_matches x 15) + (medium_matches x 5) - (fp_context x 10) +
title_bonus(20)
score = clamp(score, 0, 100)
```

A paper is classified as engaging a theory when `score >= 50`. The "strong floor" rule ensures that any paper with at least one strong-tier match receives a minimum score of 50.

Supplementary Table 4 (ST4). 222 Candidate Benchmarks: Source Signal Distribution

Candidates were identified through a five-signal detection pipeline applied to 2,099 AI ethics papers. Signal definitions: **S1** = moral psychology score >= 50 on any of 8 mp_* fields; **S2** = benchmark/eval/dataset flag with >= 2 keyword matches; **S3** = published DATA_URL with >= 3 keyword matches; **S4** = >= 5 keyword matches regardless of other metadata; **S5** = known benchmark name in title or abstract.

Signal Frequency

Signal	Description	Papers Detected	% of 221 Candidates
S1	Moral psychology score >= 50	153	69.2%
S2	Benchmark flag + keywords	118	53.4%
S3	DATA_URL + keywords	65	29.4%
S4	High keyword density (>= 5)	8	3.6%
S5	Known benchmark name	0*	–

*Note: S5 matches were absorbed into S1–S4 during deduplication. Candidates matching known benchmark names (ETHICS, MoralBench, ValueBench, Scruples, Delphi, MACHIAVELLI, Moral Machine, MoralChoice, MoralExceptQA) were all also detected by other signals.

Signal Combination Distribution

Signal Combination	Count	% of 221	Interpretation
S1 only	82	37.1%	High mp_* score but not flagged as benchmark or with data
S2 only	32	14.5%	Flagged as benchmark with keywords but mp_* score < 50
S1 + S2	31	14.0%	High mp_* score and benchmark flag
S1 + S2 + S3	29	13.1%	Full convergence: high score, benchmark flag, and public data
S2 + S3	18	8.1%	Benchmark with data but mp_* score < 50
No signal recorded	11	5.0%	Included via manual review or edge-case matching
S1 + S2 + S3 + S4	6	2.7%	Maximum signal convergence (4 signals)
S1 + S3	5	2.3%	High mp_* score with data but not flagged as benchmark
S3 only	5	2.3%	Data available with keywords but low score and no flag
S2 + S3 + S4	2	0.9%	Benchmark with data and high keyword density

Multi-Signal vs. Single-Signal

Category	Count	% of 221
Single signal only	119	53.8%
Multiple signals (2+)	91	41.2%
No signal recorded	11	5.0%

Candidate Type Distribution

Type	Count	% of 221
Benchmark	176	79.6%
Dataset	43	19.5%
Other	2	0.9%

Candidate Year Distribution

Year	Count	Year	Count
2015	1	2022	8
2016	1	2023	27
2018	5	2024	62
2019	1	2025	97
2020	6	2026	10
2021	3		

Supplementary Table 5 (ST5). Assessment Coverage Matrix (55 Benchmarks x Key Features)

All 55 benchmarks receiving full 28-feature characterization are listed below. The table reports seven key features: A1 (theories engaged), A2 (operationalization level, 1–4), A5 (human baseline), A6 (construct validity, 1–5), B1 (task format), B2 (dataset size), and primary theory. Benchmarks are grouped by source: 40 from the primary council assessment and 15 from the secondary "hidden gem" scan.

Primary Assessment (40 benchmarks)

#	Benchmark	Theories	A1	A2	A5	A6	B1	B2
1	Moral Machine Experiment (Awad et al., 2018)	Trolley	1	3	Published norms	5	Forced choice	40,000
2	Crowdsourcing Moral Machines (Awad et al., 2020)	Trolley	1	3	Published norms	4	Forced choice	40,000
3	Moral Foundations Vignettes (Clifford et al., 2015)	MFT	1	4	Published norms	5	Scenario rating	132
4	Moral Foundations Twitter Corpus (Hoover et al., 2020)	MFT	1	3	Paper-specific	4	Multi-label	35,108
5	Moral Foundations Reddit Corpus (Trager et al., 2022)	MFT	1	3	Paper-specific	4	Multi-label	16,123
6	Dutch MF Stimulus Database (Hopp et al., 2023)	MFT	1	4	Paper-specific	5	Scenario rating	240
7	Moral Foundations Weibo Corpus (Cao et al., 2024)	MFT	1	3	Paper-specific	3	Multi-label	25,671
8	Moral Foundations of LLMs (Abdulhai et al., 2024)	MFT	1	4	Published norms	5	Likert	30
9	Differences in MF of LLMs (Kirgis, 2024)	MFT	1	4	Published norms	4	Likert	116
10	MFTCXplain (Trager et al.,	MFT	1	3	Paper-	3	Multi-	3,000
[illegible]	[illegible] et al.)	Trolley	[illegible]	[illegible]	[illegible] specific	[illegible]	[illegible]	[illegible]

Hidden Gem Assessment (15 benchmarks)

#	Benchmark	Theories	A1	A2	A5	A6	B1	B2
41	EMNLP (Jiang et al., 2025)	Kohlberg	1	3	Paper-specific	3	Mixed	88
42	How Much of a Pluralist is ChatGPT? (2025)	Pluralism, MFT	2	4	Paper-specific	4	Scenario rating	120
43	The Straight and Narrow (Hu et al., 2026)	MFT	1	3	None	3	Scenario rating	0
44	Normative Eval with Everyday Moral Dilemmas (Sachdeva et al., 2025)	MFT	1	2	Published norms	3	MCQ	10,826
45	ValAct-15k (Huang et al., 2026)	Schwartz	1	4	Paper-specific	4	MCQ	15,000
46	Value-Spectrum (Li et al., 2024)	Schwartz	1	3	None	2	MCQ	50,191
47	AdAEM (Duan et al., 2025)	Schwartz	1	3	None	3	MCQ	12,310
48	Pluralistic Moral Gap (Russo et al., 2025)	Other	1	2	Published norms	3	Binary	1,618
49	Insights into Moral Reasoning (Sameer et al., 2025)	MFT, Kohlberg, Intuitionist	3	4	Paper-specific	4	Likert	2,438
50	PVQ-RR on LLMs (Hadar-Shoval et al., 2024)	Schwartz	1	4	Published norms	3	Likert	57
51	Bago et al. (2022) 45-Country Trolley	Trolley, Dual Process, Intuitionist	3	3	Published norms	3	Binary	27,502
52	[illegible]	[illegible]	4	3	None	4	Mixed	6,600
53	[illegible]	[illegible]	4	3	None	4	Mixed	6,600
54	[illegible]	[illegible]	4	3	None	4	Mixed	6,600

Feature Distribution Summary (55 benchmarks)

A2 — Construct Operationalization:

Level	Interpretation	Count	%
4	Validated instrument administered	16	29%
3	Theory-derived / adapted	25	45%
2	Theory-inspired, no validation	12	22%
1	Keyword-level engagement only	2	4%

A5 — Human Baseline Availability:

Level	Count	%
Published norms	23	42%
Paper-specific	18	33%
Partial	4	7%
None	10	18%

A6 — Construct Validity:

Score	Interpretation	Count	%
5	Multi-study validated	6	11%
4	Single-study validated	20	36%
3	Adapts validated instrument	20	36%
2	Theory-inspired, no validation	7	13%
1	Keyword-level only	2	4%

B1 — Task Format:

Format	Count	%
Forced-choice / Binary	14	25%
MCQ	8	15%
Likert	11	20%
Scenario rating / ranking	8	15%
Multi-label classification	3	5%
Free-text (incl. + MCQ)	5	9%
Mixed / Multi-turn	4	7%
N/A (survey, design tool)	2	4%

Theory Engagement (A1):

Theories Engaged	Count	%
0	1	2%
1	38	69%
2	9	16%
3	5	9%
4	2	4%

Primary Theory Distribution:

Theory	Count (primary or co-primary)
MFT	24
Trolley	13
Schwartz	11
Kohlberg	10
Dual Process	3
Social Intuitionism	2
Dyadic Morality	1
Moral Identity	1
Other / None	2

Note: Counts sum to > 55 because benchmarks engaging multiple theories are counted under each.

Supplementary Figures

Main Paper Figures (F1–F12)

Figure 1: Top-15 Benchmarks by Composite Score

File: `figures/fig1-composite-scores.png` | `figures-pdf/fig1-composite-scores.pdf` **Description:** Horizontal bar chart showing the top-15 benchmarks ranked by composite score (PD x 0.50 + PE x 0.30 + CI x 0.20), with stacked color bands indicating the contribution of each tier. Psychological Depth (blue), Practical Ease (orange), and Citation Impact (green) are shown as proportional segments. **Source:** 28-feature characterization framework applied to 55 assessed benchmarks.

Figure 2: Theory Distribution Across 222 Candidates

File: `figures/fig2-theory-distribution.png` | `figures-pdf/fig2-theory-distribution.pdf` **Description:** Bar chart showing the number of candidate benchmarks associated with each of the 8 moral psychology theories after keyword scorer expansion. MFT dominates at 101 candidates; Dyadic Morality is lowest at 7. **Source:** Five-signal detection pipeline output after keyword scorer expansion.

Figure 4: Theory Coverage Heatmap (8 Theories x 15 Benchmarks)

File: `figures/fig3-theory-heatmap.png` | `figures-pdf/fig3-theory-heatmap.pdf` **Description:** Heatmap showing which theories each of the top-15 benchmarks engages. Rows are theories (MFT, Schwartz, Trolley, Kohlberg, Social Intuitionism, Moral Identity, Dual Process, Dyadic Morality); columns are benchmarks. Cell intensity reflects engagement depth (primary, secondary, peripheral). The three empty rows (Dual Process, Social Intuitionism, Dyadic Morality with zero dedicated benchmarks) are the visual representation of Claim 3. **Source:** Expert council assessment of benchmark-theory alignment.

Figure 6: Implementation Status Grid

File: `figures/fig4-implementation-grid.png` | `figures-pdf/fig4-implementation-grid.pdf` **Description:** Matrix showing the implementation status of top-15 benchmarks across 5 evaluation harness categories (direct API, lm-eval, Inspect, custom, behavioral). Color indicates status: green (implemented), yellow (partially implemented), red (not implemented), gray (not applicable). **Source:** Implementation trial results and harness compatibility assessment.

Figure 3: Construct Validity (A6) Score Distribution

File: `figures/fig5-construct-validity.png` | `figures-pdf/fig5-construct-validity.pdf` **Description:** Histogram showing the distribution of A6 (construct validity evidence) scores across 55 assessed benchmarks. The median of 3/5 is highlighted. The right-skewed distribution shows that only 11% of benchmarks achieve multi-study validation (A6=5). **Source:** 28-feature framework assessment.

Figure 5: Human Baseline Availability (A5) Distribution

File: `figures/fig6-human-baselines.png` | `figures-pdf/fig6-human-baselines.pdf` **Description:** Pie/bar chart showing A5 (human baseline availability) distribution: Published norms (42%), Paper-specific (33%), Partial (7%), None (18%). **Source:** 28-feature framework assessment.

Figure 7: Cross-Vendor Convergence — Benchmark Discrimination

File: `figures/fig7-cross-vendor-convergence.png` | `figures-pdf/fig7-cross-vendor-convergence.pdf` **Description:** Horizontal bar chart showing the coefficient of variation (CV) for each of the 23 benchmarks across all 9 models, ordered from most convergent (CV=0, ceiling-tied) to most divergent. Color-coded by discrimination category: red (ceiling, CV=0), orange (near-invariant, CV<0.05), yellow (low variation, CV<0.10), green (discriminating, CV>=0.10). A dashed vertical line at CV=0.05 marks the threshold between near-invariant and meaningfully varying benchmarks. **Data source:** Cross-vendor comparison data (9 models x 23 benchmarks).

Figure 8: RLHF Moral Fingerprint — Three Vendors, One Profile

File: `figures/fig8-rlhf-fingerprint-radar.png` | `figures-pdf/fig8-rlhf-fingerprint-radar.pdf` **Description:** Radar (spider) chart with 6 axes representing the six universal RLHF signatures (WEIRD-liberal bias, Purity floor, Authority suppression, Ethical consistency, Post-conventional default, Deontological refusal asymmetry). Three overlapping polygons (Anthropic in blue, Google in red, OpenAI in green) demonstrate near-complete overlap, visualizing the vendor-invariant moral fingerprint. **Data source:** Normalized signature scores derived from cross-vendor analysis.

Figure 9: DIT P-Score by Model — Non-Monotonic Scale Sensitivity

File: `figures/fig9-scale-sensitivity-dit.png` | `figures-pdf/fig9-scale-sensitivity-dit.pdf` **Description:** Bar chart showing DIT P-scores for all 9 models, colored by vendor. The non-monotonic pattern within Anthropic (Haiku 60, Opus 75, Sonnet 85) is highlighted, with a horizontal reference line at human mean (~42). All models score well above human means; the paradox is that the mid-tier model outscores the flagship. **Data source:** MP3 (DIT) trial results across all 9 models.

Figure 10: The Haiku Dual-Process Effect

File: `figures/fig10-cot-effect-haiku.png` | `figures-pdf/fig10-cot-effect-haiku.pdf` **Description:** Bar chart showing the chain-of-thought (CoT) effect magnitude for each of the 9 models on the MP-DP (Dual Process) protocol. Eight models show a CoT effect of exactly 0.0; Claude Haiku shows +0.125 (12.5 percentage points more utilitarian with CoT). An annotation arrow highlights Haiku as the only model with a dual-process signature. Haiku's bar has a thick black border for emphasis. **Data source:** MP-DP trial results (with/without CoT condition).

Figure 11: MFT Foundation Profiles Across 9 Models
File: `figures/fig11-mft-profile-9models.png` | `figures-pdf/fig11-mft-profile-9models.pdf` **Description:** Grouped bar chart with 5 foundation axes (Care, Fairness, Loyalty, Authority, Purity) and 9 model bars per foundation, colored by vendor. Shows the universal WEIRD-liberal profile: high Care and Fairness, low Loyalty and Authority, and a universal purity floor (~1.0). Annotations highlight "Individualizing (high)", "Binding (low)", and "Purity floor (universal)". **Data source:** MFV (MP4) scores across 9 models.

Figure 12: Cultural Persona MFQ Shift Magnitudes (MP18)
File: `figures/fig12-cultural-persona-shift.png` | `figures-pdf/fig12-cultural-persona-shift.pdf` **Description:** Grouped bar chart showing the shift in MFQ Individualizing/Binding ratio when models are prompted with Confucian, Islamic, Ubuntu, and Western (baseline) cultural personas. Three Anthropic models shown (Opus, Sonnet, Haiku). All models show negative shifts (toward more balanced Binding) under non-Western personas, with Islamic persona producing the largest shift. **Data source:** MP18 (Cultural Persona) trial results.

Supplementary Figures (SF1–SF4)

SF1: Per-Benchmark Score Distributions Across 9 Models
Specification: Panel of 23 mini-charts (one per benchmark), each showing 9 model data points colored by vendor (blue=Anthropic, red=Gemini, green=OpenAI) overlaid on a box showing the interquartile range. Benchmarks ordered by CV (most convergent at top). Panels for MP6, MP7, MP11, MP22 show zero-width boxes. **Dimensions:** 8.5" x 14" (full page, portrait) **Status:** To be generated from cross-vendor comparison JSON.

SF2: Within-Vendor vs. Between-Vendor Variance Decomposition
Specification: Stacked bar chart for each of the 23 benchmarks showing the proportion of total variance explained by within-vendor differences (lighter shade) vs. between-vendor differences (darker shade). Sorted by between-vendor proportion. For ~8 of 13 non-ceiling benchmarks, within-Anthropic variance exceeds total between-vendor variance. **Dimensions:** 10" x 6" (landscape) **Status:** To be generated from variance decomposition analysis.

SF3: Schwartz Value Profiles Across 9 Models
Specification: Parallel coordinates plot with 10 Schwartz value types as axes, colored lines for each model (by vendor). Shows the universal pattern of high Universalism/Self-Direction and low Achievement/Power. **Dimensions:** 10" x 6" (landscape) **Status:** To be generated from MP5 (ValueCompass) data.

SF4: Dyadic Morality Attribution Patterns

Specification: Grouped bar chart showing mean structural blame, blame above zero, and dyadic effect sizes across all 9 models for the MP-DM protocol. Shows the extremely narrow range (3.208–3.292 on blame), visualizing the near-invariance of dyadic processing. **Dimensions:** 8” x 5” **Status:** To be generated from MP-DM trial data.

Figure Generation Notes

Color palette (consistent across all figures): - Anthropic: `#4A90D9` (blue) - Google/Gemini: `#E74C3C` (red) - OpenAI: `#2ECC71` (green)

Typography: - Title: 13pt bold - Axis labels: 11pt - Data labels: 10pt - Annotations: 9-10pt italic

Output formats: All figures generated as both PNG (300 DPI) and PDF.

Generation scripts: `scripts/build_figures_pdf.py` handles F1-F12. SF1-SF4 specifications are provided here for future generation from raw trial data.